

SIEMENS EDA

Algorithmic C (AC) Math Library Reference Manual

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Chapter 1: Introduction to ac_math

The Algorithmic C Math Library (*ac_math*) contains synthesizable C++ functions commonly used in Digital Signal Processing applications. The functions use the Algorithmic C data types and are meant to serve as examples on how to write parameterized models and to facilitate migrating an algorithm from using floating-point to fixed-point arithmetic where the math functions either need to be computed dynamically or via lookup tables or piecewise linear approximations.

The input and output arguments of the math functions are parameterized so that arithmetic may be performed at the desired fixed point precision and provide a high degree of flexibility on the area/performance trade-off of hardware implementations obtained during Catapult synthesis.

The hardware implementations produced by Catapult on the math functions are bit accurate. Simulation of the RTL can thus be easily compared to the C++ simulation of the algorithm. The following sections provide a summary of the *ac_math* library:

- [Summary of Functions](#)
- [Types of Approximations](#)
- [Installing the ac_math library](#)

1.1.1. Using the ac_math library

In order to utilize any of the math functions, add the following include line to the source:

```
#include <ac_math.h>
```

1.2. Summary of Functions

The following sections summarize the functions and classes currently supported in the *ac_math* library. A discussion of the approximation methods follows.

1.2.1. Basic Math Functions

Function Type/Call	Approx. Method	Supported Data Types			
		ac_fixed	ac_float	ac_complex	Other float*
Absolute Value (<i>ac_abs()</i>)	N/A	Yes	Yes	Yes	No
Division (<i>ac_div()</i>)	N/A	Yes	Yes	Yes	No
Normalization (<i>ac_normalize()</i>)	N/A	Yes	No	Yes	No
Reciprocal (<i>ac_reciprocal_pwl()</i>)	PWL	Yes	Yes	Yes	Yes
Reciprocal (<i>ac_reciprocal_pwl_ha()</i>)	PWL	Yes	Yes	Yes	Yes
Reciprocal (<i>ac_reciprocal_pwl_vha()</i>)	PWL	Yes	Yes	Yes	Yes

Function Type/Call	Approx. Method	Supported Data Types			
		ac_fixed	ac_float	ac_complex	Other float*
Logarithm Base e (<i>ac_log_pwl()</i>)	PWL	Yes	Yes	No	Yes
Logarithm Base e (<i>ac_log_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Logarithm Base 2 (<i>ac_log2_pwl()</i>)	PWL	Yes	Yes	No	Yes
Logarithm Base 2 (<i>ac_log2_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Exponent Base e (<i>ac_exp_pwl()</i>)	PWL	Yes	Yes	No	Yes
Exponent Base e (<i>ac_exp_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Exponent Base 2 (<i>ac_pow2_pwl()</i>)	PWL	Yes	Yes	No	Yes
Exponent Base 2 (<i>ac_exp2_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Generic Exponent (<i>ac_pow_pwl()</i>)	PWL	Yes	No	No	No
Generic Exponent (<i>ac_pow_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Square Root (<i>ac_sqrt_pwl()</i>)	PWL	Yes	Yes	Yes	Yes
Square Root (<i>ac_sqrt()</i>)	N/A	Yes	Yes	No	Yes
Inverse Square Root (<i>ac_inverse_sqrt_pwl()</i>)	PWL	Yes	Yes	Yes	Yes
Inverse Square Root (<i>ac_inverse_sqrt_pwl_vha()</i>)	PWL	Yes	Yes	Yes	Yes
Sine/Cosine (<i>ac_sincos_lut()</i>)	LUT	Yes	No	No	No
Sine/Cosine (<i>ac_sincos_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Cosine (<i>ac_cos_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Sine (<i>ac_sin_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Tangent (<i>ac_tan_pwl()</i>)	PWL	Yes	Yes	No	Yes
Arctangent (<i>ac_atan_pwl()</i>)	PWL	Yes	Yes	No	Yes
Arctangent (<i>ac_atan_pwl_ha()</i>)	PWL	Yes	Yes	No	Yes
Arctangent (<i>ac_atan_pwl_vha()</i>)	PWL	Yes	Yes	Yes	Yes
Arctangent (<i>ac_arctan_cordic()</i>)	CORDIC	Yes	No	No	No
Arccosine (<i>ac_arccos_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Arcsine (<i>ac_arcsin_cordic()</i>)	CORDIC	Yes	Yes	No	Yes
Shift Left (<i>ac_shift_left()</i>)	N/A	Yes	No	Yes	No
Shift Right (<i>ac_shift_right()</i>)	N/A	Yes	No	Yes	No
Hyperbolic Tangent (<i>ac_tanh_pwl()</i>)	PWL	Yes	Yes	No	Yes
Sigmoid (<i>ac_sigmoid_pwl()</i>)	PWL	Yes	Yes	No	Yes
Softmax (<i>ac_softmax_pwl()</i>)	PWL	Yes	No	No	No
Softplus (<i>ac_softplus_pwl()</i>)	PWL	Yes	No	No	No
Softsign (<i>ac_softsign_pwl()</i>)	PWL	Yes	No	No	No

Function Type/Call	Approx. Method	Supported Data Types			
		ac_fixed	ac_float	ac_complex	Other float*
Elu (ac_elu_pwl())	PWL	Yes	No	No	No
Selu (ac_selu_pwl())	PWL	Yes	No	No	No
Gelu (ac_gelu_pwl())	PWL	Yes	No	No	No
LeakyReLU (ac_leakyrelu())	N/A	Yes	No	No	No
ReLU (ac_relu())	N/A	Yes	No	No	No
PReLU (ac_prelu())	N/A	Yes	No	No	No
AC Float Adder Tree (add_tree(), add_tree_ptr)	N/A	No	Yes	No	No
AC Float Block Adder Tree (block_add_tree(), block_add_tree_ptr())	N/A	No	Yes	No	No
Optimized ac_fixed*ac_float Multiplication (ac_ffx_mul())	N/A	Yes	Yes	No	No
Barrel Shift (ac_barrel_shift())	N/A	*ac_int support only			
Padded Divison (ac_div_v2())	N/A	*Currently ac_int support only			

* Note: “Other float” represents ac_ieee_float and ac_std_float support.

1.2.2. AC Matrix Class

The class ac_matrix implements a 2-D container class with a template parameter to specify the data type of the internal storage.

The class has member functions to implement some common operations including

- Assignment: operator=()
- Read-Only and Read-Write Element Access: *this(<row>,<col>)
- Comparison: operator!=(), operator==()
- Piecewise Addition: operator+(), operator+=()
- Piecewise Subtraction: operator-(), operator-=()
- Piecewise Multiplication: pwisemult()
- Matrix Multiplication (nested loops): operator*()
- Matrix Transpose: transpose()

- Sum All Elements: `sum()`
- Scale All Elements: `scale(value)`
- Formatted Stream Output: `ostream &operator<<()`

When using the computational functions with AC Datatypes, the form that returns a value is designed in such a way as to determine the full precision required in the output type in order to preserve accuracy during the operation. So using `operator+` between two 10 bit `ac_fixed` matrices will return an 11 bit `ac_fixed` matrix. If you wish to prevent the bit growth and accept the truncation, you can use the compound operators `+=`, `-=`, etc. so that the target object receives the truncated values.

In addition to the built-in member functions, the `ac_math` library also includes stand-alone functions for more complicated linear algebra operations as described in the next section.

1.2.3. Linear Algebra Functions

The `ac_math` library includes several linear algebra functions that operate on either `ac_matrix` or plain C-style arrays. These functions, when used with AC Datatypes, are designed to give the user greater control over the bit precision of internal variables and the return value.

- Matrix Multiplication
- Matrix Determinant
- Cholesky Decomposition
- Cholesky Inverse
- QR Decomposition

1.3. Types of Approximations

The following sections discuss the calculation methods used by the functions in the `ac_math` library and the trade-offs of using one method over another in your design.

- Piecewise Linear
- Lookup Table (LUT)
- CORDIC
- Miscellaneous Math Functions

1.3.1. Piecewise Linear

Some of the functions available in the `ac_math` library are implemented as piecewise linear (PWL) approximations.

A PWL approximation of a function essentially attempts to replicate a function as a set of line segments that are the closest fit to the actual function curve.

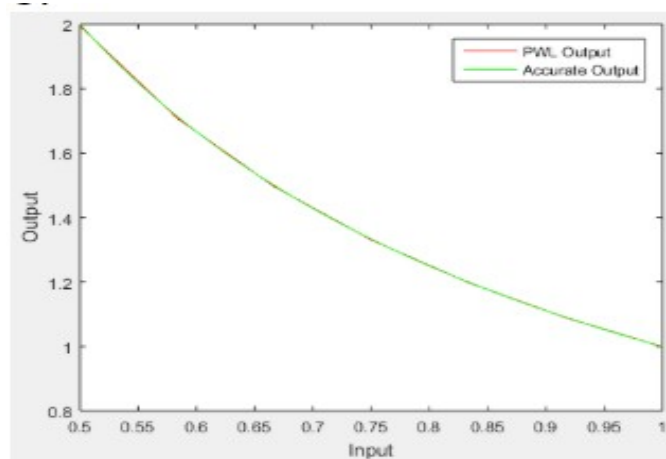


Illustration 1: Reciprocal PWL approximation

An example of a PWL approximation would be the *ac_reciprocal_pwl()* function, for the interval of $[0.5, 1)$. A graphical depiction is given above. The interval is divided into 6 segments. As is evident from the graph and the close nature of the fit, where the red line (PWL output) is virtually indistinguishable from the green one (Accurate output), the approximate output is very close to the accurate output.

Constructing the PWL Lookup Table

In order to construct the PWL approximation, we need information on the accurate function output for the end points of each segment. Once that information is obtained, we calculated the slope and intercept value for that segment based on the input and output values corresponding to the segment end points. The segments are then shifted slightly in the direction opposite to the direction of concavity of the function in the interval of a particular segment, in order to further optimize the fit obtained. This is done by appropriately changing the intercept values for that segment. The slope and intercept values are then stored in separate lookup tables.

The errors obtained are generally small and tolerable for approximate applications. For instance, the reciprocal PWL approximation just mentioned has a maximum absolute error of 0.005511 over the interval of $[0.5, 1)$, which corresponds to at least 7 error-free fractional bits in case of a fixed-point PWL output.

Pitfalls

However, it is important to note that if PWL function calls are cascaded, this error can build up and exceed tolerances. Such a situation is encountered, for instance, if the PWL functions are used for linear algebra functions that operate on matrices. Hence, the user must be aware of this pitfall and take steps to avoid it if necessary.

C++ Compiler

The PWL functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 as the standard for their compilation, or a later standard, failing which a compile-time error is thrown.

Rounding Mode for PWL Output

The internal variable to store the PWL output for all the functions is set to have rounding turned off by default (*AC_TRN*). This is because the default bitwidth of the PWL output operates at full precision, and no rounding is required. If, however, the user wishes to reduce the number of bits and use another rounding mode, they can pass it as a template argument. For an example on how to pass this argument for all the PWL functions that are implemented, please refer to the “Function Prototypes” and “Example Function Calls” sections for the various functions as explained later in the documentation.

1.3.2. Lookup Table (LUT)

Some of the functions can be efficiently implemented as a lookup table. For example, the sine and cosine functions can be defined as a lookup table where the symmetry of the functions can be exploited to minimize the size of the table. In most cases the table is described using literal values expressed as C++ double precision values. The context in which the function is then used (determined by the fixed-point output precision) will determine the amount of error in the function.

1.3.3. CORDIC

Some of the hyperbolic and trigonometric functions have implementations based on the CORDIC algorithm. These implementations use an iterative approach that typically converges with one digit per iteration. The iterations may involve addition, subtraction, bit shifts and table lookups. Given the iterative nature of these implementations the resulting hardware will be larger and/or slower than the PWL or LUT implementations but offer the best accuracy.

1.3.4. Miscellaneous Math Functions

A number of math functions that do not fall in the categories listed above are also provided. These are the absolute value, division, square root and shifting functions. These functions give an exact or very accurate output.

1.4. Installing the *ac_math* library

The library consists of three directories, shown here:

```
.  
  
|-- include  
  
| |-- ac_math  
  
| | |-- ac_abs.h  
  
| | |-- ac_arccos_cordic.h  
  
| | |-- ac_arcsin_cordic.h  
  
| | |-- ac_atan2_cordic.h
```

| | |-- ac_atan_pwl.h
| | |-- ac_atan_pwl_ha.h
| | |-- ac_atan_pwl_vha.h
| | |-- ac_barrel_shift.h
| | |-- ac_chol_d.h
| | |-- ac_cholinv.h
| | |-- ac_determinant.h
| | |-- ac_div.h
| | |-- ac_hcordic.h
| | |-- ac_inverse_sqrt_pwl.h
| | |-- ac_inverse_sqrt_pwl_vha.h
| | |-- ac_log_pwl.h
| | |-- ac_matrixmul.h
| | |-- ac_normalize.h
| | |-- ac_pow_pwl.h
| | |-- ac_qrd.h
| | |-- ac_random.h
| | |-- ac_reciprocal_pwl.h
| | |-- ac_reciprocal_pwl_ha.h
| | |-- ac_reciprocal_pwl_vha.h
| | |-- ac_shift.h
| | |-- ac_sigmoid_pwl.h
| | |-- ac_sincos_cordic.h
| | |-- ac_sincos_lut.h


```
| | |-- ac_softmax_pwl.h
| | |-- ac_sqrt.h
| | |-- ac_sqrt_pwl.h
| | |-- ac_tan_pwl.h
| | |-- ac_tanh_pwl.h
| | |-- ac_softplus_pwl.h
| | |-- ac_softsign_pwl.h
| | |-- ac_elu_pwl.h
| | |-- ac_selu_pwl.h
| | |-- ac_gelu_pwl.h
| | |-- ac_leakyrelu.h
| | |-- ac_relu.h
| | |-- ac_flx_mul.h
| | `-- ac_prelu.h
| |-- ac_float_add_tree.h
| |-- ac_math.h
| `-- ac_matrix.h
|-- lutgen
| |-- ac_atan_pwl_lutgen.cpp
| |-- ac_atan_pwl_ha_lutgen.cpp
| |-- ac_atan_pwl_vha_lutgen.cpp
| |-- ac_inverse_sqrt_pwl_lutgen.cpp
| |-- ac_inverse_sqrt_pwl_vha_lutgen.cpp
| |-- ac_log_pwl_lutgen.cpp
```

```
| |-- ac_pow_pwl_lutgen.cpp
| |-- ac_reciprocal_pwl_lutgen.cpp
| |-- ac_reciprocal_pwl_ha_lutgen.cpp
| |-- ac_reciprocal_pwl_vha_lutgen.cpp
| |-- ac_sigmoid_pwl_lutgen.cpp
| |-- ac_sincos_lut_lutgen.cpp
| |-- ac_sqrt_pwl_lutgen.cpp
| |-- ac_tan_pwl_lutgen.cpp
| |-- ac_tanh_pwl_lutgen.cpp
| `-- helper_functions.h

|-- pdfdocs

| `-- ac_math_ref.pdf

`-- tests

    |-- Makefile

    |-- rtest_ac_abs.cpp

    |-- rtest_ac_arccos_cordic.cpp

    |-- rtest_ac_arcsin_cordic.cpp

    |-- rtest_ac_atan2_cordic.cpp

    |-- rtest_ac_atan_pwl.cpp

    |-- rtest_ac_atan_pwl_ha.cpp

    |-- rtest_ac_atan_pwl_vha.cpp

    |-- rtest_ac_chol_d.cpp

    |-- rtest_ac_cholinv.cpp

    |-- rtest_ac_determinant.cpp
```

```
|-- rtest_ac_div.cpp  
  
|-- rtest_ac_exp2_cordic.cpp  
  
|-- rtest_ac_exp_cordic.cpp  
  
|-- rtest_ac_exp_pwl.cpp  
  
|-- rtest_ac_inverse_sqrt_pwl.cpp  
  
|-- rtest_ac_inverse_sqrt_pwl_vha.cpp  
  
|-- rtest_ac_log2_cordic.cpp  
  
|-- rtest_ac_log2_pwl.cpp  
  
|-- rtest_ac_log_cordic.cpp  
  
|-- rtest_ac_log_pwl.cpp  
  
|-- rtest_ac_matrix.cpp  
  
|-- rtest_ac_matrixmul.cpp  
  
|-- rtest_ac_normalize.cpp  
  
|-- rtest_ac_pow2_pwl.cpp  
  
|-- rtest_ac_pow_cordic.cpp  
  
|-- rtest_ac_pow_pwl.cpp  
  
|-- rtest_ac_qrd.cpp  
  
|-- rtest_ac_reciprocal_pwl.cpp  
  
|-- rtest_ac_reciprocal_pwl_ha.cpp  
  
|-- rtest_ac_reciprocal_pwl_vha.cpp  
  
|-- rtest_ac_shift.cpp  
  
|-- rtest_ac_sigmoid_pwl.cpp  
  
|-- rtest_ac_sincos_cordic.cpp  
  
|-- rtest_ac_sincos_lut.cpp
```

```
|-- rtest_ac_softmax_pwl.cpp  
  
|-- rtest_ac_sqrt.cpp  
  
|-- rtest_ac_sqrt_pwl.cpp  
  
|-- rtest_ac_tan_pwl.cpp  
  
|-- rtest_ac_tanh_pwl.cpp  
  
|-- rtest_ac_softplus_pwl.cpp  
  
|-- rtest_ac_softsign_pwl.cpp  
  
|-- rtest_ac_elu_pwl.cpp  
  
|-- rtest_ac_selu_pwl.cpp  
  
|-- rtest_ac_gelu_pwl.cpp  
  
|-- rtest_ac_leakyrelu.cpp  
  
|-- rtest_ac_relu.cpp  
  
|-- rtest_ac_prelu.cpp  
  
|-- rtest_ac_flfx_mul.cpp  
  
`-- rtest_ac_float_add_tree.cpp
```

In order to utilize this library you must have the AC Datatypes package installed and configure your software environment to provide the path to the AC Datatypes “include” directory as part of your C++ compilation arguments.

Testing and Error Calculation

The **ac_math** library includes a series of unit tests that exercise each of the approximation functions across various fixed-point bit widths to ensure that the accuracy of the approximation is within a certain tolerance of the standard C++ math library equivalent (under the same input and output bit-width constraints). To exercise these tests from a Linux shell, use the following GNU make command line:

```
gmake all AC_TYPES_INC=<path to AC Datatypes include directory>
```

where the variable **AC_TYPES_INC** specifies the path to the install location of the AC Datatypes package. The results of the tests look something like this:

```
TEST: ac_inverse_sqrt_pwl()  INPUT: ac_float< 5, 3, 3, AC_RND>  OUTPUT:  
ac_float<64,32,10, AC_RND>  RESULT: PASSED , max error (0.083916)
```

```
TEST: ac_inverse_sqrt_pwl() INPUT: ac_float< 5, 1, 3, AC_RND> OUTPUT:
ac_float<64,32,10, AC_RND> RESULT: PASSED , max error (0.083916)
TEST: ac_inverse_sqrt_pwl() INPUT: ac_float< 5, 0, 3, AC_RND> OUTPUT:
ac_float<64,32,10, AC_RND> RESULT: PASSED , max error (0.083916)
TEST: ac_inverse_sqrt_pwl() INPUT: ac_float< 5,-2, 3, AC_RND> OUTPUT:
ac_float<64,32,10, AC_RND> RESULT: PASSED , max error (0.083916)
TEST: ac_inverse_sqrt_pwl() INPUT: ac_float< 5, 9, 3, AC_RND> OUTPUT:
ac_float<60,30,11, AC_RND> RESULT: PASSED , max error (0.083916)
```

The output shows the function under test, the input and output bit-widths and the maximum error observed over the tested range of minimum and maximum values expressible in the input type stepping by the smallest value expressible in the input type.

In addition to the PWL approximations, the LUT and CORDIC implementations are also subject to error calculations on their output, in order to evaluate whether the error is tolerable.

For comparison, the input to the function being tested is converted to a double value (or `ac_complex<double>`, in case of complex inputs), and this double value is passed to the equivalent C++ math library function. The output, which is a double (or `ac_complex<double>`, in case of complex outputs), is subject to quantization according the output type of the function being tested.

It is this quantized output that is compared to the output of the function being tested. The comparison is done either in terms of relative error or absolute error. For real outputs, the relative/absolute error between the accurate (C++ math library) output and approximate output is calculated, based upon whether the accurate output lies above or below a pre-defined threshold; if the accurate output lies above the threshold, the relative error is calculated, if it lies below the threshold, the absolute error is calculated. For complex outputs, the error is calculated in a manner similar to the real output, with the differences being that (a) the quantization is carried out on the real and imaginary parts of the accurate output with respect to the type of the real and imaginary parts of the output of the function being tested and (b) the magnitude of the relative/absolute error is what is reported. This approach for comparison is followed for all the *ac_math* functions except for the:

- *ac_log_pwl*, *ac_log2_pwl*, *ac_log_cordic*, *ac_log2_cordic*, *ac_atan_pwl*, *ac_atan_pwl_ha*, *ac_atan_pwl_vha*, *ac_tanh_pwl*, *ac_sigmoid_pwl*, *ac_sincos_lut* and *ac_softmax_pwl* functions where only the absolute error is calculated.
- *ac_abs*, *ac_shift_left*, *ac_shift_right*, *ac_barrel_shift* and *ac_normalize* and *ac_ffx_mul* functions, which return an exact value at the output. The testing for these functions involves making sure that the expected values are the same as the corresponding actual values that the design returns.

Chapter 2: Piecewise Linear Functions

The `ac_math` package includes the following piecewise linear functions:

- [Logarithm \(ac_log_pwl / ac_log2_pwl\)](#)
- [Power \(ac_exp_pwl / ac_pow2_pwl / ac_pow_pwl\)](#)
- [Reciprocal \(ac_reciprocal_pwl\)](#)
- [High-accuracy Reciprocal \(ac_reciprocal_pwl_ha\)](#)
- [Very High-accuracy Reciprocal \(ac_reciprocal_pwl_vha\)](#)
- [Square Root \(ac_sqrt_pwl\)](#)
- [2.7Inverse Square Root \(ac_inverse_sqrt_pwl\)](#)
- [Very High-Accuracy Inv Sqrt \(ac_inverse_sqrt_pwl_vha\)](#)
- [Tangent \(ac_tan_pwl\)](#)
- [Arctangent \(ac_atan_pwl\)](#)
- [High-accuracy Arctangent \(ac_atan_pwl_ha\)](#)
- [Very High-accuracy Arctangent \(ac_atan_pwl_vha\)](#)
- [Sigmoid \(ac_sigmoid_pwl\)](#)
- [Hyperbolic Tangent \(ac_tanh_pwl\)](#)
- [Softmax \(ac_softmax_pwl\)](#)
- [Softplus \(ac_softplus_pwl\)](#)
- [Softsign \(ac_softsign_pwl\)](#)
- [Elu \(ac_elu_pwl\)](#)
- [Selu \(ac_selu_pwl\)](#)
- [Gelu \(ac_gelu_pwl\)](#)

Every function above, except for the softmax, softplus, softsign and elu functions, contains its own unique PWL implementation.

Every function normalizes the function input to a value that is within the domain of the PWL approximation. The PWL approximation produces an output for this normalized value, and then applies a sort of “denormalization” that cancels out the effect of the previous normalization on the PWL output. This gives us an approximate output for the input that was supplied to the function. For each PWL approximation except for the softmax operation, the details are given in the table below:

PWL Function	Domain	Segments	Max. Abs. Error	Min. No. of Error-free Fractional Bits
Logarithm	[0.5, 1)	8	0.001251	9
Power	[0, 1)	4	0.003718	8
Reciprocal	[0.5, 1)	8	0.003274	8
High-accuracy Reciprocal	[0.5, 1)	32	0.000236429	12
Very High-accuracy Reciprocal	[0.5, 1)	64	6.33158e-05	13
Square Root	[0.5, 1)	4	0.000582	10
Inverse Square Root	[0.5, 1)	8	0.000893	10
Very High-accuracy Inv. Sqrt.	[0.5, 1)	32	6.56307e-05	13
Tangent	[0, $\pi/4$)	8	0.002300	8
Arctangent	[0, 1)	4	0.002885	8
High-accuracy Arctangent	[0, 1)	32	8.9351e-05	13
Very high-accuracy Arctangent	[0, 1)	64	4.59311e-05	14
Sigmoid	[0, 5)	8	0.002755	10
Softmax	N/A	N/A	N/A	N/A
Softplus	N/A	N/A	N/A	N/A
Softsign	N/A	N/A	N/A	N/A
Elu	N/A	N/A	N/A	N/A
Selu	N/A	N/A	N/A	N/A
Gelu	N/A	N/A	N/A	N/A

The following subsections describes the implementation and usage of these functions in more detail.

2.1. Logarithm (ac_log_pwl / ac_log2_pwl)

The *ac_log_pwl* library provides a piecewise linear implementation for the base 2 and base e logarithmic functions, optimized to provide high performance with quick results. The following datatypes are supported by this IP design: (a) *ac_fixed* (b) *ac_float* (c) *ac_std_float* (d) *ac_ieee_float*. The *ac_fixed* version of the *ac_log2_pwl* function is the one that performs the actual PWL computation. The natural logarithm is computed using the log2 output and the change of base property.

2.1.1. The *ac_log2_pwl* Implementation

The fixed point *ac_log2_pwl* implementation normalizes the function input to the PWL domain, and performs a PWL approximation to find log2 of the normalized input. This normalized value is then denormalized to produce the final output.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the actual, accurate implementation of the base 2 logarithm function, the following graph compares the PWL output against the accurate function output, where the red line (PWL output) is virtually indistinguishable from the green one (Accurate output):

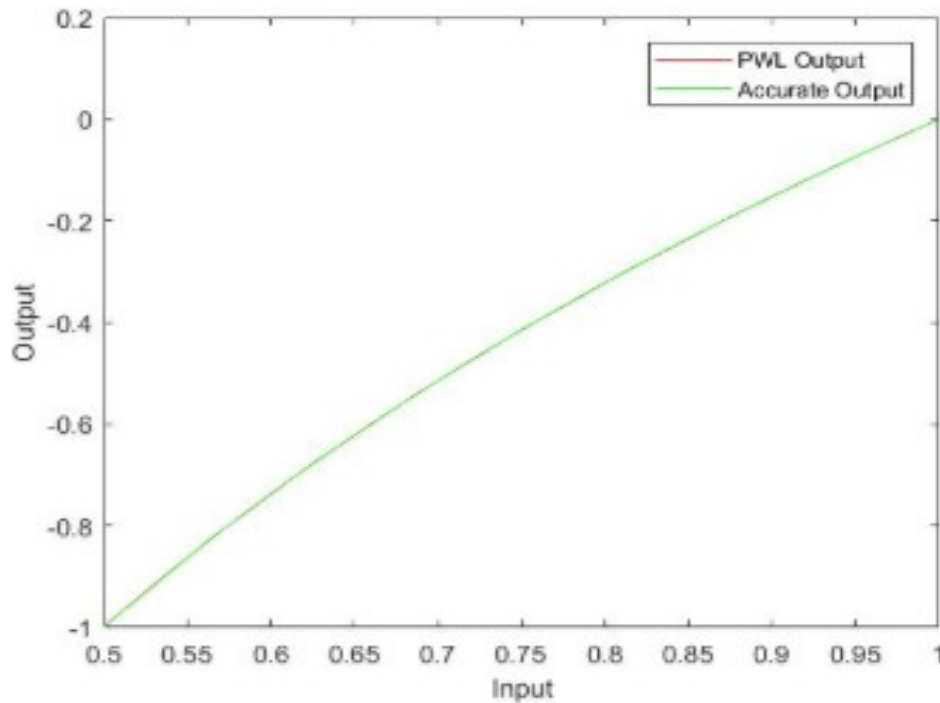


Illustration 2: PWL Output vs. Accurate Output for Base2 Logarithm

Floating Point Support

The design supports *ac_float*, *ac_std_float* and *ac_ieee_float* datatypes. The *ac_float* version uses the *ac_fixed* version to compute log2 of the fixed point input mantissa. The corresponding output is denormalized by taking into account the input exponent and integer width of the input mantissa.

It should be noted that, since the input to the fixed point version (i.e. the input mantissa) is already normalized, the fixed point version does not need to perform a normalization on the input. Therefore, to avoid unnecessary hardware, floating point version uses the default *call_normalize* parameter. For more details on this default parameter, refer to Function Prototypes.

The *ac_std_float* and *ac_ieee_float* versions are essentially wrappers around the *ac_float* version that perform a conversion between *ac_std_float/ac_ieee_float* and the intermediate *ac_float* values that are passed to the *ac_float* implementation.

Handling Zero and Negative Inputs

A macro-enabled *AC_ASSERT* is provided to alert the user if a zero input is passed to the PWL function. In addition, functionality is also provided to ensure that the *ac_log2_pwl* function passes the minimum negative value that can be represented with the output type when a zero input is encountered, to mimic an output of negative infinity.

Similarly, a macro-enabled *AC_ASSERT* is also provided with the floating point implementation to alert the user if a negative input is passed to the PWL function.

2.1.1. The *ac_log_pwl* Implementation

In order to calculate the natural logarithm of *ac_fixed* inputs, we rely upon the *ac_log2_pwl* implementation. As mentioned earlier, the change of base property is used. It can be represented by the following equation

$$\log(x) = \log_2(x) * \log(2)$$

The output of the *ac_log2_pwl* implementation is multiplied by $\log(2)$, a constant value. The product is the output of the *ac_log_pwl* function.

Floating Point Support

The interaction between floating point and fixed point *ac_log_pwl* functions is the same as that outlined in for the corresponding *ac_log2_pwl* functions, with the exception being that an additional multiplication is required to perform a change of base as explained above, for *ac_log_pwl* functions.

2.1.2. Function Prototypes

The following code shows the template prototypes for the different implementations:

```
template<ac_q_mode pwlQ = AC_TRN,
        int W, int I, ac_q_mode Q,  ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO>
void ac_log2_pwl(
    const ac_fixed<W, I, false, Q, O> input,
    ac_fixed<outW, outS, outS, outQ, outO> &output,
    const bool call_normalize = true
)
```

The default *call_normalize* argument shown above allows the user to selectively call the *ac_normalize* function. As mentioned earlier, the floating point function calls to the fixed point *ac_log2_pwl* version set *call_normalize* to false in order to bypass the usage of *ac_normalize* and the generation of normalization hardware.

Bear in mind that to completely bypass the usage of *ac_normalize*, the *call_normalize* argument must stay false for every function call. This is done by passing a constant boolean parameter for *call_normalize*, as can be seen in Bypassing Normalization.

```
template <ac_q_mode pwlQ = AC_TRN,
```

```

        int W, int I, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO>
void ac_log_pwl(
    const ac_fixed <W, I, false, Q, O> input,
    ac_fixed <outW, outI, outS, outQ, outO> &output
)

```

Note that the fixed point functions only accept unsigned inputs, which corresponds to the behavior of the actual logarithm function, which only accepts positive values for its domain. This is different than the behavior of the C math library logarithm functions, which use signed double datatypes for their inputs.

```

template <ac_q_mode pwlQ = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_log2_pwl(
    const ac_float<W, I, E, Q> input,
    ac_float<outW, outI, outE, outQ> &output
)

```

```

template <ac_q_mode pwlQ = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_log_pwl(
    const ac_float<W, I, E, Q> input,
    ac_float<outW, outI, outE, outQ> &output
)

```

```

template <ac_q_mode pwl_Q = AC_TRN, int W, int E, int outW, int outE>
void ac_log2_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
)

```

```

template <ac_q_mode pwl_Q = AC_TRN, int W, int E, int outW, int outE>
void ac_log_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
)

```

```
template <ac_q_mode pwl_Q = AC_TRN,
          ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_log2_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
)
```

```
template <ac_q_mode pwl_Q = AC_TRN,
          ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_log_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
)
```

Returning by Value

The *ac_log2_pwl* and *ac_log_pwl* functions can return their output by value as well as by reference. In order to return the output by value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the Example Function Calls section below. The prototypes for the function call to return by value for both functions is shown below:

```
template<class T_out,
          ac_q_mode q_mode_temp = AC_TRN,
          class T_in>
T_out ac_log2_pwl(
    const T_in &input
)
```

```
template<class T_out,
          ac_q_mode q_mode_temp = AC_TRN,
          class T_in>
T_out ac_log_pwl(
    const T_in &input
)
```

2.1.3. Example Function Calls

An example of a function call to store the value of the base 2 and base e logarithms of a sample *ac_fixed* variable *x* in the variables *y_log2* and *y_log* is shown below:

```
ac_fixed<20, 11, false, AC_RND, AC_SAT> x = 2.875;
typedef ac_fixed<30, 15, true, AC_RND, AC_SAT> output_type;
output_type y_log, y_log2;
```

```
ac_log2_pwl(x, y_log2); // Approximates y_log2 = log2(x)
ac_log_pwl(x, y_log);   // Approximates y_log = log(x)
```

Returning by Value

In order to have the function return by value, the user must pass the output type information as a template parameter. This is done as follows, for the base 2 and base e logarithmic functions:

```
y = ac_log2_pwl<output_type>(x);
y = ac_log_pwl<output_type>(x);
```

If the user wants to change the default template parameters, e. g. they want to round the output of the PWL implementation, they can do so by using the following function calls as guidelines:

```
y = ac_log2_pwl<output_type, AC_RND>(x);
y = ac_log_pwl<output_type, AC_RND>(x);
```

Bypassing Normalization

As mentioned earlier in Function Prototypes, the calls to *ac_normalize* and hence the synthesis of normalization hardware can be bypassed by changing the *call_normalize* argument. An example on how to do so, for *ac_fixed* inputs that are already normalized, is shown below:

```
ac_fixed<16, 5, false> x = 3.5;
ac_fixed<18, 2, true> y;
const bool call_normalize = false;
ac_log2_pwl(x, y, call_normalize);
```

2.2. Power (*ac_exp_pwl / ac_pow2_pwl / ac_pow_pwl*)

The *ac_pow_pwl* library is designed to provide a quick approximation of the base 2, natural exponentials and generic power functions for real numbers using a piecewise linear (PWL) implementation of the base 2 exponential function with 5 points/4 segments along with the *ac_log2_pwl* function for the generic power function. This frees us of the burden of having to calculate the output using methods such as Taylor series expansion, which requires loop unrolling, a large number of adders and hence a comparatively large area, to give 100% throughput. By contrast, the *ac_pow_pwl* function can give 100% throughput by merely pipelining it with an II of 1, which results in a lower area for the hardware.

2.2.1. The *ac_pow2_pwl* Implementation

The *ac_pow_pwl* library provides three functions, one which calculates the base 2 exponential, one which calculates the natural exponential, and one more that calculates the generic power function (a^b). The natural exponential version (henceforth called the *ac_exp_pwl* implementation) depends upon the base 2 exponential version (henceforth called the *ac_pow2_pwl* implementation) for its computation, as will be explained later. The generic power function (henceforth called the *ac_pow_pwl* implementation) also depends on the *ac_pow2_pwl* implementation, in addition to the *ac_log2_pwl* function. All three functions only accept *real*, *ac_fixed* inputs and calculate *ac_fixed* outputs.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the actual, accurate implementation of the base 2 exponential function, the following graph compares the PWL output against the accurate function output:

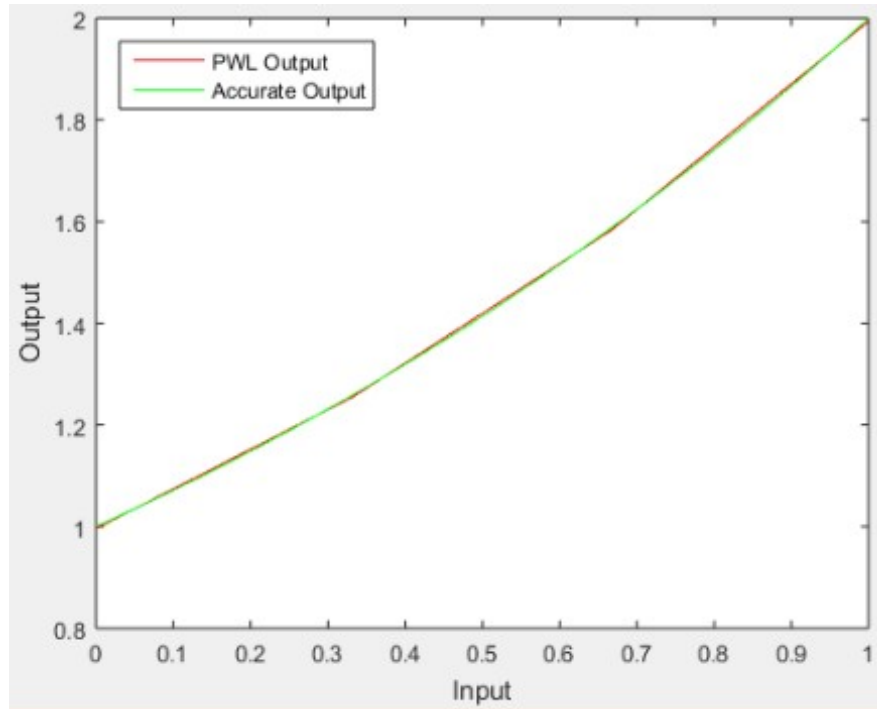


Illustration 3: PWL Output vs. Accurate Output for Base 2 Exponential

Floating Point Support

There are *ac_pow2_pwl* implementations designed to also support *ac_float*, *ac_std_float* and *ac_ieee_float* datatypes. The *ac_float* version of *ac_pow2_pwl* converts the floating point input to a fixed point input, the fractional part of which is then passed to the fixed point implementation. The output corresponding to the fractional part is then denormalized with the integer part of the input to produce the final output.

The *ac_std_float* and *ac_ieee_float* versions are essentially wrappers around the *ac_float* version that perform a conversion between *ac_std_float*/*ac_ieee_float* and the intermediate *ac_float* values that are passed to the *ac_float* implementation. If the *AC_POW_PWL_NAN_SUPPORTED* macro is defined, *ac_std_float* and *ac_ieee_float* versions will produce a nan output when supplied with a nan input.

2.2.2. The *ac_exp_pwl* Implementation

In order to calculate the natural exponential of *ac_fixed* inputs, we rely upon the *ac_pow2_pwl* implementation. The following relation is used:

$$\exp(x) = 2^{(x * \log_2(e))}$$

Hence, all that needs to be done is to multiply the input by $\log_2(e)$, store the product in a temporary variable, and then pass that to the `ac_pow2_pwl` implementation.

Changing Intermediate Type

The fixed-point `ac_exp_pwl` function has a default template parameter called `n_f_b` which can adjust the minimum number of fractional bits in the variable used for storing the multiplication of the input and $\log_2(e)$. To enable this adjustment, the user must define the macro `AC_POW_PWL_CHANGE_FRAC_BITS`. If that macro is defined, the function sets the number of fractional bits in the intermediate type to `n_f_b` or the number of fractional bits in the input, whichever one is greater. If the macro is not defined, the minimum number of fractional bits will be 12 or the number of fractional bits in the input, whichever one is greater.

Even if the macro is not defined, `n_f_b` will still be present in the template parameter-list but not actually used in the function itself, so as to preserve backward compatibility with calls to previous versions of the fixed-point `ac_exp_pwl` function, which always used the template parameter.

Refer to Function Prototypes for the prototype and template parameter-list of the `ac_exp_pwl` functions.

Floating Point Support

The floating point support for `ac_exp_pwl` is almost identical to that provided for the `ac_pow2_pwl` implementation. The only difference is that the input mantissa is multiplied by $\log_2(e)$ before converting to a fixed point input and passing that to the fixed point implementation.

2.2.3. The `ac_pow_pwl` Implementation

In order to calculate the generic power function output for `ac_fixed` inputs, we use a relation similar to that used in the `ac_exp_pwl` implementation, namely

$$a^x = 2^{(x * \log_2(a))}$$

The difference being that in this case, the base, i.e. a , is a dynamic input received from the user, hence we cannot use a constant value to represent $\log_2(a)$. We instead use the `ac_log2_pwl` function to calculate this value, and multiply the calculated value with the exponent, i.e. x . The product of this multiplication is passed to the `ac_pow2_pwl` implementation.

Floating Point Support

The `ac_pow_pwl` library does not have floating point support, due to difficulties involved in bounding the output error for floating point inputs. The user is invited to use the generic cordic exponential function instead, i.e. `ac_pow_cordic`.

2.2.4. Function Prototypes

The following code shows the function prototypes for the different implementations:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_pow2_pwl(
```

```
const ac_fixed<W, I, S, Q, O> &input,
ac_fixed<outW, outI, false, outQ, outO> &output
)
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_pow2_pwl(
    const ac_float<W, I, E, Q> input,
    ac_float<outW, outI, outE, outQ> &output
)
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int E, int outW, int outE>
void ac_pow2_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
)
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_pow2_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
)
```

```
template<int n_f_b = 11, ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_exp_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

```
template <ac_q_mode pwl_Q = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
```

```
void ac_exp_pwl(
    const ac_float<W, I, E, Q> input,
    ac_float<outW, outI, outE, outQ> &output
)
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int E, int outW, int outE>
void ac_exp_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
)
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_exp_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
)
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int baseW, int baseI, ac_q_mode baseQ, ac_o_mode baseO,
        int exponW, int exponI, bool exponS, ac_q_mode exponQ, ac_o_mode exponO,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_pow_pwl(
    const ac_fixed<baseW, baseI, false, baseQ, baseO> &base,
    const ac_fixed<exponW, exponI, exponS, exponQ, exponO> &expon,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Note that the function only accepts unsigned `ac_fixed` datatypes for the output and base (in case of the `ac_pow_pwl` implementation). This corresponds to the behavior of the actual power functions, which have a range that covers only all positive real values, and whose base only covers the domain of positive real values. The behavior of our PWL functions in this sense is different than the behavior of the standard C math library functions, which accept signed doubles for both the input(s) and the output.

Returning by Value

All the power functions can return their output by value as well as by reference. In order to return the output by value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the *Example Function Calls* section below. The prototypes for the function call to return by value for both functions is shown below:


```
template<class T_out, ac_q_mode pwl_Q = AC_TRN, class T_in>
T_out ac_pow2_pwl(
    const T_in &input
)
```

```
template<class T_out, ac_q_mode pwl_Q = AC_TRN, class T_in>
T_out ac_exp_pwl(
    const T_in &input
)
```

Since the ac_fixed ac_exp_pwl has an additional template parameter, i.e. *n_f_b*, the prototype for the return-by-value version of it is different than the generic version shown above:

```
template <class T_out, int n_f_b = 11, ac_q_mode pwl_Q = AC_TRN,
          int W, int I, bool S, ac_q_mode Q, ac_o_mode O>
T_out ac_exp_pwl(
    const ac_fixed<W, I, S, Q, O> &input
)
```

```
template<class T_out, ac_q_mode pwl_Q = AC_TRN, class T_in_base, class T_in_expon>
T_out ac_pow_pwl(
    const T_in_base &base,
    const T_in_expon &expon
)
```

2.2.5. Example Function Calls

An example of a function call to store the value of the natural exponential of a sample *ac_fixed* variable *x* in a variable *y* is shown below:

```
typedef ac_fixed<21, 12, false, AC_RND, AC_SAT> base_type;
typedef ac_fixed<20, 11, true, AC_RND, AC_SAT> exp_type;
typedef ac_fixed<30, 15, false, AC_RND, AC_SAT> output_type;

base_type a = 2.5;
exp_type x = 2.875;
output_type y_exp, y_pow2, y_pow;
ac_pow2_pwl(x, y_pow2); //Approximates y_pow2 = pow(2, x.to_double())
ac_exp_pwl(x, y_exp);   //Approximates y_exp = exp(x)
ac_pow_pwl(a, x, y_pow); //Approximates y_pow = pow(a, x)
```

The variable *y* hereafter contains the approximate value of the natural exponential of *x*.

Returning by Value

In order to have the function return by value, the user must pass the output type information as a template parameter. This is done as follows, for the base 2 and base e exponential functions:

```
y_pow2 = ac_pow2_pwl<output_type>(x);
y_exp  = ac_exp_pwl<output_type>(x);
y_pow  = ac_pow_pwl<output_type>(a, x);
```

2.3. Reciprocal (*ac_reciprocal_pwl*)

The *ac_reciprocal_pwl* function is designed to provide a quick approximation of the reciprocal of real and complex numbers using a Piecewise Linear (PWL) implementation with 9 points/8 segments. Many hardware division operations are calculated indirectly by first obtaining the value of the reciprocal of the denominator, and then multiplying that with the numerator. The calculation of the reciprocal can be done using PWL approximation, which is faster and requires lesser area than actual division hardware.

2.3.1. The *ac_reciprocal_pwl* Implementation

The *ac_reciprocal_pwl* library provides four overloaded functions for the calculation of the reciprocal of real and complex inputs. Each overloaded function handles a different input datatype. The six datatypes hence handled are (a) *ac_fixed*, (b) *ac_float*, (c) *ac_complex<ac_fixed>*, (d) *ac_complex<ac_float>*, (e) *ac_std_float* and (f) *ac_ieee_float*. It is the *ac_fixed* function that actually contains the code required for the PWL implementation. All the other functions rely upon the *ac_fixed* PWL implementation.

Handling Zero Input

The *ac_reciprocal_pwl* library provides a macro-enabled AC_ASSERT which will produce a run-time assert when a zero input is encountered. If this assert fails to kick in, additional functionality is provided that ensures output saturation when a zero input is encountered.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the “real thing”, i.e. the actual, accurate implementation of the reciprocal function, the following graph compares the PWL output against the accurate function output:

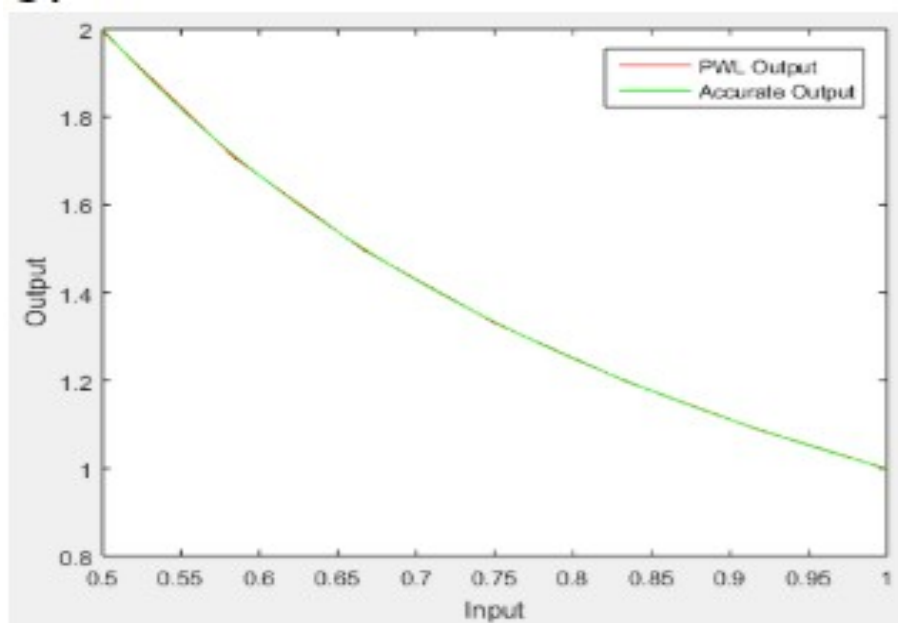


Illustration 4: PWL Output vs. Accurate Output for Reciprocal

ac_std_float and ac_ieee_float Support

The functions for *ac_std_float* and *ac_ieee_float* support serve as wrappers around the *ac_float* reciprocal designs, which perform conversions between the *ac_std_float/ac_ieee_float* float inputs/outputs and the intermediate *ac_float* variables that are passed to the *ac_float* design.

2.3.2. Function Templates

The following are the overloaded function prototypes for the *ac_reciprocal_pwl* function for different datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO>
void ac_reciprocal_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, outS, outQ, outO> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_reciprocal_pwl(
    const ac_float<W, I, E, Q> &input,
```

```
ac_float<outW, outI, outE, outQ> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO>
void ac_reciprocal_pwl(
    const ac_complex<ac_fixed<W, I, S, Q, O> > &input,
    ac_complex<ac_fixed<outW, outI, outS, outQ, outO> > &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_reciprocal_pwl(
    const ac_complex<ac_float<W, I, E, Q> > &input,
    ac_complex<ac_float<outW, outI, outE, outQ> > &output
);
```

```
template <ac_q_mode pwl_Q = AC_TRN, int W, int E, int outW, int outE>
void ac_reciprocal_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        ac_ieee_float_format Format,
        ac_ieee_float_format outFormat>
void ac_reciprocal_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
);
```

Returning by Value

The *ac_reciprocal_pwl* functions can return their output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the *Example Function Calls* section below. The prototype for the function call to return by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_reciprocal_pwl(
    const T_in &input
);
```

2.3.3. Example Function Calls

An example of a function call to store the value of the reciprocal of a sample *ac_fixed* variable *x* in a variable *y* is shown below:

```
typedef ac_fixed<20, 11, true, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 15, true, AC_RND, AC_SAT> output_type;
input_type x = -1.75;
output_type y;
ac_reciprocal_pwl(x, y); //Approximates  $y = 1 / x$ 
```

The variable *y* hereafter stores the approximate value of the reciprocal of *x*.

Returning by Value

As mentioned earlier, the *ac_reciprocal_pwl* functions can also return by value. In order to do so, the type information for the output must be passed explicitly to function as shown below.

```
y = ac_reciprocal_pwl<output_type>(x);
```

If the user also wishes to change the rounding mode for the temporary variable and have the function return by value, they can call the function as follows:

```
y = ac_reciprocal_pwl<output_type, AC_RND>(x);
```

2.4. High-accuracy Reciprocal (*ac_reciprocal_pwl_ha*)

The high-accuracy reciprocal library is designed to be a higher-accuracy version of the *ac_reciprocal_pwl* function. The implementation, function prototypes and example function calls for the *ac_reciprocal_pwl_ha* library are almost the same as that for the *ac_reciprocal_pwl* library, with the only two major differences being as follows:

- The function name is different; the high-accuracy reciprocal library has the suffix *_ha* attached to denote that it is a *high-accuracy* implementation.
- The high-accuracy version uses 32 segments to enable a higher-accuracy output, as compared to 8 segments for the original implementation. Hence, the high-accuracy version will consume more area in hardware. The 32-element LUTs for the high-accuracy version might be too big to be mapped to a register bank, and hence be mapped to memories instead. The user might want to consider changing the memory mapping threshold if they do not wish for this to happen.

Keeping these differences in mind, The user can consult the documentation for *ac_reciprocal_pwl* for more details on the working of the high-accuracy version.

2.5. Very High-accuracy Reciprocal (**ac_reciprocal_pwl_vha**)

For applications where an ever higher accuracy output is desired, the user can utilize the *ac_reciprocal_pwl_vha* library, which uses a 64-segment PWL. Like its high-accuracy reciprocal counterpart, this library is also very similar to the *ac_reciprocal_pwl* library, with the only major differences being the name and the number of segments used. This library is designed to have a percentage relative error below 0.005%.

2.6. Square Root (**ac_sqrt_pwl**)

The *ac_sqrt_pwl* function is a piecewise linear implementation of a square root function that has been optimized to provide a fast, low-area implementation with minimal error, making it useful as a basic building block in high speed IP blocks.

2.6.1. The **ac_sqrt_pwl** Implementation

The *ac_sqrt_pwl* library provides three overloaded functions for the calculation of the square root of real and complex inputs, each handling a different input datatype. The three datatypes hence handled are (a) *ac_fixed*, (b) *ac_float* (c) *ac_std_float* (d) *ac_ieee_float* and (e) *ac_complex<ac_fixed>*. It is the *ac_fixed* implementation that actually contains the code required for the PWL approximation.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the “real thing”, i.e. the accurate implementation of the square root function, the following graph is provided:

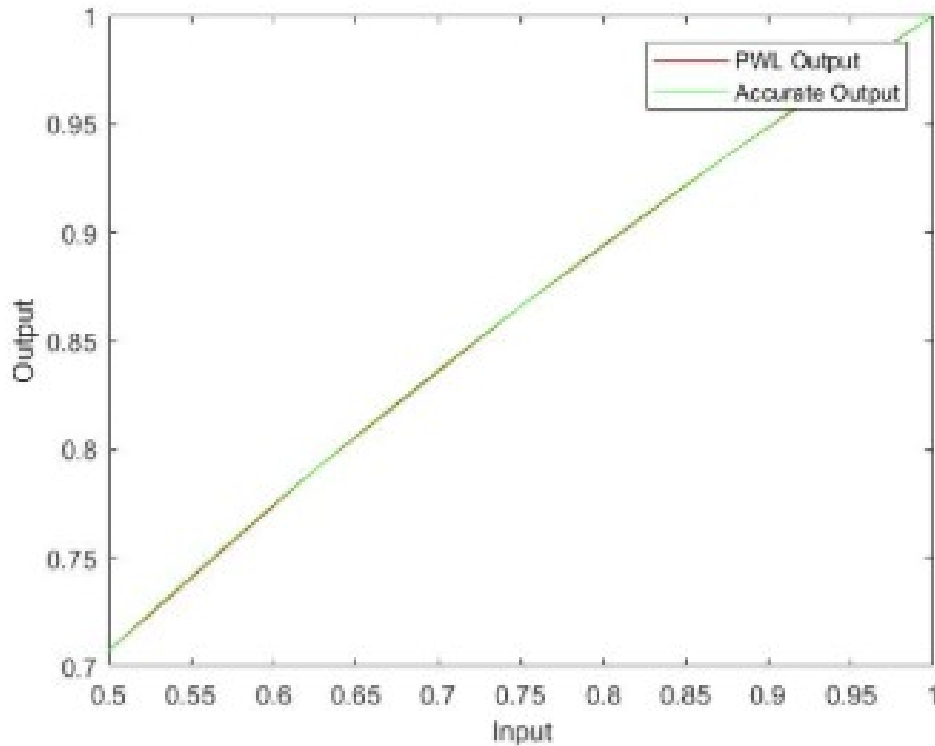


Illustration 5: PWL Output vs. Accurate Output for Square Root

Floating Point Support

The functions for *ac_std_float* and *ac_ieee_float* support serve as a wrapper around the *ac_float* square root design, and perform conversions between the *ac_std_float*/*ac_ieee_float* inputs/outputs and the intermediate *ac_float* variables that are passed to the *ac_float* design.

Handling Negative Inputs

A macro-enabled *AC_ASSERT* is provided in the *ac_float* implementation to alert the user to negative values that may accidentally be passed to the design.

Toggling Normalization

The floating point functions use the *ac_fixed* implementation to calculate the square root of the input mantissa. As the mantissa is already normalized, the fixed point function call is configured such that *ac_normalize* is not called and the generation of normalization hardware is bypassed. This is done by changing the value passed for the default *call_normalize* argument in the fixed point function interface. The interface, along with the *call_normalize* argument, is displayed in Function Templates. Refer to Example Function Calls for a usage example.

2.6.2. Function Templates

The following are the overloaded function prototypes for the *ac_sqrt_pwl* function for different datatypes:

```
template <ac_q_mode pwlQ = AC_TRN,
```

```

        int W, int I, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_sqrt_pwl(
    const ac_fixed <W, I, false, Q, O> input,
    ac_fixed <outW, outI, false, outQ, outO> &output,
    const bool call_normalize = true
);

```

```

template <ac_q_mode pwlQ = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_sqrt_pwl(
    const ac_float <W, I, E, Q> input,
    ac_float <outW, outI, outE, outQ> &output
);

```

```

template <ac_q_mode pwl_Q = AC_TRN, int W, int E, int outW, int outE>
void ac_sqrt_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
);

```

```

template <ac_q_mode pwl_Q = AC_TRN,
        ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_sqrt_pwl(
    const ac_ieee_float<Format> input,
    ac_ieee_float<outFormat> &output
);

```

```

template <ac_q_mode pwlQ = AC_TRN,
        int W, int I, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_sqrt_pwl(
    const ac_complex <ac_fixed <W, I, true, Q, O> > input,
    ac_complex <ac_fixed <outW, outI, true, outQ, outO> > &output
);

```


Returning by Value

The *ac_sqrt_pwl* functions can return their output by value as well as by reference. To return the value, the user must pass the information of the output type to the function as a template argument. Refer to Example Function Calls for a usage example. The prototype for the function call to return by value is shown below:

```
template<class T_out, ac_q_mode pwlQ = AC_TRN, class T_in>
T_out ac_sqrt_pwl(
    const T_in &input
);
```

2.6.3. Example Function Calls

An example of a function call to store the value of the square root of a sample *ac_fixed* variable *x* in a variable *y*, by using both return by reference and return by value, is shown below. A line of code that illustrates the method to change the rounding mode of the intermediate PWL variable is also included.

```
typedef ac_fixed<20, 11, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 15, false, AC_RND, AC_SAT> output_type;

input_type x = 1.75;
output_type y;

// Returns y = sqrt(x), and returns by reference.
ac_sqrt_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_sqrt_pwl<AC_RND> (x, y);

// The following line, returns by value instead of by reference.
y = ac_sqrt_pwl <output_type> (x);
```

If the input to the *ac_fixed* function is already normalized, the user can bypass normalization and save on hardware that way. To do so, they can pass *false* for the *call_normalize* argument, as shown below:

```
const bool call_normalize = false;
ac_sqrt_pwl(x, y, false);
```

2.7. Inverse Square Root (ac_inverse_sqrt_pwl)

The *ac_inverse_sqrt_pwl* function is a piecewise linear implementation of the inverse square root function ($1/\sqrt{x}$) that has been optimized to provide a high-performance implementation with minimal error, making it useful as a basic building block in high speed IP blocks. This function is implemented as an overloaded function for the different datatypes it supports. It provides a more accurate and lower-area output as compared to using the square root and reciprocal PWL approximations together to calculate the inverse square root of a number.

2.7.1. The `ac_inverse_sqrt_pwl` Implementation

The `ac_inverse_sqrt_pwl` library provides four overloaded functions for the calculation of the inverse square root of real and complex inputs. Each overloaded function handles a different input datatype. The five datatypes hence handled are (a) `ac_fixed`, (b) `ac_float` (c) `ac_std_float`, (d) `ac_ieee_float` and (e) `ac_complex<ac_fixed>`. It is the `ac_fixed` function that actually contains the code required for the PWL implementation. All the other functions rely upon the `ac_fixed` PWL implementation.

Handling Invalid Inputs

The `ac_inverse_sqrt_pwl` library provides a macro-enabled `AC_ASSERT` which will produce a run-time assert when a zero input is encountered. If this assert fails to kick in, additional functionality is provided that ensures output saturation when a zero input is encountered.

A macro-enabled `AC_ASSERT` is also provided that is triggered in case of a negative floating point input.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the “real thing”, i.e. the accurate implementation of the inverse square root function, the following graph is provided:

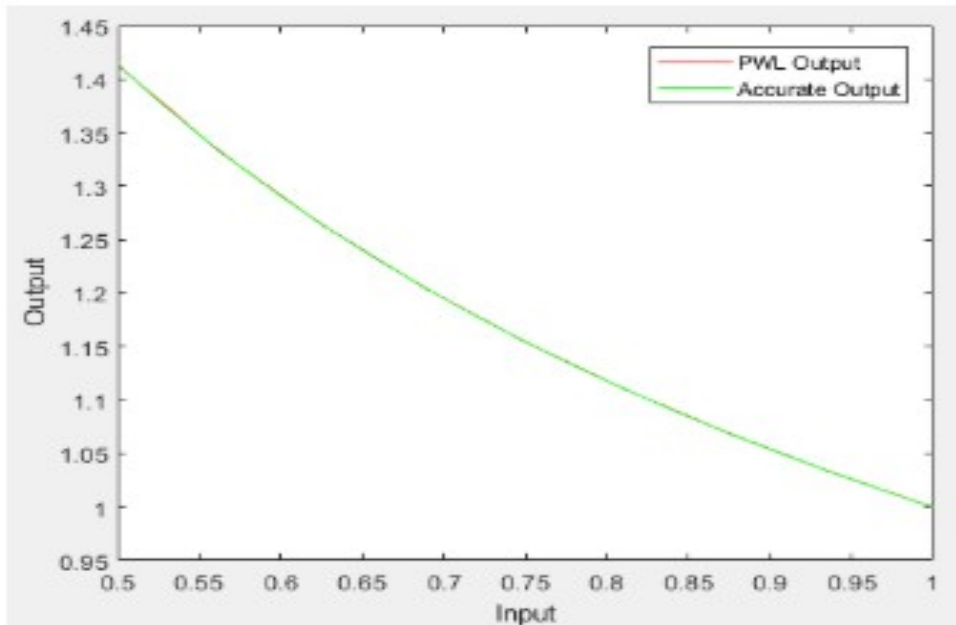


Illustration 6: PWL Output vs. Accurate Output for Inverse Square Root

Floating Point Support

The functions for `ac_std_float` and `ac_ieee_float` support serves as a wrapper around the `ac_float` inverse square root design, which performs conversions between the standard/IEEE float inputs/outputs and the intermediate `ac_float` variables that are passed to the `ac_float` design.

Toggling Normalization

The floating point functions use the *ac_fixed* implementation to calculate the square root of the input mantissa. As the mantissa is already normalized, the fixed point function call is configured such that *ac_normalize* is not called and the generation of normalization hardware is bypassed. This is done by changing the value passed for the default *call_normalize* argument in the fixed point function interface. The interface, along with the *call_normalize* argument, is displayed in Function Templates. Refer to Example Function Calls for a usage example.

2.7.2. Function Templates

The following are the overloaded function prototypes for the *ac_inverse_sqrt_pwl* function for different datatypes:

```
template<ac_q_mode q_mode_temp = AC_TRN,
        int W1, int I1, ac_q_mode q1, ac_o_mode o1,
        int W2, int I2, ac_q_mode q2, ac_o_mode o2>
void ac_inverse_sqrt_pwl(
    const ac_fixed <W1,I1, false, q1, o1> &input,
    ac_fixed <W2, I2, false, q2, o2> &output,
    const bool call_normalize = true
);
```

```
template <ac_q_mode q_mode_temp = AC_TRN,
        int W1, int I1, int E1, ac_q_mode q1,
        int W2, int I2, int E2, ac_q_mode q2>
void ac_inverse_sqrt_pwl(
    const ac_float <W1, I1, E1, q1> &input,
    ac_float <W2, I2, E2, q2> &output
);
```

```
template <ac_q_mode q_mode_temp = AC_TRN, int W1, int E1, int W2, int E2>
void ac_inverse_sqrt_pwl(
    const ac_std_float<W1, E1> &input,
    ac_std_float<W2, E2> &output
);
```

```
template<ac_q_mode q_mode_temp = AC_TRN,
        ac_ieee_float_format Format,
        ac_ieee_float_format outFormat>
void ac_inverse_sqrt_pwl(
```

```
const ac_ieee_float<Format> input,
ac_ieee_float<outFormat> &output
);
```

```
template<ac_q_mode q_mode_temp = AC_TRN,
        int W1, int I1, ac_q_mode q1, ac_o_mode o1,
        int W2, int I2, ac_q_mode q2, ac_o_mode o2>
void ac_inverse_sqrt_pwl(
    const ac_complex <ac_fixed <W1,I1,true, q1, o1> > &input,
    ac_complex <ac_fixed <W2, I2, true, q2, o2> > &output
);
```

Returning by Value

The *ac_inverse_sqrt_pwl* functions can return their output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the Example Function Calls section below. The prototype for the function call to return by value is as follows:

```
template<class T_out, ac_q_mode q_mode_temp = AC_TRN, class T_in>
T_out ac_inverse_sqrt_pwl(
    const T_in &input
)
```

2.7.3. Example Function Calls

An example of a function call to store the value of the inverse square root of a sample *ac_fixed* variable *x* in a variable *y*, by using both return by reference and return by value, is shown below.

```
typedef ac_fixed<20, 11, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 15, false, AC_RND, AC_SAT> output_type;

input_type x = 1.75;
output_type y;

// Returns y = 1/sqrt(x), and returns by reference.
ac_inverse_sqrt_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_inverse_sqrt_pwl<AC_RND> (x, y);

// The following line, returns by value instead of by reference.
y = ac_inverse_sqrt_pwl <output_type> (x);
```

If the input to the `ac_fixed` function is already normalized, the user can bypass normalization and save on hardware that way. To do so, they can pass `false` for the `call_normalize` argument, as shown below:

```
const bool call_normalize = false;
ac_inverse_sqrt_pwl(x, y, false);
```

2.8. Very High-Accuracy Inv Sqrt (`ac_inverse_sqrt_pwl_vha`)

The very high-accuracy inverse square root library is designed to be a higher-accuracy version of the `ac_inverse_sqrt_pwl` function. The implementation, function prototypes and example function calls for the `ac_inverse_sqrt_pwl_vha` library are almost the same as that for the `ac_inverse_sqrt_pwl_vha` library, with the two major differences being as follows:

- The function name is different; the very high-accuracy inverse square root library has the suffix `_vha` attached to denote that it is a very *high*-accuracy implementation.
- The VHA version uses 64 segments to enable a higher-accuracy output, as compared to 8 segments for the original implementation. Hence, the high-accuracy version will consume more area in hardware. The 64-element LUTs for the high-accuracy version might be too big to be mapped to a register bank, and hence be mapped to memories instead. The user might want to consider changing the memory mapping threshold if they do not wish for this to happen.

Keeping these differences in mind, The user can consult the documentation for `ac_inverse_sqrt_pwl` for more details on the working of the very high-accuracy version.

2.9. Tangent (`ac_tan_pwl`)

The `ac_tan_pwl` library provides a fast, PWL implementation of the tangent function of first-quadrant angles with minimal inaccuracy. The implementation is faster than other algorithms such as the CORDIC algorithm, which require a significant number of iterations and stored table values to reach an acceptable accuracy in the output. The tangent PWL library supports (a) `ac_fixed` (b) `ac_float` (c) `ac_std_float` and (d) `ac_ieee_float` datatypes.

Note: The LUT generator for `ac_tan_pwl` does not guarantee a monotonically decreasing output by default. If the user wants guaranteed monotonicity, they must either manually tweak the LUT values or use the `-m` command line option while running the LUT generator executable. In addition, they'll need to set the `extra_monotonicity_check` template parameter to `true`.

2.9.1. The `ac_tan_pwl` Implementation

Each supported datatype is handled separately through overloaded functions. The `ac_fixed` implementation contains the actual PWL implementation, and the `ac_float` implementation depends on the `ac_fixed` implementation to perform the PWL calculation.

Normalization

The actual tangent PWL implementation only covers the domain of $[0, \pi/4)$. In order to cover a larger range of first-quadrant angle values, i. e. in order to extend the domain of the *ac_tan_pwl* function to $[\pi/4, \pi/2)$, the input angle in radians is halved using a shift operation, and the half-angle equation is used:

$$\tan(x) = 2 * \tan(x/2) * (1 / (1 - \tan(x/2)^2))$$

In order to calculate the $1 / (1 - \tan(x/2)^2)$ factor in the above equation, we use the *ac_reciprocal_pwl* library for *ac_fixed* inputs and outputs. Hence, the user must keep in mind that any changes made to that library will also affect the accuracy of the tangent approximation for values that equal or exceed $\pi/4$.

Saturation

It is important to note that the output will saturate to the maximum possible value once the input has crossed the value of 1.556640625 radians, which is roughly 89.2 degrees.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the “real thing”, i.e. the accurate implementation of the tangent function, the following graph is provided for the domain $[0, \pi/4)$:

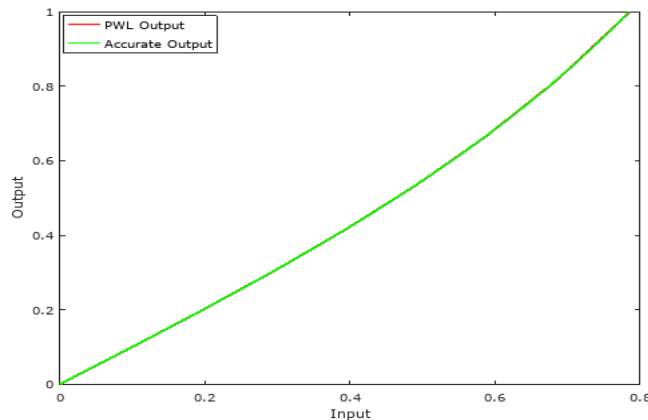


Illustration 7: PWL Output vs. Accurate Output for Tangent

Floating Point Support

The floating point designs use the *ac_shift_left* library to convert the values of input mantissa and exponent into a fixed point value that can be passed to the *ac_fixed* implementation. The bitwidth of the intermediate fixed point value can be reduced due to the fact that the input cannot exceed $\pi/2$ and because the *ac_fixed* PWL function truncates inputs that exceed a certain number of fractional bits.

The *ac_std_float* and *ac_ieee_float* functions serves as a wrapper around the *ac_float* function that converts *ac_std_float* and *ac_ieee_float* inputs/outputs to/from their equivalent *ac_float* representations.

Handling Invalid Inputs

Macro-enabled `AC_ASSERTs` are provided to alert the users to inputs that exceed $\pi/2$ or, in case of floating point implementations, negative inputs.

2.9.2. Function Templates

The following are the function prototypes for the `ac_tan_pwl` function for different datatypes:

```
// half_angle_prec: Adjusts the number of bits used to compute the
// tangent through the half angle formula, if input > pi/4.
//
// extra_monotonicity_check: Makes sure that the input is at least equal
// to 1 if input >= pi/4. Might be necessary for certain PWL implementations
// if monotonicity is desired. The default implementation does not need it.
template<ac_q_mode pwl_Q = AC_TRN, int half_angle_prec = 16,
        bool extra_monotonicity_check = false,
        int W, int I, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_tan_pwl(
    const ac_fixed<W, I, false, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN, int half_angle_prec = 16,
        bool extra_monotonicity_check = false,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_tan_pwl(
    const ac_float<W, I, E, Q> &input,
    ac_float<outW, outI, outE, outQ> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN, int half_angle_prec = 16,
        bool extra_monotonicity_check = false,
        int W, int E, int outW, int outE>
void ac_tan_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN, int half_angle_prec = 16,
        bool extra_monotonicity_check = false,
        ac_ieee_float_format Format,
        ac_ieee_float_format outFormat>
void ac_tan_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
);
```

Returning by Value

The *ac_tan_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the Example Function Calls section below. The prototype for the function call to return by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN, int half_angle_prec = 16,
        bool extra_monotonicity_check = false, class T_in>
T_out ac_tan_pwl(const T_in &input)
```

2.9.3. Example Function Calls

An example of a function call to store the value of the tangent of a sample *ac_fixed* variable *x* in a variable *y*, by using both return by reference and return by value, is shown below.

```
typedef ac_fixed<20, 11, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 15, false, AC_RND, AC_SAT> output_type;

input_type x = 0.8;
output_type y;

// Returns y = tan(x), and returns by reference.
ac_tan_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_tan_pwl<AC_RND> (x, y);

// Adjusting the number of bits used to compute the half-angle value for
// angles > pi/4. The new number is 30.
// In addition, the extra monotonicity check is enabled.
ac_tan_pwl<AC_TRN, 30, true>(x, y);

// The following line, returns by value instead of by reference.
y = ac_tan_pwl <output_type> (x);
```


2.10. Arctangent (**ac_atan_pwl**)

The *ac_atan_pwl* library provides a PWL based approximation of the arctangent function for positive (a) *ac_fixed* (b) *ac_float* (c) *ac_std_float* and (d) *ac_ieee_float* inputs.

2.10.1. The **ac_atan_pwl** Implementation

The *ac_atan_pwl* has overloaded functions to support the different datatypes listed earlier. The *ac_fixed* version is the one that contains the actual PWL implementation, and the floating point versions depend on it for their functioning.

Normalization

The actual arctangent PWL implementation only covers the domain of [0, 1). This is done not only because the domain from [1, ∞) can be covered with the reciprocal function, but because attempting to cover a larger domain with saturation for input values that cross a certain limit means that:

- It's harder to obtain an adequate fit to the actual arctangent function, due to the nature of the function curve, necessitating more segments and a larger area to store LUT values.
- A comparator must be used for saturation, resulting in a further increase in area.

The formula used for any values that exceed unity is as follows:

$$\text{atan}(x) = \pi/2 - \text{atan}(1/x)$$

Where the value of 1/x is obtained using the reciprocal PWL function.

An input of unity is handled by storing it in a variable which saturates to a value slightly less than unity, hence ensuring that the normalized input value is still within the domain of the PWL function.

Floating Point Support

The functions for *ac_std_float* and *ac_ieee_float* support serves as a wrapper around the *ac_float* inverse square root design, which performs conversions between the standard/IEEE float inputs/outputs and the intermediate *ac_float* variables that are passed to the *ac_float* design.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the “real thing”, i.e. the accurate implementation of the arctangent function, the following graph is provided for the domain [0, 1):

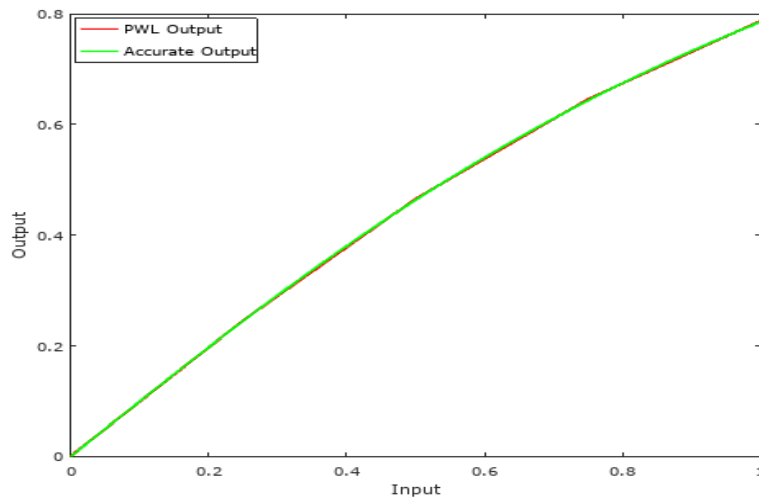


Illustration 8: PWL vs Accurate Output for Arctangent

2.10.2. Function Templates

The following is the function prototype for the *ac_atan_pwl* function for different datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_atan_pwl(
    const ac_fixed<W, I, false, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
);
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ>
void ac_atan_pwl(
    const ac_float<W, I, E, Q> &input,
    ac_float<outW, outI, outE, outQ> &output
);
```

```
template <ac_q_mode pwl_Q = AC_TRN, int W, int E, int outW, int outE>
void ac_atan_pwl(
    const ac_std_float<W, E> &input,
    ac_std_float<outW, outE> &output
);
```

```
);
```

```
template<ac_q_mode pwl_Q = AC_TRN,
        ac_ieee_float_format Format,
        ac_ieee_float_format outFormat>
void ac_atan_pwl(
    const ac_ieee_float<Format> &input,
    ac_ieee_float<outFormat> &output
);
```

Returning by Value

The *ac_atan_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the Example Function Calls section below. The prototype for the function call to return by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_atan_pwl(const T_in &input);
```

2.10.3. Example Function Calls

An example function call for *ac_fixed* variables is given below.

```
typedef ac_fixed<20, 11, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 15, false, AC_RND, AC_SAT> output_type;

input_type x = 0.5;
output_type y;

// Returns y = atan(x), and returns by reference.
ac_atan_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_atan_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_atan_pwl <output_type> (x);
```

2.11. High-accuracy Arctangent (**ac_atan_pwl_ha**)

The high-accuracy arctangent library is designed to be a higher-accuracy version of the *ac_atan_pwl* function. The implementation, function prototypes and example function calls for the *ac_atan_pwl_ha* library are almost the same as that for the *ac_atan_pwl* library, with the only three major differences being as follows:

- The function name is different; the high-accuracy arctangent library has the suffix *_ha* attached to denote that it is a *high-accuracy* implementation.
- The high-accuracy version uses 32 segments to enable a higher-accuracy output, as compared to 4 segments for the original implementation. Hence, the high-accuracy version will consume more area in hardware. The 32-element LUTs for the high-accuracy version might be too big to be mapped to a register bank, and hence be mapped to memories instead. The user might want to consider changing the memory mapping threshold if they do not wish for this to happen.
- The high-accuracy reciprocal PWL is called to enable the calculation of $\text{atan}(1/x)$.

Keeping this differences in mind, the user can consult the documentation for *ac_atan_pwl* for more details on the working of the high-accuracy version.

2.12. Very High-accuracy Arctangent (**ac_atan_pwl_vha**)

For applications where an ever higher accuracy output is desired, the user can utilize the *ac_atan_pwl_vha* library, which uses a 64-segment PWL. Like its high-accuracy arctangent counterpart, this library is also very similar to the *ac_atan_pwl* library, with the only major differences being the name, the number of segments used, and the reciprocal PWL library called (*ac_reciprocal_pwl_vha*). This library is designed to have an absolute error below $5e-5$.

2.13. Sigmoid (**ac_sigmoid_pwl**)

The *ac_sigmoid_pwl* library provides a PWL based approximation of the sigmoid function for *ac_fixed* inputs with minimal inaccuracy.

Normalization

The actual sigmoid PWL implementation only covers the domain of $[0, 5)$. This pwl domain was selected because attempting to cover a larger domain makes it harder to obtain an adequate fit to the actual sigmoid function necessitating more segments and a larger area to store LUT values. As the input approaches infinity, the sigmoid function saturates and thus for inputs greater than 5, the output is saturated to the maximum possible pwl output. For negative inputs, the output will simply be 1 - the output of its positive counterpart taking advantage of the symmetry of the sigmoid function.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the “real thing”, i.e. the accurate implementation of the sigmoid function, the following graph is provided for the domain $[0, 5)$:

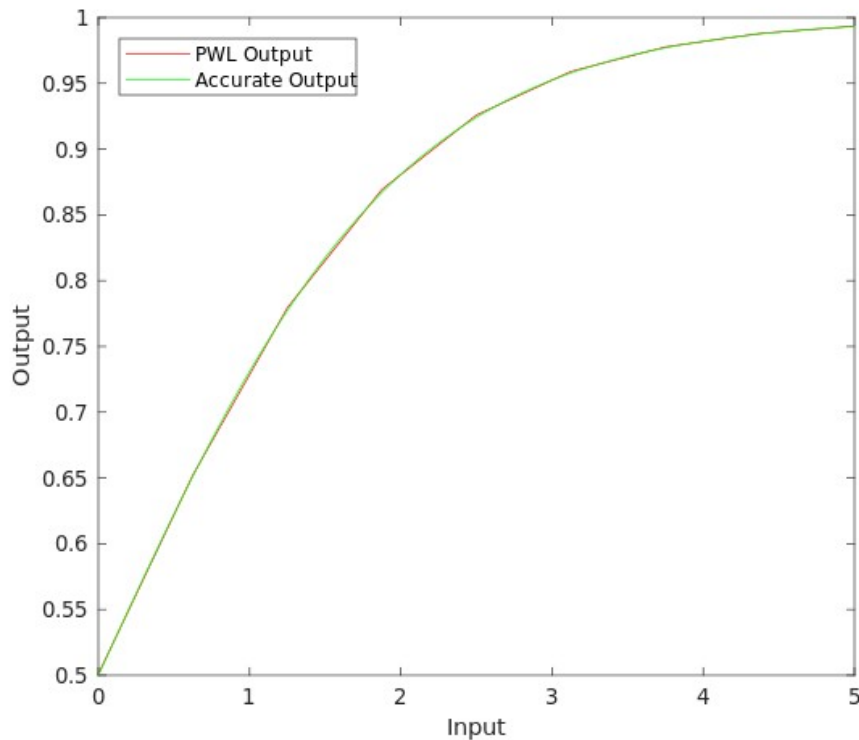


Illustration 9: PWL Output vs. Accurate Output for Sigmoid

2.13.2. Function Templates

The following is the function prototype for the `ac_sigmoid_pwl` function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_sigmoid_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
);
```

Returning by Value

The `ac_sigmoid_pwl` function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an

example of how to do this, please refer to the 2.13.3 Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_sigmoid_pwl(const T_in &input);
```

2.13.3. Example Function Calls

An example of a function call to store the value of sigmoid of a sample *ac_fixed* variable *x* in a variable *y*, by using both return by reference and return by value, is shown below.

```
typedef ac_fixed<15, 7, true, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 3, false, AC_RND, AC_SAT> output_type;

input_type x = 2.5;
output_type y;

// Returns y = sigmoid(x), and returns by reference.
ac_sigmoid_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_sigmoid_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_sigmoid_pwl <output_type> (x);
```

2.14. Hyperbolic Tangent (*ac_tanh_pwl*)

The *ac_tanh_pwl* library provides a PWL based approximation of the hyperbolic tangent function for *ac_fixed* inputs with minimal inaccuracy.

Normalization

The actual hyperbolic tangent PWL implementation only covers the domain of [0, 3). This pwl domain was selected because it gives a good fit to the actual hyperbolic tangent function. As the input approaches infinity, the hyperbolic tangent function saturates and thus for inputs greater than 3, the output is saturated to the maximum possible pwl output. For negative inputs, the output will simply be the negation of the output of its positive counterpart taking advantage of the symmetry of the hyperbolic tangent function.

PWL Approximation Graph

To explain the closeness of the PWL approximation to the accurate implementation of the hyperbolic tangent function, the following graph is provided for the domain [0, 3) :

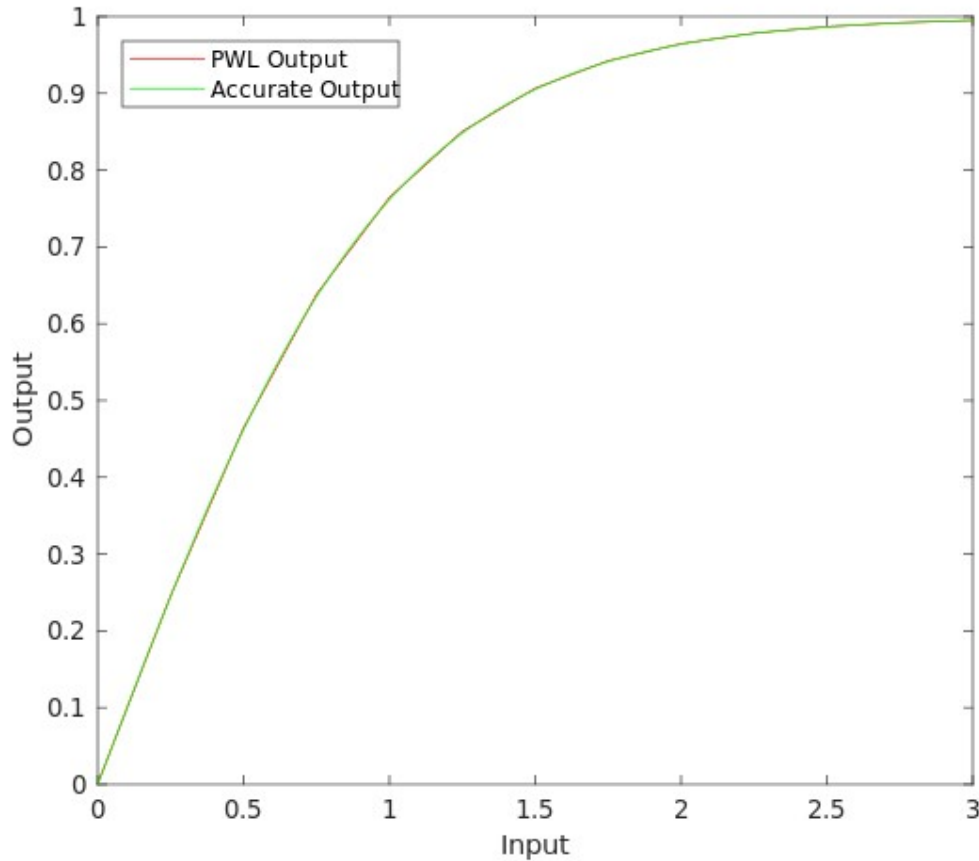


Illustration 10: PWL Output vs. Accurate Output for Hyperbolic Tangent

2.14.2. Function Templates

The following is the function prototype for the *ac_tanh_pwl* function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO>
void ac_tanh_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, outS, outQ, outO> &output
);
```

Returning by Value

The *ac_tanh_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the 2.13.3 Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_tanh_pwl(const T_in &input);
```

2.14.3. Example Function Calls

An example of a function call to store the value of hyperbolic tan of a sample *ac_fixed* variable *x* in a variable *y*, by using both return by reference and return by value, is shown below.

```
typedef ac_fixed<15, 7, true, AC_RND, AC_SAT> input_type;
typedef ac_fixed<30, 3, false, AC_RND, AC_SAT> output_type;

input_type x = 2.5;
output_type y;

// Returns y = tanh(x), and returns by reference.
ac_tanh_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_tanh_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_tanh_pwl <output_type> (x);
```

2.15. Softmax (`ac_softmax_pwl`)

The *ac_softmax_pwl* library provides a PWL based approximation of the softmax function for an array of *ac_fixed* inputs with minimal inaccuracy.

Intermediate Datatypes

Three intermediate datatypes are important for the computation of the softmax output:

- *T_exp*: Determines the precision for the exponent values of each input, calculated through the *ac_exp_pwl* function. Depends on the input bitwidth, by default. The default derived bitwidth, specifically the derived integer bitwidth, can be very large, hence a *static_assert* is provided in the function to alert the user if it exceeds 64 bits, at compile time and thereby limit the area.
- *T_sum*: Determines the precision for the sum of exponent values. Depends on the bitwidth of *T_exp*.
- *T_recip*: Determines the precision of the reciprocal of the sum above, calculated through the *ac_reciprocal_pwl* function. Depends on the bitwidth of *T_sum*, by default.

All the above datatypes are unsigned, fixed-point datatypes.

Loops and Architectural Exploration

The function contains three loops with the following labels:

- **CALC_EXP_LOOP**: Loops through all input array elements and calculates exponent value of each.
- **SUM_EXP_LOOP**: Loops through all exponent values calculated above and finds their sum.
- **CALC_SOFTMAX_LOOP**: Finds final softmax value by multiplying the exponent values of each input with the reciprocal of their sum.

All these loops can be pipelined and/or unrolled, contingent upon other factors like clock constraints and datatypes involved. If the main function as well as all loops in the design are pipelined with an II of 1, one can achieve a throughput of $2K$ clock cycles, where K is the length of the input/output vector. If the main function is pipelined with an II of 1 and all loops fully unrolled, a throughput of one clock cycle can be achieved. Care should be taken in doing the latter, especially in unrolling the **CALC_SOFTMAX_LOOP**. Doing means that the design now uses K multipliers which may be very large, depending on the size of T_{recip} and T_{exp} . If the area is very large, the user is advised to either consider partially or completely rolling **CALC_SOFTMAX_LOOP**, or to use template parameters to override and reduce the default T_{recip} and T_{exp} bitwidths and hence reduce the area. Refer to the Changing Intermediate Bitwidths section below, for more details.

2.15.2. Function Templates

The following is the function prototype for the `ac_softmax_pwl` function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q=AC_TRN,
        bool or_e=false,
        int iW_e=0, int iI_e=0, ac_q_mode iQ_e=AC_TRN, ac_o_mode iO_e=AC_WRAP,
        bool or_r=false,
        int iW_r=0, int iI_r=0, ac_q_mode iQ_r=AC_TRN, ac_o_mode iO_r=AC_WRAP,
        int K,
        int W, int I, bool S, int Q, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO
        >
void ac_softmax_pwl(
    const ac_fixed<W, I, S, Q, O> (&input)[K],
    const ac_fixed<outW, outI, false, outQ, outO> (&output)[K]
);
```

Changing Intermediate Bitwidths

As mentioned above, the user might want to modify the bitwidths of T_{exp} or T_{recip} . To do so, they must set the `or_e` (for T_{exp}) or `or_r` (for T_{recip}) parameters to true, and assign values to the template parameters with the suffix of `_e` (for T_{exp}) or the suffix of `_r` (for T_{recip}). Please refer to the Example Function Calls section for an example.

2.15.3. Example Function Calls

The following code snippet shows the user how to call the *ac_softmax_pwl* function in various different configurations, for an array of 4 inputs/outputs.

```
ac_fixed<20, 4, true> input[4];
ac_fixed<17, 1, false> output[4];
// Initialize input array.
input[0] = 0.5;
input[1] = -2.4;
input[2] = 3.2;
input[3] = -1.9;
// Call ac_softmax_pwl function with default configuration
ac_softmax_pwl(input, output);
// Call ac_softmax_pwl function, with AC_RND as the rounding type for the PWL
output,
// ac_fixed<32, 16, false, AC_TRN, AC_SAT> as the type for T_exp and the de-
fault
// T_recip type.
ac_softmax_pwl<AC_RND, true, 32, 16, AC_TRN, AC_SAT>(input, output);
```

2.16. Softplus (**ac_softplus_pwl**)

The *ac_softplus_pwl* library provides a PWL based approximation of the softplus function for *ac_fixed* inputs with minimal inaccuracy.

Intermediate Datatypes

Two intermediate datatypes are important for the computation of the softplus output:

- `out_pow_type`: Datatype for the output coming out after calling the *ac_exp_pwl* function.
- `inp_log_type`: Datatype for the input going to the *ac_log_pwl* function.

All the above datatypes are unsigned, fixed-point datatypes.

2.16.2. Function Templates

The following is the function prototype for the *ac_softplus_pwl* function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_softplus_pwl(
```

```
const ac_fixed<W, I, S, Q, O> &input,
ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The *ac_softplus_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_softplus_pwl(const T_in &input);
```

2.16.3. Example Function Calls

The following code snippet shows the user how to call the *ac_softplus_pwl* function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;

input_type x = 2.5;
output_type y;

// Returns y = softplus(x), and returns by reference.
ac_softplus_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_softplus_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_softplus_pwl <output_type> (x);
```

2.17. Softsign (ac_softsign_pwl)

The *ac_softsign_pwl* library provides a PWL based approximation of the softsign function for *ac_fixed* inputs with minimal inaccuracy.

Intermediate Datatypes

Two intermediate datatypes are important for the computation of the softsign output:

- *rec_inp_type*: Datatype for the input to the *ac_reciprocal_pwl* function.

- `rec_out_type`: Datatype for the output to the `ac_reciprocal_pwl` function.

All the above datatypes are unsigned, fixed-point datatypes.

2.17.2. Function Templates

The following is the function prototype for the `ac_softsign_pwl` function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_softsign_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The `ac_softsign_pwl` function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_softsign_pwl(const T_in &input);
```

2.17.3. Example Function Calls

The following code snippet shows the user how to call the `ac_softsign_pwl` function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;

input_type x = 2.5;
output_type y;

// Returns y = softsign(x), and returns by reference.
ac_softsign_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_softsign_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_softsign_pwl <output_type> (x);
```

2.18. Elu (ac_elu_pwl)

The *ac_elu_pwl* library provides a PWL based approximation of the *elu* function for *ac_fixed* inputs with minimal inaccuracy.

Intermediate Datatypes

There is one intermediate datatype which is important for the computation of the elu output:

- *out_pow_type*: Datatype for the output to the *ac_pow_pwl* function.

The above datatype is an unsigned, fixed-point datatype.

2.18.2. Function Templates

The following is the function prototype for the *ac_elu_pwl* function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO,
        int alphaW, int alphaI, bool alphaS, ac_q_mode alphaQ, ac_o_mode al-
phaO>
void ac_elu_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output,
    const ac_fixed<alphaW, alphaI, alphaS, alphaQ, alphaO> &alpha
)
```

Returning by Value

The *ac_elu_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in,
        class T_alpha>
T_out ac_elu_pwl(const T_in &input, const T_alpha &alpha);
```

2.18.3. Example Function Calls

The following code snippet shows the user how to call the *ac_elu_pwl* function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;
```

```
typedef ac_fixed<10, 3, false> alpha_type;

input_type x = 2.5;
output_type y;
alpha_type alpha;

// Returns y = elu(x), and returns by reference.
ac_elu_pwl (x, y, alpha);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_elu_pwl<AC_RND> (x, y, alpha);

// The following line returns by value instead of by reference.
y = ac_elu_pwl <output_type> (x, alpha);
```

2.19. Selu (ac_selu_pwl)

The *ac_selu_pwl* library provides a PWL based approximation of the selu function for *ac_fixed* inputs with minimal inaccuracy.

2.19.1. Function Templates

The following is the function prototype for the *ac_selu_pwl* function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_selu_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The *ac_selu_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
T_out ac_selu_pwl(const T_in &input);
```

2.19.2. Example Function Calls

The following code snippet shows the user how to call the *ac_selu_pwl* function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;

input_type x = 2.5;
output_type y;

// Returns y = selu(x), and returns by reference.
ac_selu_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_selu_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_selu_pwl <output_type> (x);
```

2.20. Gelu (**ac_gelu_pwl**)

The *ac_gelu_pwl* library provides a PWL based approximation of the gelu function for *ac_fixed* inputs with minimal inaccuracy.

2.20.1. Function Templates

The following is the function prototype for the *ac_gelu_pwl* function for *ac_fixed* datatypes:

```
template<ac_q_mode pwl_Q = AC_TRN,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_gelu_pwl(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The *ac_gelu_pwl* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        ac_q_mode pwl_Q = AC_TRN,
        class T_in>
```

```
T_out ac_gelu_pwl(const T_in &input);
```

2.20.2. Example Function Calls

The following code snippet shows the user how to call the *ac_selu_pwl* function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;

input_type x = 2.5;
output_type y;

// Returns y = gelu(x), and returns by reference.
ac_gelu_pwl (x, y);

// Change the rounding mode for intermediate PWL variable to AC_RND
ac_gelu_pwl<AC_RND> (x, y);

// The following line returns by value instead of by reference.
y = ac_gelu_pwl <output_type> (x);
```

2.21.

Chapter 3: CORDIC Math Functions

The CORDIC-based functions in the *ac_math* library include the following operations:

- [Sine/Cosine \(ac_sin_cordic / ac_cos_cordic / ac_sincos_cordic\)](#)
- [Arcsin/Arccos \(ac_arcsin_cordic/ ac_arccos_cordic\)](#)
- [Arctangent \(ac_atan2_cordic\)](#)
- [Exponential \(ac_exp_cordic/ ac_exp2_cordic\)](#)
- [Logarithm \(ac_log_cordic/ ac_log2_cordic\)](#)
- [Power \(ac_pow_cordic\)](#)

3.1. Sine/Cosine (ac_sin_cordic / ac_cos_cordic / ac_sincos_cordic)

There are algorithms that require the computation of the *ac_sin_cordic()* and/or *ac_cos_cordic()* functions for dynamic angles (as opposed to a set of predefined angles as in FFTs which are best implemented as a table lookup). In such cases, the CORDIC algorithm is often used as a way to compute the *ac_sin_cordic()*, *ac_cos_cordic()* or *ac_sincos_cordic()* functions. The math library contains a sample CORDIC implementation using fixed-point data types. The available functional interfaces compute sin, cos or both scaled sin and cos with one call (more efficient hardware than having separate calls to sin and cos).

The first argument is the angle scaled by ($1/\pi$). The advantage of the prescaled angle is that it makes the quadrant computation inside the CORDIC algorithm trivial and it also simplifies the complexity of the caller as in most cases the calls to sin and cos are made with angles that are multiples of π .

AC: ac_fixed

```
void ac_sin_cordic (
    ac_fixed<AW,AI,true,AQ,AO> angle_over_pi,
    ac_fixed<OW,OI,true,OQ,OO> &sin
)
```

```
void ac_cos_cordic (
    ac_fixed<AW,AI,true,AQ,AO> angle_over_pi,
    ac_fixed<OW,OI,true,OQ,OO> &cos
)
```

```
void ac_sincos_cordic (
```

```
ac_fixed<AW,AI,true,AQ,AO> angle_over_pi,
ac_fixed<OW,OI,true,OQ,OO> C,
ac_fixed<OW,OI,true,OQ,OO> &sin,
ac_fixed<OW,OI,true,OQ,OO> &cos
)
```

SystemC: sc_fixed

```
void ac_sin_cordic (
    sc_fixed<AW,AI,AQ,AO> angle_over_pi,
    sc_fixed<OW,OI,OQ,OO> &sin
)
```

```
void ac_cos_cordic (
    sc_fixed<AW,AI,AQ,AO> angle_over_pi,
    sc_fixed<OW,OI,OQ,OO> &cos
)
```

```
void ac_sincos_cordic (
    sc_fixed<AW,AI,AQ,AO> angle_over_pi,
    sc_fixed<OW,OI,OQ,OO> C,
    sc_fixed<OW,OI,OQ,OO> &sin,
    sc_fixed<OW,OI,OQ,OO> &cos
)
```

Table 1: Functions for Sine, Cosine and Combined Sine and Cosine with Amplitude

Note that the interface that computes both *sine* and *cosine*, the data types for *C*, *sin*, *cos* parameters must be identical.

3.2. Arcsin/Accos (ac_arcsin_cordic/ ac_arccos_cordic)

The *ac_math* library contains sample CORDIC implementation using fixed-point data types. The available functional interfaces compute the inverse trigonometric functions arcsin and arccos.

The result of these functions is scaled by (1/PI), which is often the case in practice, maintaining consistency with the sin/cos implementations.

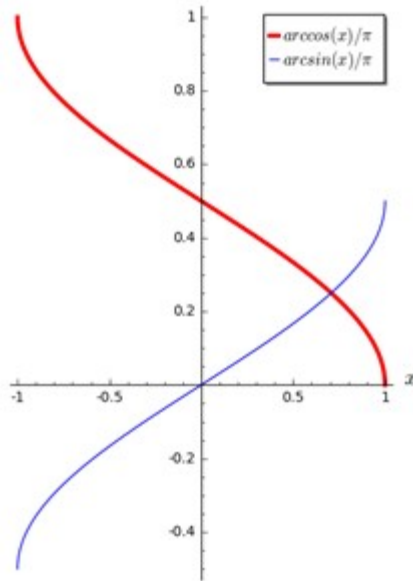


Illustration 11: Graph of scaled arcsin and arccos functions

AC: ac_fixed

```
void ac_arccos_cordic(
    ac_fixed<AW,AI,true,AQ,AO> x,
    ac_fixed<OW,OI,false,OQ,OO> &arccos_x_over_pi
);

void ac_arcsin_cordic(
    ac_fixed<AW,AI,true,AQ,AO> x,
    ac_fixed<OW,OI,true,OQ,OO> &arcsin_x_over_pi
);
```

SystemC: sc_fixed*

```
void ac_arccos_cordic(
    sc_fixed<AW,AI,AQ,AO> x,
    sc_ufixed<OW,OI,OQ,OO> &arccos_x_over_pi
);

void ac_arcsin_cordic(
    sc_fixed<AW,AI,AQ,AO> x,
    sc_fixed<OW,OI,OQ,OO> &arcsin_x_over_pi
);
```

*Note: only supports up to 20 fractional bits, i.e., AW - AI <= 20.

3.3. Arctangent (ac_atan2_cordic)

The *ac_atan2_cordic* function is a fixed-point CORDIC implementation of the functionality of the corresponding *ac_math.h* function. It takes two signed fixed-point arguments called *y* and *x* and it returns the arc tangent in the range $-PI$ to PI . The accuracy of the computation is directly dependent on the precision of the fixed-point *atan* variable passed to the function.

AC: ac_fixed	SystemC: sc_fixed
<pre>void ac_atan2_cordic(ac_fixed<YW,YI,true,YQ,YO> y, ac_fixed<XW,XI,true,XQ,XO> x, ac_fixed<OW,OI,true,OQ,OO> &atan)</pre>	<pre>void ac_atan2_cordic(sc_fixed<YW,YI,YQ,YO> y, sc_fixed<XW,XI,XQ,XO> x, sc_fixed<OW,OI,OQ,OO> &atan)</pre>

Table 2: Functions for Atan2

The behavior when one or both of the arguments is zero is identical with that of the *ac_atan2_cordic* in *ac_math.h*:

- 0 when $x == 0$ and $y == 0$
- $-PI/2$ when $x == 0$ and $y < 0$
- $+PI/2$ when $x == 0$ and $y > 0$

3.4. Exponential (ac_exp_cordic/ ac_exp2_cordic)

The *ac_math* library contains sample CORDIC implementations of exponentials. The *exp* function evaluates the exponential of an argument *x*, i.e., the transcendental number 'e' raised to the power of *x*. The *exp2* function evaluates the number 2 raised to the power of an argument *x*. The input and argument arguments can be *ac_fixed*, *ac_float*, *ac_std_float* or *ac_ieee_float* variables.

3.4.1. Function Declarations

The following code shows the function declarations for the AC Numerical Datatypes. There are no SystemC variants.

```
template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, bool AS, ac_q_mode AQ, ac_o_mode AV,
          int ZW, int ZI, ac_q_mode ZQ, ac_o_mode ZV>
void ac_exp_cordic(
    const ac_fixed<AW,AI,AS,AQ,AV> &x,
    ac_fixed<ZW,ZI,false,ZQ,ZV> &z
)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
```

```

        int AW, int AI, bool AS, ac_q_mode AQ, ac_o_mode AV,
        int ZW, int ZI, ac_q_mode ZQ, ac_o_mode ZV>
void ac_exp2_cordic(
    const ac_fixed<AW,AI,AS,AQ,AV> &x,
    ac_fixed<ZW,ZI,false,ZQ,ZV> &z
)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        int AW, int AI, int AE, ac_q_mode AQ,
        int ZW, int ZI, int ZE, ac_q_mode ZQ>
void ac_exp_cordic(const ac_float<AW,AI,AE,AQ> &x, ac_float<ZW,ZI,ZE,ZQ> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        int AW, int AI, int AE, ac_q_mode AQ,
        int ZW, int ZI, int ZE, ac_q_mode ZQ>
void ac_exp2_cordic(const ac_float<AW,AI,AE,AQ> &x, ac_float<ZW,ZI,ZE,ZQ> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        int AW, int AE, int ZW, int ZE>
void ac_exp_cordic(const ac_std_float<AW, AE> &x, ac_std_float<ZW, ZE> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        int AW, int AE, int ZW, int ZE>
void ac_exp2_cordic(const ac_std_float<AW, AE> &x, ac_std_float<ZW, ZE> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_exp_cordic(const ac_ieee_float<Format> &x, ac_ieee_float<outFormat> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_exp2_cordic(const ac_ieee_float<Format> &x, ac_ieee_float<outFormat> &z)

```

The OR_TF, TF_ and TQ parameters influence the type of certain intermediate fixed point variables. In particular, OR_TF and TF_ are important for implementations that may have large input or output fractional bitwidths, because they prevent the intermediate variables passed to the internal functions from becoming too big in size and hence triggering compiler errors, which can happen if the input or output have a very large bitwidth. The TQ parameter influences the rounding mode of some of the intermediate variables. The purpose of each of the variables can be summarized as follows:

- OR_TF : Overrides default fractional bitwidth of some intermediate variables if set to true. Set to false by default.
- TF_ : Defines the bitwidth to override with. Set to 32 by default.
- TQ : Defines the rounding mode for some intermediate variables. Set to AC_TRN by default.

3.4.2. NaN Handling

The *ac_std_float* and *ac_ieee_float* versions of the exponential cordic libraries can also produce NaN outputs. If the `AC_HCORDIC_NAN_SUPPORTED` macro is defined, these versions will output nan values for nan inputs.

3.5. Logarithm (ac_log_cordic/ ac_log2_cordic)

The *ac_math* library contains sample CORDIC implementations of logarithms. The 'log' function evaluates the natural logarithm of a fixed point argument x, i.e., the logarithm with transcendental number 'e' as its base. The log2 function evaluates the base 2 logarithm of argument x.

3.5.1. Function Declarations

The following code shows the function declarations for the AC Numerical Datatypes. There are no SystemC variants.

```
template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, ac_q_mode AQ, ac_o_mode AV,
          int ZW, int ZI, bool ZS, ac_q_mode ZQ, ac_o_mode ZV>
void ac_log_cordic(
    const ac_fixed<AW,AI,false,AQ,AV> &x,
    ac_fixed<ZW,ZI,ZS,ZQ,ZV> &z
);

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, ac_q_mode AQ, ac_o_mode AV,
          int ZW, int ZI, bool ZS, ac_q_mode ZQ, ac_o_mode ZV>
void ac_log2_cordic(
    const ac_fixed<AW,AI,false,AQ,AV> &x,
    ac_fixed<ZW,ZI,ZS,ZQ,ZV> &z
);

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, int AE, ac_q_mode AQ,
          int ZW, int ZI, int ZE, ac_q_mode ZQ>
void ac_log_cordic(const ac_float<AW,AI,AE,AQ> &x, ac_float<ZW,ZI,ZE,ZQ> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, int AE, ac_q_mode AQ,
          int ZW, int ZI, int ZE, ac_q_mode ZQ>
void ac_log2_cordic(const ac_float<AW,AI,AE,AQ> &x, ac_float<ZW,ZI,ZE,ZQ> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AE, int ZW, int ZE>
void ac_log_cordic(const ac_std_float<AW, AE> &x, ac_std_float<ZW, ZE> &z)
```

```
template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AE, int ZW, int ZE>
void ac_log2_cordic(const ac_std_float<AW, AE> &x, ac_std_float<ZW, ZE> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_log_cordic(const ac_ieee_float<Format> &x, ac_ieee_float<outFormat> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_log2_cordic(const ac_ieee_float<Format> &x, ac_ieee_float<outFormat> &z)
```

The OR_TF, TF_ and TQ parameters influence the type of certain intermediate fixed point variables. In particular, OR_TF and TF_ are important for implementations that may have a large input/output bitwidth, because they prevent the intermediate variables passed to the internal functions from becoming too big in size and hence triggering compiler errors, which can happen if the input or output have a very large bitwidth. The TQ parameter influences the rounding mode of one of the intermediate variables, i.e. the one which stores the normalized input value. The purpose of each of the variables can be summarized as follows:

- OR_TF : Overrides default fractional bitwidth of some intermediate variables if set to true. Set to false by default.
- TF_ : Defines the bitwidth to override with. Set to 32 by default.
- TQ : Defines the rounding mode for an intermediate variable. Set to AC_TRN by default.

3.5.2. NaN Handling

The *ac_std_float* and *ac_ieee_float* versions of the logarithmic cordic libraries can also produce NaN outputs. If the AC_HCORDIC_NAN_SUPPORTED macro is defined, these versions will output nan values for nan and negative inputs.

3.6. Power (ac_pow_cordic)

The *ac_pow_cordic* function evaluates the result of raising a variable *a* to the power of another variable *b*. The supported variable types are *ac_fixed*, *ac_float*, *ac_std_float* and *ac_ieee_float*.

3.6.1. Function Declarations

The following code shows the function declarations for the AC Numerical Datatypes supported. There are no SystemC variants.

```
template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, ac_q_mode AQ, ac_o_mode AV,
          int BW, int BI, bool BS, ac_q_mode BQ, ac_o_mode BV,
          int ZW, int ZI, ac_q_mode ZQ, ac_o_mode ZV>
```

```
void ac_pow_cordic(const ac_fixed<AW,AI,AS,AQ,AO> &a,
                  const ac_fixed<BW,BI,BS,BQ,BO> &b,
                  ac_fixed<ZW,ZI,false,ZQ,ZV> &z);

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AI, int AE, ac_q_mode AQ,
          int BW, int BI, int BE, ac_q_mode BQ,
          int ZW, int ZI, int ZE, ac_q_mode ZQ>
void ac_pow_cordic(const ac_float<AW, AI, AE, AQ> &a,
                  const ac_float<BW, BI, BE, BQ> &b,
                  ac_float<ZW, ZI, ZE, ZQ> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          int AW, int AE, int BW, int BE, int ZW, int ZE>
void ac_pow_cordic(const ac_std_float<AW, AE> &a,
                  const ac_std_float<BW, BE> &b,
                  ac_std_float<ZW, ZE> &z)

template <bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
          ac_ieee_float_format aFormat, ac_ieee_float_format bFormat,
          ac_ieee_float_format outFormat>
void ac_pow_cordic(const ac_ieee_float<aFormat> &a,
                  const ac_ieee_float<bFormat> &b,
                  ac_ieee_float<outFormat> &z)
```

The OR_TF, TF_ and TQ parameters influence the type of certain intermediate fixed point variables. In particular, OR_TF and TF_ are important for implementations that may have a large input/output bitwidth, because they prevent the intermediate variables passed to the internal functions from becoming too big in size and hence triggering compiler errors, which can happen if the input or output have a very large bitwidth. The TQ parameter influences the rounding mode of some of the intermediate variables. The purpose of each of the variables can be summarized as follows:

- OR_TF : Overrides default fractional bitwidth of some intermediate variables if set to true. Set to false by default.
- TF_ : Defines the bitwidth to override with. Set to 32 by default.
- TQ : Defines the rounding mode for some intermediate variables. Set to AC_TRN by default.

Note: The fixed point logarithm and exponential cordic functions have limitations on the types they can accept for their input/output. The logarithm functions will only accept unsigned inputs, as the actual logarithm functions operate on the positive real-valued domain, while the exponential functions only accept unsigned types for the output, as they output values in the range of positive real values. This is different than the standard C math functions for logarithm and exponential functions, which accept signed doubles at their input and output.

Chapter 4: Lookup Table (LUT) Functions

The *ac_math* package includes the following Lookup Table based functions:

- [Sine/Cosine \(*ac_sincos_lut*\)](#)

The following subsections describes the implementation and usage of these functions in more detail.

4.1. Sine/Cosine (*ac_sincos_lut*)

The *ac_sincos_lut* function is designed to provide the sine and cosine values using a lookup table (LUT).

4.1.1. The *ac_sincos_lut* Implementation

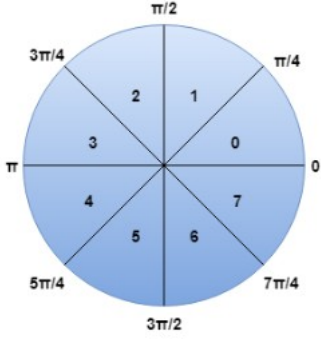
The *ac_sincos_lut* function accepts *ac_fixed* datatypes as input. The domain for the input is (0,1) radians/2PI. The function returns an *ac_complex<ac_fixed>* variable where the real part represents the *cosine* value and the imaginary part represents the *sine* value. The number of table entries is 512 values. A prototype of the *ac_sincos_lut* function is shown below:

```
template<class T_in, class T_out>
void ac_sincos_lut(const T_in &input, ac_complex<T_out> &output)
```

In the above prototype, the class *T_in* represents the datatype of the input which should be an *ac_fixed* datatype. The class *T_out* represents the datatype of the output which should also be an *ac_fixed* datatype. Here, *input* represents the input to the *ac_sincos_lut* function, the *output* is an *ac_complex* variable whose real part is the output *cosine* value and whose imaginary part is the output *sine* value.

Algorithm

All the entries for the sine and cosine lookup tables were generated using the C++ math library *sin()* and *cos()* functions and the values are represented as constant doubles. The table contains only those values for the range of angles from (0-PI/4] radians and symmetry is used to adjust the output accordingly. By leveraging the symmetry the table can have more entries for greater precision while still covering the full range of angle inputs. To compute the index into the table, the first 3 bits are extracted from the MSB side of the input to determine in which octant the input lies. The octants can be defined as:

Input Angle (1/2PI radians)	Octant	
[0.000 – 0.125)	0	
[0.125 – 0.250)	1	
[0.250 – 0.375)	2	
[0.375 – 0.500)	3	
[0.500 – 0.625)	4	
[0.625 – 0.750)	5	
[0.750 – 0.875)	6	
[0.875 – 1.000)	7	

The look up table index in this implementation uses the input bits (MSB-3 : LSB). When the input bitwidth is exactly 12 bits, this results in 512 possible indexes. If the input bitwidth is greater than 12 bits, then the closest table entry is found. If the input bitwidth is less than 12 bits, then a stride is implemented.

Handling negative inputs

If the input angle is a negative value (for example -0.125) it would be represented in 2's complement form as 1.1110 0000. Ignoring the integer portion of the result, the fractional portion is 0.875 which is identical to -0.125.

Returning by Value

The implementations of the ac_sincos_lut() function can return the output by value as well as by reference. See the sample code below for examples of each form. The prototype for the function to return by value is shown below:

```
template<class T_out, class T_in>
T_out ac_sincos_lut(const T_in &input);
```

4.1.2. Example Function Call

An example of a function call is as follows where x represents the input angle and y represents a complex variable whose real part is the cos value and imaginary part is the sin value:

```
typedef ac_fixed<12, 1, true, AC_RND, AC_SAT> input_type;
typedef ac_complex<ac_fixed<23, 1, true, AC_RND, AC_SAT> > output_type;

input_type x = 0.37;
output_type y;

// returning value by reference
ac_sincos_lut(x, y);
cout << "Sine: " << y.i() << endl;
cout << "Cosine: " << y.r() << endl;
```

```
// returning by value  
y = ac_sincos_lut<output_type>(x);
```

4.1.3. Increasing look up table entries

In the current implementation, the number of lookup table entries for the sine and cosine lookup tables is 512. This means that the implementation gives an accurate output for the cases when input bitwidth is less than or equal to 12 bits. So for cases when input bitwidth is greater than 12 bits, the closest table entry is found (as mentioned above) which might incur some error (maximum error is approximately equal to the largest difference between consecutive lookup table entries) and therefore if a more accurate implementation is desired, then the current lookup table entries can be replaced. The formula for the number of lookup table entries for sine and cosine is as follows:-

Number of lookup table entries = $2^{(\text{input bitwidth} - 3)}$

For example, the number of lookup table entries to get an accurate output for the following input bitwidths are:-

- 13 bits – 1024
- 14 bits – 2048
- 15 bits – 4096

The *lutgensincos.cpp* file under the *.../src/examples/Math/ac_sincos_lut* directory can be referred for generating lookup table entries. Also the *ac_sincos_lut.h* library header file has to be accordingly modified if the number of lookup table entries are changed.

Synthesizing the ac_sincos_lut function

In High Level Synthesis, the *ac_sincos_lut* function can be completely pipelined as there are no data dependencies thus ensuring high throughput and low latency.

Chapter 5: Linear Algebra Functions

The `ac_math` package includes the following linear algebra functions:

- [Cholesky Decomposition \(`ac\_chol\_d`\)](#)
- [Cholesky Decomposition: Version 2.0 \(`ac\_cholesky\_decomposition\_v2`\)](#)
 - Right Looking Cholesky Decomposition - Generic Architecture
 - Left Looking Cholesky Decomposition - Generic Architecture
 - Right Looking Cholesky Decomposition - Systolic Architecture
- [Cholesky Inverse \(`ac\_cholinv`\)](#)
- [Cholesky Matrix Inverse \(Systolic\) \(`ac\_cholesky\_inv\_1d\_systolic\_rl`\)](#)
- [Cholesky Matrix Inversion \(`ac\_cholesky\_matrix\_inverse`\)](#)
- [Cholesky Matrix Inversion Version 2.0 \(`ac\_cholesky\_matrix\_inverse\_v2`\)](#)
- [Systolic matrix multiplication \(`ac\_matmult\_systolic`\)](#)
- [QR Decomposition \(`ac\_qrd`\)](#)

The following subsections describes the implementation and usage of these functions in more detail.

5.1. Cholesky Decomposition (`ac_chol_d`)

The `ac_chol_d` library is designed to provide a Cholesky Decomposition of a square, positive definite input matrix using the Cholesky-Crout algorithm. The user can utilize either accurate math functions or the piece-wise linear (pwl) math library for the internal calculations involved. Cholesky Decomposition is an important linear algebra operation that has applications in solving linear equations.

5.1.1. The `ac_chol_d` Implementation

The `ac_chol_d` library provides ten overloaded functions for computing the Cholesky Decomposition of real and complex matrices, and returns the lower triangular matrix result of the Cholesky Decomposition. Combined, the ten functions handle five datatypes, which are (a) `ac_fixed` (b) `ac_float` (c) `ac_std_float` (d) `ac_ieee_float` and (e) `ac_complex<ac_fixed>`. The matrix of data elements of each type can either be passed as standard two-dimensional C++ arrays, or can be packaged in the `ac_matrix` class.

Algorithm

As discussed earlier, the Cholesky-Crout algorithm is used to compute the Cholesky Decomposition. The computation for this algorithm is done in a column-wise manner. We first compute the diagonal element for each matrix either using the accurate or the approximate (PWL) *sqrt* function. After we calculate the diagonal element, we store its inverse in a separate variable, which is then reused for the computation of the non-diagonal elements below the diagonal. This inverse square root value can be calculated using the accurate *ac_div* function, or the approximate PWL version from the *ac_inverse_sqrt_pwl* library.

The Cholesky-Crout algorithm is used due to its simplicity and the reusability of the reciprocal value in calculating the non-diagonal elements.

Accurate Math Functions vs. PWL Approximations

The user has the option of being able to choose the accurate versions of the reciprocal and the sqrt functions, or their PWL approximations, as mentioned earlier, depending upon how much accuracy they may desire. Both have their advantages and disadvantages. The accurate math functions, while providing high precision, can also add a lot of overhead in terms of throughput/area. The PWL functions, while providing higher throughput at a lesser cost in area, are also imprecise.

To use PWL approximations, the user has to override a default template parameter (*use_pwl*) for the *ac_chol_d* function call. An example of doing so is giving in the Example Function Calls section.

Floating Point Implementations

As mentioned earlier, the library provides support for *ac_float*, *ac_std_float* and *ac_ieee_float* intermediate variables. The *ac_std_float* and *ac_ieee_float* implementations serve as a wrapper around and depend on the *ac_float* implementation for the actual calculations, with temporary arrays provided in both these wrapper implementations to allow for interfacing with the *ac_float* implementation.

Type of Intermediate Variables

Intermediate variables are used within the function to store the result of repeated subtractions in the process of calculating each element, as well as the reciprocal of diagonal elements. The default precision of these intermediate variables is set to be equal to the precision of the output variables. However, the user can choose to add any number of bits to these default precision values (in case of complex intermediate variables, these extra bits are used for the real/imaginary parts). The precision can be changed by overriding the default *delta_** template parameters. The two possible sets of *delta_** parameters are given as follows:

- For real/complex fixed point variables: *delta_w* and *delta_i* (added to word and integer width of fixed point intermediate variables, respectively).
- For real floating point variables: *delta_w*, *delta_i* and *delta_e* (added to word, integer and exponent width of the *ac_float* intermediate variables, respectively)

The user can also choose the rounding mode of the intermediate variables used for both fixed and floating point intermediate variables by overriding the *imd_Q* default template parameter (set to *AC_RND* by default). Similarly, the *imd_O* template parameter (set to *AC_SAT* by default) can be overridden to change the saturation mode of fixed-point intermediate types.

Note that, because the *ac_std_float* and *ac_ieee_float* implementations are a wrapper around the *ac_float* implementation, any information passed through the *delta_** and *imd_** template parameters for either is in turn passed to the *ac_float* implementation.

Examples on how to override these parameters are given in the Example Function Calls section below.

Input Checking

As explained earlier, the input matrix must be positive definite. While calculating the values for the diagonal elements of the matrix, a square root operation is performed, either using a PWL approximation or the accurate math function. If the input to the *sqrt* function is negative/zero for any value, that means that the input matrix is not positive definite, and a macro-enabled *AC_ASSERT* is provided that will throw a run-time error in such a case. In case the *AC_ASSERT* does not kick in, additional, synthesizable functionality is also provided to output a matrix of zeros.

For certain input matrices when the PWL implementations are used for calculation, the inaccuracy incurred during computations might be large enough to result in the input matrix being wrongly perceived as not positive definite even when it is such. This particularly occurs when the diagonal values of the output matrix are very small and hence, as a result, even a small absolute error in the calculation of the reciprocal value of this diagonal quickly blows up and results in a large error when the remaining elements of that column are calculated. As this error builds up during the calculation of the value for the next diagonal element, the value calculated as the input to the square root function can turn out to be negative, hence resulting in the input checking failing for this particular case even though the input matrix is positive definite.

5.1.2. Function Prototypes

The following are the overloaded function prototypes for the *ac_chol_d* function to handle different datatypes:

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0,
        ac_q_mode imd_Q = AC_RND, ac_o_mode imd_O = AC_SAT,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ,
        ac_o_mode outO,
        unsigned M>
void ac_chol_d(
    const ac_fixed<W, I, S, Q, O> A[M][M],
    ac_fixed<outW, outI, outS, outQ, outO> L[M][M]
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0, int delta_e = 0,
        ac_q_mode imd_Q = AC_RND,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ,
        unsigned M>
```

```
void ac_chol_d(
    const ac_float<W, I, E, Q> A[M][M],
    ac_float<outW, outI, outE, outQ> L[M][M]
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0, int delta_e = 0,
        ac_q_mode imd_Q = AC_RND,
        int W, int E,
        int outW, int outE,
        unsigned M>
void ac_chol_d(
    const ac_std_float<W, E> A[M][M],
    ac_std_float<outW, outE> L[M][M]
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0, int delta_e = 0,
        ac_q_mode imd_Q = AC_RND,
        ac_ieee_float_format Format,
        ac_ieee_float_format outFormat,
        unsigned M>
void ac_chol_d(
    const ac_ieee_float<Format> A[M][M],
    ac_ieee_float<outFormat> L[M][M]
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0,
        ac_q_mode imd_Q = AC_RND, ac_o_mode imd_O = AC_SAT,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO,
        unsigned M>
void ac_chol_d(
    const ac_complex<ac_fixed<W, I, S, Q, O> > A[M][M],
    ac_complex<ac_fixed<outW, outI, outS, outQ, outO> > L[M][M]
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0,
        ac_q_mode imd_Q = AC_RND, ac_o_mode imd_O = AC_SAT,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO,
        unsigned M>
void ac_chol_d(
    const ac_matrix<ac_fixed<W, I, S, Q, O>, M, M> &A,
    ac_matrix<ac_fixed<outW, outI, outS, outQ, outO>, M, M> &L
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0, int delta_e = 0,
        ac_q_mode imd_Q = AC_RND,
        int W, int I, int E, ac_q_mode Q,
        int outW, int outI, int outE, ac_q_mode outQ,
        unsigned M>
void ac_chol_d(
    const ac_matrix<ac_float<W, I, E, Q>, M, M> &A,
    ac_matrix<ac_float<outW, outI, outE, outQ>, M, M> &L
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0, int delta_e = 0,
        ac_q_mode imd_Q = AC_RND,
        int W, int E,
        int outW, int outE,
        unsigned M>
void ac_chol_d(
    const ac_matrix<ac_std_float<W, E>, M, M> &A,
    ac_matrix<ac_std_float<outW, outE>, M, M> &L
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0, int delta_e = 0,
        ac_q_mode imd_Q = AC_RND,
        ac_ieee_float_format Format,
        ac_ieee_float_format outFormat,
        unsigned M>
void ac_chol_d(
```



```
const ac_matrix<ac_ieee_float<Format>, M, M> &A,
ac_matrix<ac_ieee_float<outFormat>, M, M> &L
)
```

```
template<bool use_pwl = false,
        int delta_w = 0, int delta_i = 0,
        ac_q_mode imd_Q = AC_RND, ac_o_mode imd_O = AC_SAT,
        int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO,
        unsigned M>
void ac_chol_d(
    const ac_matrix<ac_complex<ac_fixed<W, I, S, Q, O> >, M, M> &A,
    ac_matrix<ac_complex<ac_fixed<outW, outI, outS, outQ, outO> >, M, M> &L
)
```

5.1.3. C++ Compiler

The functions use default template arguments. This requires using a C++ compiler that supports the C++11 standard. Failing to use such a compiler will result in compilation errors.

5.1.4. Example Function Calls

An example of a function call to store the lower triangular Cholesky Decomposition matrix of a sample 8x8 fixed point, positive definite matrix A in a matrix L is shown below:

```
typedef ac_matrix<ac_fixed<20,11,true,AC_RND,AC_SAT>,8,8> i_type;
typedef ac_matrix<ac_fixed<30,15,true,AC_RND,AC_SAT>,8,8> o_type;
i_type A;
o_type L;
//Hypothetical function that generates a positive definite matrix
//and stores the output in A.
gen_pos_def_matrix(A);
ac_chol_d(A, L);
```

The ac_matrix L hereafter stores the lower triangular matrix result of the Cholesky decomposition.

Choosing PWL approximation

By default, accurate math functions are used for the calculation of internal variables. However, as mentioned earlier, the user can override the default. They can use PWL approximation functions by giving a Boolean template argument, as follows:

```
ac_chol_d<true>(A, L);
```

Configuring the Type for Temporary Variables

The user can add extra bits to the default precision of the temporary variables in the function, by overriding the default template parameters. They can also add other rounding and saturation modes for the temporary variables for fixed point variables. They can do it as follows:

```
ac_chol_d<false, 10, 5, AC_TRN, AC_WRAP>(A, L);
```

By doing this, the user will add 10 bits to the default value for the temporary variable word width, 5 bits to the default value for the temporary variable integer width and switch off rounding and saturation for the temporary variables. Note that the user must also explicitly specify whether they want to use PWL approximation or the accurate math functions in such a case. (In this case, the user specifies that they want to use the accurate math functions)

Similarly, the user can also subtract bits from the default precision values. To do so, they merely need to pass negative parameters for the same. If the user wants to subtract 3 bits from word width and 2 bits from integer width, they can pass the following template parameters:

```
ac_chol_d<false, -3, -2>(A, L);
```

An example function call where the user changes the type configuration for floating point intermediate variables is shown below:

```
ac_chol_d<false, 10, 5, 2, AC_TRN>
```

Doing so will add 10 bits to the bitwidth, 5 bits to the integer width and 2 bits to the exponent width of temporary variables while setting their rounding type to `AC_TRN` and using accurate math functions.

5.2. Cholesky Decomposition: Version 2.0 (ac_cholesky_decomposition_v2)

The **ac_cholesky_decomposition_v2** HDIP provides a PPA-optimized (Power, Performance, Area) hardware solution for Cholesky factorization, offering significant advancements over previous implementations (e.g., `ac_chol_d.h`). This second-generation IP introduces enhanced architectural flexibility by supporting generic Right-Looking, generic Left-Looking, and 1-D Systolic Right-Looking Cholesky decomposition variants. This allows designers to tailor the implementation to specific application needs, balancing latency, throughput, and resource utilization more effectively. Version 2 is engineered to deliver superior Quality of Results (QoR) and reduced HLS compilation time, making it a robust and production-ready solution for hardware acceleration on FPGA and ASIC platforms.

5.2.1. The ac_cholesky_decomposition_v2 HDIP Implementation:

This library performs the fundamental Cholesky factorization of a symmetric positive-definite (SPD) matrix, decomposing it into a lower triangular matrix L . The implementation is highly modular and designed for seamless integration into larger matrix inversion pipelines or linear system solvers.

The core functionality is realized through a configurable architecture that can adapt to diverse requirements:

- **Architectural Flexibility:** Supports three distinct Cholesky decomposition algorithms (Generic Right-Looking, Generic Left-Looking, or 1-D Systolic Right-Looking), allowing users to select the optimal approach based on matrix size, performance targets, and resource constraints.
- **Pipelined Processing:** Leverages deeply pipelined processing elements (PEs) and structured data movement to achieve high throughput and efficient hardware utilization.
- **Optimized Arithmetic Primitives:** Integrates hardware-optimized square root, reciprocal, and multiplication operations, implemented with pipelined data paths to balance latency and resource usage.

5.2.2. Architecture Supported

The ac_cholesky_decomposition_v2 IP's strength lies in its multi-architectural approach to Cholesky decomposition, providing explicit control over the algorithmic strategy to best suit the target application and hardware platform. This flexibility is a direct improvement over earlier, less adaptable implementations.

Generic Architecture based Right-Looking Cholesky:

- **Application Type:** Ideal for small to moderate matrix sizes, often encountered in embedded signal processing, real-time control, or small-scale MIMO detection.
- **Mechanism:** This algorithm proceeds column-by-column. After computing each column of the lower triangular factor L, it immediately updates the remaining (trailing) sub-matrix to its right. This "forward-only" data propagation minimizes data hazards.
- **Benefit:** Offers simple data dependencies and highly localized computations, leading to faster HLS compilation, reduced control complexity, and efficient utilization of on-chip memory. It excels in latency-sensitive applications.

Generic Architecture based Left-Looking Cholesky:

- **Application Type:** Optimized for larger matrix sizes, common in massive MIMO systems, RADAR/SONAR processing, machine learning kernels, and scientific computing where memory bandwidth and data reuse are critical.
- **Mechanism:** This approach incrementally builds each column of the factorized matrix L by accumulating contributions from previously computed columns. It avoids repeated updates to large trailing sub-matrices, maximizing data reuse.
- **Benefit:** Significantly reduces external memory access, improves memory efficiency, and scales effectively with hierarchical memory architectures.

1-D Systolic Architecture based Right-Looking Cholesky:

- **Application Type:** Suited for streaming data and applications demanding maximum continuous throughput, such as high-bandwidth FPGA designs and ASIC accelerators in HPC, real-time analytics, and continuous sensor processing.

- **Mechanism:** Implements the Right-Looking algorithm on a 1-D systolic array, a linear chain of Processing Elements (PEs). Data flows in a regular, pipelined manner, with each PE performing local computations and passing results to its neighbor.
- **Benefit:** Provides a regular, highly parallel structure with deterministic dataflow and localized inter-PE communication. This results in very high, continuous throughput and excellent scalability for streaming workloads.

All algorithmic variants rely on hardware-optimized arithmetic primitives:

- **Square Root:** Implemented using `ac_math::ac_sqrt`.
- **Reciprocal:** Utilizes `ac_math::ac_reciprocal_pwl_vha` for efficient piecewise-linear approximation.
- **Multiply-Accumulate (MAC):** Core operation for matrix updates.

The IP ensures that the upper triangular portion of the output matrix is explicitly set to zero, yielding a pure lower triangular Cholesky factor L .

Key Features:

- **Multi-Architectural Cholesky:** Configurable selection among Generic Right-Looking, Generic Left-Looking, and 1-D Systolic Right-Looking implementations for optimal PPA.
- **Reduced Compile Time & PPA:** Optimized for improved Power, Performance, and Area efficiency with reduced HLS compilation time compared to previous versions.
- **Configurable Matrix Size:** Parameterized M for compile-time configuration of matrix dimensions.
- **Configurable Data Types:** Compile-time fixed-point/floating-point precision, with Value Range Analysis (VRA) for optimized word lengths.
- **Tolerance Level Control:** User-defined `TOL_CODE` (1e-3, 1e-4, 1e-5) for managing numerical stability and error bounds.
- **Platform Independence & Scalability:** `PLATFORM` parameter enables target-specific tuning for ASIC or FPGA.
- **High Quality of Results (QoR):** Pipelined architectures optimized for area, latency, or throughput, ensuring scalable and predictable performance.
- **Comprehensive Verification:** Robust infrastructure including cycle-accurate testbenches, golden reference models, and automated nightly regression.
- **Multi-Level Verification:** Rigorously validated with Catapult C Design Checker (CDC), SCVerify, and CFormal (LEC, stall behavior, assertions).
- **Coverage Metrics:** Achieves 100% code coverage across branch, statement, and functional metrics.
- **RTL Lint Compliance:** Generates industry-standard, lint-clean RTL for seamless SoC integration.

5.2.3. Configuration Parameters:

The IP is highly configurable via template parameters at HLS compile time, with design-time assertions (AC_ASSERT) for validation.

- **TOL_CODE**: An unsigned integer (3, 4, 5) defining the numerical tolerance (1e-3, 1e-4, 5e-5). This automatically tunes internal fixed-point word lengths and fractional precision.
- **M**: Defines the matrix dimension (M X M).
- **NUM_SETS**: Specifies the number of randomized input test vectors for simulation.
- **ALGO**: Selects the Cholesky decomposition algorithm: SYSTOLIC, RIGHT, or LEFT (e.g., -DUSE_LEFT_ALGO).
- **PLATFORM**: Configures platform-specific optimizations for either ASIC or FPGA targets (e.g., -DUSE_ASIC_PLAT).

Internal data types (Input_T, Output_T, SqrRoot_T, Reciprocal_T, Division_T, etc.) are automatically selected based on TOL_CODE and M to balance accuracy and hardware resources.

5.2.4. Function Prototypes

The ac_cholesky_decomposition_v2 functionality is accessed via the ac_cholesky_decomposition_v2_dut wrapper class.

```
#include <ac_math/ac_cholesky_decomposition_v2.h>

template<int TOL_CODE, unsigned M, int ALGO, int PLATFORM>
class ac_cholesky_decomposition_v2_dut {
public:
    void choldecomtop(
        ac_channel<ac_array<Input_T, M, M>> &A_in,
        ac_channel<ac_array<Output_T, M, M>> &A_out
    );
    // 'if' statements with ALGO selection template parameters.
    if (ALGO == SYSTOLIC) {
        CHOL_SYST.systolic_cholesky_decomposition(Data_In, Data_Out);
    }

    if (ALGO == RIGHT) {
        CHOL_RIGHT.right_cholesky_decomposition(Data_In, Data_Out);
    }

    if (ALGO == LEFT) {
        CHOL_LEFT.left_cholesky_decomposition(Data_In, Data_Out);
    }
}
```

```
private:
//SYSTOLIC CHOLESKY FUNCTION CALLING
ac_math::ac_systolic_cholesky_decomposition<Input_T, Output_T, SqrRoot_T, Sqr-
Root_OutT,Division_T, Reciprocal_T, Reciprocal_OutT,Mat_size_T, M,PLATFORM>
CHOL_SYST;

//RIGHT CHOLESKY FUNCTION CALLING
ac_math::ac_right_cholesky_decomposition<Input_T, Output_T, SqrRoot_T, Sqr-
Root_OutT,Division_T, Reciprocal_T, Reciprocal_OutT,Mat_size_T,M,PLATFORM>
CHOL_RIGHT;

//LEFT CHOLESKY FUNCTION CALLING
ac_math::ac_left_cholesky_decomposition<Input_T, Output_T, SqrRoot_T, Sqr-
Root_OutT,Division_T, Reciprocal_T, Reciprocal_OutT,Mat_size_T, M,PLATFORM>
CHOL_LEFT;
};
```

Note: Input_T and Output_T are internally defined fixed-point types, automatically configured based on TOL_CODE and M.

5.2.5. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

5.2.6. Example Function Calls:

To use the IP:

```
#include <ac_math/ac_cholesky_decomposition_v2.h>

// Define matrix size, tolerance, algorithm, and platform
const int MY_TOL_CODE = 5; // For 5e-5 tolerance
const unsigned MY_MATRIX_SIZE = 8; // For an 8x8 matrix
const int MY_ALGO = SYSTOLIC; // Example: Use Systolic algorithm
const int MY_PLATFORM = ASIC; // Example: Target ASIC platform

// Instantiate the top-level class
ac_math::ac_cholesky_decomposition_v2_dut<MY_TOL_CODE, MY_MATRIX_SIZE, MY_ALGO,
MY_PLATFORM> choldecom_instance;

// Define input and output channels (types derived internally)
ac_channel<ac_array<Input_T, MY_MATRIX_SIZE, MY_MATRIX_SIZE>> A_input_channel;
```

```
ac_channel<ac_array<Output_T, MY_MATRIX_SIZE, MY_MATRIX_SIZE>> L_output_channel;

// Populate A_input_channel with a positive-definite matrix

// Call the decomposition function
choldecom_instance.choldecomtop(A_input_channel, L_output_channel);

// L_output_channel now contains the lower triangular Cholesky factor.
```

5.2.7. CHOLESKY DECOMPOSITION ANALYSIS RESULTS (ASIC)

Frequency=400MhZ, Technology: 45nm

TOL	Size	Archit.	Latency	Throughput	Reg. Area	MUX A	Func A	Total	Runtime
5	4x4	Right	41	42	4379+230	754	18026	23485	37.13
		Left	38	39	4532+214	1285	140201	20305	37.57
		Systolic	37	38	4532+209	1349	13986	20193	38.10
	8x8	Right	95	96	19362+633	5629	15667	42783	54.72
		Left	116	117	19694+633	6655	15799	43784	58.72
		Systolic	115	116	20001+628	5629	15667	42783	54.69
	16x16	Right	243	244	92695+1315	68954	60684	227103	254.96
		Left	237	238	87671+1283	83390	60306	241208	348.28
		Systolic	222	223	88341+1203	82257	79820	259999	298.92

5.3. Cholesky Inverse (ac_cholinv)

The *ac_cholinv* function provides matrix inversion of a positive definite matrix using forward substitution and Cholesky Decomposition which uses the Cholesky-Crout algorithm. The user can utilize either accurate math functions or the PWL math library for the internal calculations involved.

5.3.1. The ac_cholinv Implementation

The *ac_cholinv* library provides two overloaded functions for computing the inverse of input matrices and returns the inverted matrix. Each overloaded function handles a different input datatype. The two datatypes hence handled are *ac_fixed* and *ac_complex<ac_fixed>*. The matrix of data elements of each type are packaged in the *ac_matrix* class.

Algorithm

As mentioned earlier, the Cholesky-Crout algorithm is used to compute the Cholesky Decomposition which is done in the *ac_chol_d.h* library and it returns a lower triangular matrix. Please refer to the documentation of the library *ac_chol_d.h* for Cholesky Decomposition. Then the inverse of the lower triangular matrix is computed using forward substitution and then this inverse is multiplied with its conjugate transpose to obtain the inverse matrix.

Accurate Math Functions vs. PWL Approximations

The user has the option of being able to choose the accurate versions of the reciprocal and the *sqrt* functions, or their PWL approximations, as mentioned earlier, depending upon how much accuracy they may desire. Both have their advantages and disadvantages. The accurate math functions, while accurate, can also add a lot of overhead in terms of throughput/area. The PWL functions, while providing higher throughput, are a bit imprecise.

To use the pwl functions, the user has to override a default template parameter (*use_pwl1*) for the *ac_chol_d* function call in the *ac_cholinv* function and (*use_pwl2*) for the *ac_cholinv* function call. An example of doing so is giving in the *Example Function Calls* section.

Type of Intermediate Variables

Intermediate variables are used within the function to store the result of repeated additions in the process of calculating each element, as well as the reciprocal of diagonal elements. The default precision of these temporary variables is set to be equal to the precision of the output variables. However, the user can choose to add any number of bits to these default values for word width as well as the integer width of the temporary variables (in case of complex temporary variables, these extra bits are used for the real/imaginary parts). They can do this by overriding the default template parameters (*add2w* and *add2i* for word and integer width, respectively). Furthermore, the user can also choose the rounding and saturation modes for the temporary variables by overriding the appropriate template parameters (*temp_Q* for rounding mode and *temp_O* for saturation). An example of how to do so is giving in the *Example Function Calls* section.

5.3.2. Function Prototypes

The following are the overloaded function prototypes for the *ac_chol_d* function to handle different datatypes:

```
template<bool use_pwl1 = false, bool use_pwl2 = false,
        int add2w = 0, int add2i = 0,
        ac_q_mode temp_Q = AC_RND, ac_o_mode temp_O = AC_SAT,
        unsigned M, class T_in, class T_out>
void ac_chol_d(
    const ac_matrix<T_in, M, M> &A,
    ac_matrix<T_out, M, M> &L
)

template<bool use_pwl1 = false, bool use_pwl2 = false, int add2w = 0, int
add2i = 0, ac_q_mode temp_Q = AC_RND, ac_o_mode temp_O = AC_SAT, unsigned M,
class T_in, class T_out>
void ac_chol_d(
    const ac_matrix<ac_complex<T_in>, M, M> &A,
    ac_matrix<ac_complex<T_out>, M, M> &L
)
```


5.3.3. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

5.3.4. Example Function Calls

An example of a function call to store the lower triangular Cholesky Decomposition matrix of a sample 8x8 positive definite matrix A in a matrix L is shown below:

```
typedef ac_matrix<ac_fixed<20,11,true,AC_RND,AC_SAT>,8,8> i_type;
typedef ac_matrix<ac_fixed<30,15,true,AC_RND,AC_SAT>,8,8> o_type;
i_type A;
o_type Ainv;
//Hypothetical function that generates a positive definite matrix
//and stores the output in A.
gen_pos_def_matrix(A);
ac_cholinv(A, Ainv);
```

Choosing PWL approximation

By default, accurate math functions are used for the calculation of internal variables. However, as mentioned earlier, the user can override the default. The first Boolean template parameter if true uses PWL approximation functions for calculation of internal variables and if false uses accurate math functions in the ac_chol_d library and similarly if the second Boolean template parameter if true uses PWL approximation functions for calculation of internal variables and if false uses accurate math functions in the ac_cholinv library.

1. If the user wants to use the PWL approximation functions for both the ac_chol_d and ac_chol_inv libraries, then the following is the way the function should be called

```
ac_chol_inv<true, true>(A, L);
```

2. If the user wants to use the PWL approximation functions for the ac_chol_d library and accurate math functions for the ac_chol_inv library, then the following is the way the function should be called.

```
ac_chol_inv<true, false>(A, L);
```

Note: If the user wants to specify the template parameter for choosing the PWL approximation functions only for the ac_cholinv library, then the user has to explicitly specify whether they want to use PWL approximation or the accurate math functions for the ac_chol_d library as well.

Configuring the Type For Temporary Variables

The user can add extra bits to the default precision of the temporary variables in the function, by overriding the default template parameters. They can also add their own quantization and overflow modes for the temporary variables. For example, they can do it as follows:

```
ac_chol_inv<false, false, 10, 5, AC_RND, AC_SAT>(A, Ainv);
```

By doing this, the user will add 10 bits to the default value for the temporary variable word width, 5 bits to the default value for the temporary variable integer width and switch on rounding and saturation for the temporary variables. Note that the user must also explicitly specify whether they want to use PWL approximation or the accurate math functions in such a case. (In this case, the user specifies that they want to use the accurate math functions)

Similarly, the user can also subtract bits from the default precision values. To do so, they merely need to pass negative parameters for the same. If the user wants to subtract 3 bits from word width and 2 bits from integer width, they can pass the following template parameters:

```
ac_chol_d<false, false, -3, -2>(A, Ainv);
```

5.4. Cholesky Matrix Inverse (Systolic) (ac_cholesky_inv_1d_systolic_rl)

The ac_cholesky_inv_1d_systolic_rl HDIP provides a high performance matrix inversion for square, symmetric positive-definite (SPD) input matrices. It uses a three-stage pipelined architecture (Cholesky decomposition, forward substitution, and backward substitution) based on the Right-Looking Cholesky Decomposition algorithm, implemented with a 1D systolic array architecture for low-latency and high-throughput hardware acceleration.

The ac_cholesky_inv_1d_systolic_rl Implementation:

The right-looking Cholesky decomposition algorithm is strategically chosen for this hardware implementation due to its inherent compatibility with pipelined, dataflow-oriented architectures. This approach enables efficient forward-only data propagation, allowing immediate updates to the trailing sub-matrix after processing each diagonal element. This effectively eliminates data hazards, simplifies control logic, and maximizes concurrency, making it ideal for systolic array implementations and resulting in significantly improved throughput, higher resource utilization, and a streamlined hardware design. This IP implements matrix inversion process through a three-stage pipeline:

1. **Cholesky Factorization:** Decomposes the input SPD matrix into a lower triangular matrix L .
2. **Forward Substitution:** Solves $L \cdot y = I$.
3. **Backward Substitution:** Solves $L^T \cdot x = y$ to obtain the inverse matrix.

The implementation leverages a 1D systolic array architecture, chosen for its balance of latency, throughput, area efficiency, and simplified control logic. It supports fixed-point arithmetic, with internal data types and precision pre-tuned based on user-defined tolerance.

Key Features:

- **Matrix Inversion:** Efficiently inverts SPD matrices using a 3-stage pipelined Cholesky approach.
- **Configurable Matrix Size:** Supports parameterized dimensions ($M \times M$) at compile-time.

- **Configurable Data Types:** Compile-time data type selection (e.g., fixed-point) for precision tuning.
- **Tolerance Level Control:** User-defined numerical tolerance levels ($1e^{-3}$, $1e^{-4}$, $1e^{-5}$ for error-bound control).
- **High Quality of Results (QoR):** Optimized 1D systolic hardware architecture for area, latency, and throughput.
- **Comprehensive Verification:** Includes test bench, golden reference model, and validation via Catapult CDC, SCVerify, and CFormal.
- **RTL Lint Compliance:** Generates lint-clean RTL adhering to industry standards.

Algorithm:

The library uses the Right-Looking Cholesky Decomposition algorithm, which is well-suited for pipelined, dataflow-oriented architectures due to its forward-only data propagation. This approach minimizes data hazards and maximizes concurrency in the systolic array.

The core operations involve:

- **Diagonal Element Computation:** Using square root functions (`ac_math::ac_sqrt`).
- **Reciprocal Calculation:** For normalization, using piecewise-linear approximations (`ac_math::ac_reciprocal_pwl_vha`).
- **Off-Diagonal Element Scaling and Matrix Updates:** To progressively form the Cholesky factor.

These steps are deeply pipelined and mapped to Processing Elements (PEs) in the 1D systolic array, reducing communication latency.

5.4.1. Configuration Parameters

The HDIP is highly configurable via template parameters, allowing users to customize the design at HLS compile time.

- **TOL_CODE:** An unsigned integer (3, 4, 5) defining the numerical tolerance level ($1e^{-3}$, $1e^{-4}$, $1e^{-5}$). This parameter automatically tunes internal fixed-point word lengths and fractional precision.
- **M:** Defines the matrix dimension (M X M).
- **NUM_SETS:** Specifies the number of randomized input test vectors for simulation.

Design-time assertions (`AC_ASSERT`) are included to validate template configurations early in the design flow.

5.4.2. Function Prototypes

The ac_cholesky_inv_1d_systolic_rl functionality is typically accessed through a top-level wrapper class, ac_cholesky_inv_1d_systolic_rl_top.

```
template<int TOL_CODE, unsigned M>
class ac_cholesky_inv_1d_systolic_rl_top {
public:
    void cholinvtop(
        ac_channel<ac_array<Input_T, M, M>> &A_in,
        ac_channel<ac_array<Output_T, M, M>> &A_inv_out
    );
};
```

Note: Input_T and Output_T are internally defined fixed-point types, automatically configured based on TOL_CODE.

5.4.3. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

5.4.4. Example Function Calls

An example of instantiating and calling the Cholesky Inversion IP:

```
// Assuming ac_cholesky_inv_1d_systolic_rl_top is in ac_linalg.h

#include <ac_math/ac_linalg.h>

// Define matrix size and tolerance

const int TOL_CODE = 4; // For 1e-4 tolerance
const unsigned MATRIX_SIZE = 8; // For an 8x8 matrix

// Instantiate the top-level class

ac_math::ac_cholesky_inv_1d_systolic_rl_top<TOL_CODE, MATRIX_SIZE> cholinv_instance;

// Define input and output channels (types are derived internally based on MY_TOL_CODE)

ac_channel<ac_array<Input_T, MATRIX_SIZE, MATRIX_SIZE>> A_input_channel;
```

```
ac_channel<ac_array<Output_T, MATRIX_SIZE, MATRIX_SIZE>> A_inverse_output_channel;  
  
// Populate A_input_channel with a positive-definite matrix  
// ... (e.g., using a hypothetical gen_pos_def_matrix function)  
// Call the inversion function  
  
cholin_instance.cholinvtop(A_input_channel, A_inverse_output_channel);  
  
// A_inverse_output_channel now contains the inverted matrix.
```

5.5. Cholesky Matrix Inversion (`ac_cholesky_matrix_inverse`)

The `ac_cholesky_matrix_inverse` HDIP provides a hardware-efficient solution for inverting square, symmetric positive-definite (SPD) matrices. It employs a three-stage pipelined architecture (Cholesky decomposition, forward substitution, and backward substitution) and dynamically selects between Right-Looking and Left-Looking Cholesky algorithms based on matrix size for optimal compile-time and runtime efficiency. This IP is designed for high-performance hardware acceleration on FPGA and ASIC platforms.

5.5.1. The `ac_cholesky_matrix_inverse` Implementation

This library facilitates matrix inversion through a meticulously designed three-stage pipeline:

- **Cholesky Factorization:** The initial stage involves decomposing the input SPD matrix into its lower triangular Cholesky factor L . This factorization adaptively employs either a Right-Looking or Left-Looking algorithmic variant, determined by the input matrix dimensions to optimize performance and resource utilization.
- **Forward Substitution:** Following factorization, this stage solves the linear system $L \cdot Y = I$, where I is the identity matrix, to compute an intermediate matrix Y .
- **Backward Substitution:** The final stage addresses the system $L^T \cdot X = Y$, utilizing the transpose of the Cholesky factor L^T and the intermediate matrix Y to yield the final inverse matrix X .

The architecture is realized as a linear array of Processing Elements (PEs) and supports fixed-point arithmetic. To further enhance efficiency and manage compilation overhead for larger matrices, the forward and backward substitution stages are segmented into smaller, optimized computational chunks.

5.5.2. Algorithm

The core of the `ac_cholesky_matrix_inverse` HDIP is its adaptive Cholesky decomposition algorithm, which strategically selects between Right-Looking and Left-Looking approaches to maximize hardware efficiency.

- **Right-Looking Cholesky:** This variant is primarily utilized for smaller matrix dimensions. Its strength lies in its simpler data dependency structure and highly localized computations, which enable

aggressive pipelining and faster synthesis. It progresses column-wise, updating the trailing submatrix immediately after each column's factorization, minimizing inter-column dependencies.

- **Left-Looking Cholesky:** Conversely, the Left-Looking approach is optimized for larger matrix sizes. It excels by incrementally building each column of the factorized matrix through the reuse of previously computed data. This strategy significantly improves memory efficiency and reduces external data movement, making it robust for large-scale operations where memory bandwidth and data locality are critical.

Both decomposition methods involve fundamental arithmetic operations:

- **Diagonal Element Computation:** Achieved through square root functions, typically `ac_math::ac_sqrt`.
- **Reciprocal Calculation:** Essential for normalization, implemented using hardware-efficient piece-wise linear approximations such as `ac_math::ac_reciprocal_pwl_vha`.
- **Iterative Updates:** Off-diagonal elements are scaled, and sub-matrices are progressively updated to form the Cholesky factor.

The subsequent forward and backward substitution stages systematically solve the respective linear systems, leveraging the computed Cholesky factor and its transpose to derive the final inverse matrix.

5.5.3. Configuration Parameters

The IP is highly configurable via template parameters at HLS compile time, with design-time assertions (`AC_ASSERT`) for validation.

- **TOL_CODE:** An unsigned integer (3, 4, 5) defining the numerical tolerance ($1e^{-3}$, $1e^{-4}$, $5e^{-5}$). This auto-tunes internal fixed-point word lengths and fractional precision.
- **M:** Defines the matrix dimension (M X M).
- **NUM_SETS:** Specifies the number of randomized input test vectors for simulation.
- **MODE:** Selects between CCORE-optimized and standard versions of the operator (default false enables CCORE).
- **ALGO:** Automatically switches between Right-Looking and Left-Looking algorithms based on matrix size (e.g., when $M > 16$).

Internal data types (`Input_T`, `Output_T`, `SqrRoot_T`, `Reciprocal_T`, `Division_T`, etc.) are automatically selected based on `TOL_CODE` and `M` to balance accuracy and hardware resources.

5.5.4. Function Prototypes

The `ac_cholesky_matrix_inverse` functionality is accessed via the `ac_cholesky_matrix_inverse_dut` wrapper class.

```
#include <ac_math/ac_cholesky_matrix_inverse.h>
template<int TOL_CODE, unsigned M>
class ac_cholesky_matrix_inverse_dut {
public:
    void cholinvtop(
        ac_channel<ac_array<Input_T, M, M>> &A_in,
        ac_channel<ac_array<Output_T, M, M>> &A_inv_out
    );
};
```

Note: Input_T and Output_T are internally defined fixed-point types, automatically configured based on TOL_CODE and M.

5.5.5. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

5.5.6. Example Function Calls

To use the IP:

```
#include <ac_math/ac_cholesky_matrix_inverse.h>

// Define matrix size and tolerance
const int MY_TOL_CODE = 5; // For 5e-5 tolerance
const unsigned MY_MATRIX_SIZE = 16; // For a 16x16 matrix

// Instantiate the top-level class
ac_math::ac_cholesky_matrix_inverse_dut<MY_TOL_CODE, MY_MATRIX_SIZE> cholinv_instance;

// Define input and output channels (types derived internally)
ac_channel<ac_array<Input_T_derived, MY_MATRIX_SIZE, MY_MATRIX_SIZE>> A_input_channel;
ac_channel<ac_array<Output_T_derived, MY_MATRIX_SIZE, MY_MATRIX_SIZE>> A_inverse_output_channel;

// Populate A_input_channel with a positive-definite matrix
// Call the inversion function
cholinv_instance.cholinvtop(A_input_channel, A_inverse_output_channel);

// A_inverse_output_channel now contains the inverted matrix.
```

5.6. Cholesky Matrix Inversion Version 2.0 (ac_cholesky_matrix_inverse_v2)

The **ac_cholesky_matrix_inverse_v2** HDIP delivers a highly optimized, hardware-efficient solution for inverting square, symmetric positive-definite (SPD) matrices. This second-generation HDIP significantly improves upon prior implementations by offering enhanced architectural flexibility and superior Power, Performance, Area (PPA) and compile time requirement efficiency, particularly for larger matrix dimensions. It integrates a three-stage pipelined architecture (Cholesky decomposition, forward substitution, and backward substitution) and provides configurable selection among generic Right-Looking, generic Left-Looking, and 1-D Systolic Cholesky algorithms. This allows users to precisely tailor the implementation to specific matrix sizes, performance targets, and resource budgets, ensuring optimal hardware acceleration on FPGA and ASIC platforms.

The ac_cholesky_matrix_inverse_v2 Implementation:

This library facilitates matrix inversion through a meticulously designed three-stage pipeline, leveraging the ac_cholesky_decomposition_v2 HDIP for its initial stage:

- **Cholesky Factorization:** This foundational stage decomposes the input SPD matrix into its lower triangular Cholesky factor L . It is implemented using the ac_cholesky_decomposition_v2 HDIP, which offers configurable architectural variants (Right-Looking, Left-Looking, or 1-D Systolic) to optimize for matrix size, latency, throughput, and resource utilization.
- **Forward Substitution:** Following factorization, this stage solves the lower triangular linear system $L \cdot Y = I$, where I is the identity matrix, to compute an intermediate matrix Y .
- **Backward Substitution:** The final stage addresses the system $L^T \cdot X = Y$, utilizing the transpose of the Cholesky factor L^T and the intermediate matrix Y to yield the final inverse matrix X . This stage employs an optimized chunked Multiply-Accumulate (MAC) approach for efficient computation.

The architecture is realized as a linear array of Processing Elements (PEs) and supports fixed-point arithmetic. To further enhance efficiency and manage compilation overhead for larger matrices, the forward and backward substitution stages are segmented into smaller, optimized computational chunks.

5.6.1. Algorithm

The core of the ac_cholesky_matrix_inverse_v2 IP is its multi-architectural Cholesky decomposition, which allows explicit selection among three distinct approaches to optimize hardware efficiency across various application requirements. This flexibility is a key improvement over previous versions, enabling fine-grained control over PPA characteristics.

Generic Right-Looking Cholesky:

- **Application:** Best suited for small to medium matrix sizes (e.g., 2X2 to 16 X 16).
- **Mechanism:** Processes the matrix column-wise. After factorizing each column, it immediately updates the trailing sub-matrix.

- **Benefits:** Simple control logic, highly localized computations, fast HLS compilation, low area overhead, and efficient for data resident in on-chip memory. Ideal for latency-sensitive applications.

Generic Left-Looking Cholesky:

- **Application:** Optimized for medium to large matrix sizes (e.g., 8X8 to 32 X 32).
- **Mechanism:** Builds each column of the factorized matrix incrementally by reusing previously computed data.
- **Benefits:** Maximizes data reuse, significantly reduces external memory access, improves memory efficiency, and scales effectively with hierarchical memory systems. Preferred for throughput-dominated, large-scale workloads.

1-D Systolic Right-Looking Cholesky:

- **Application:** Designed for large and streaming matrix workloads (e.g., 2X2 to 32 X 32), where maximum throughput is critical.
- **Mechanism:** Employs a regular, highly parallel 1-D systolic array structure with deterministic dataflow and localized inter-PE communication.
- **Benefits:** Achieves very high, continuous throughput, predictable performance, and excellent scalability for streaming data. While potentially incurring higher initial compilation effort, it delivers superior sustained performance.

All decomposition methods rely on fundamental arithmetic operations:

- **Diagonal Element Computation:** Utilizes hardware-optimized square root functions (e.g., `ac_math::ac_sqrt`).
- **Reciprocal Calculation:** Essential for normalization, implemented via efficient piecewise-linear approximations (e.g., `ac_math::ac_reciprocal_pwl_vha`).
- **Iterative Updates:** Off-diagonal elements are scaled, and sub-matrices are progressively updated to form the Cholesky factor L.

The subsequent forward and backward substitution stages systematically solve their respective linear systems, leveraging the computed Cholesky factor L and its transpose LT to derive the final inverse matrix. These stages are designed with chunked processing to maintain high throughput and reduce compilation time.

5.6.2. Key Features

- **Multi-Architectural Cholesky:** Configurable selection among Generic Right-Looking, Generic Left-Looking, and 1-D Systolic implementations for optimal PPA.
- **PPA Optimization:** Achieves superior Power, Performance, and Area efficiency with reduced HLS compilation time compared to previous versions.

- **Scalable Matrix Dimension Support:** Parameterized M for compile-time configuration of matrix sizes.
- **Configurable Precision & Data Types:** Compile-time fixed-point/floating-point precision, with Value Range Analysis (VRA) for optimized word lengths.
- **Numerical Tolerance Control:** User-defined TOL_CODE ($1e^{-3}$, $1e^{-4}$, $1e^{-5}$) for error management.
- **Platform-Aware Optimization:** PLATFORM parameter enables target-specific tuning for ASIC or FPGA.
- **High Throughput & Predictable Scaling:** Deeply pipelined architectures ensure continuous throughput and stable performance scaling.
- **End-to-End Verification:** Comprehensive infrastructure including cycle-accurate test-benches, golden reference models, and automated nightly regression.
- **Multi-Level Formal & Dynamic Verification:** Rigorously validated with Catapult CDC, SCVerify, CFormal (LEC, stall behavior, assertions).
- **Comprehensive Coverage:** Achieves 100% code coverage (branch, statement, functional).
- **Lint-Clean RTL:** Generates industry-standard, lint-clean RTL for seamless SoC integration.

5.6.3. Configuration Parameters

The IP is highly configurable via template parameters at HLS compile time, with design-time assertions (AC_ASSERT) for validation.

- **TOL_CODE:** An unsigned integer (3, 4, 5) defining the numerical tolerance ($1e^{-3}$, $1e^{-4}$, $5e^{-5}$). This automatically tunes internal fixed-point word lengths and fractional precision.
- **M:** Defines the matrix dimension (M X M).
- **NUM_SETS:** Specifies the number of randomized input test vectors for simulation.
- **PLATFORM:** Configures platform-specific optimizations for either ASIC or FPGA targets (e.g., -DUSE_ASIC_PLAT).
- **ALGO:** Selects the Cholesky decomposition algorithm: SYSTOLIC, RIGHT, or LEFT (e.g., -DUSE_RIGHT_ALGO).

Internal data types (Input_T, Output_T, SqrRoot_T, Reciprocal_T, Division_T, etc.) are automatically selected based on TOL_CODE and M to balance accuracy and hardware resources.

5.6.4. Function Prototypes

The ac_cholesky_matrix_inverse_v2 functionality is accessed via the ac_cholesky_matrix_inverse_v2_dut wrapper class.

```
#include <ac_math/ac_cholesky_matrix_inverse_v2.h>
template<int TOL_CODE, unsigned M>
class ac_cholesky_matrix_inverse_v2_dut {
public:
    void cholinvtop(
        ac_channel<ac_array<Input_T, M, M>> &A_in,
        ac_channel<ac_array<Output_T, M, M>> &A_inv_out
    );
};
```

Note: Input_T and Output_T are internally defined fixed-point types, automatically configured based on TOL_CODE and M.

5.6.5. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

5.6.6. Example Function Calls

To use the HDIP:

```
#include <ac_math/ac_cholesky_matrix_inverse_v2.h>

// Define matrix size and tolerance
const int MY_TOL_CODE = 5; // For 5e-5 tolerance
const unsigned MY_MATRIX_SIZE = 16; // For a 16x16 matrix

// Instantiate the top-level class
ac_math::ac_cholesky_matrix_inverse_v2_dut<MY_TOL_CODE, MY_MATRIX_SIZE> cholinv_instance;

// Define input and output channels (types derived internally)
ac_channel<ac_array<Input_T_derived, MY_MATRIX_SIZE, MY_MATRIX_SIZE>> A_input_channel;
ac_channel<ac_array<Output_T_derived, MY_MATRIX_SIZE, MY_MATRIX_SIZE>> A_inverse_output_channel;

// Populate A_input_channel with a positive-definite matrix
// Call the inversion function
cholinvc_instance.cholinvtop(A_input_channel, A_inverse_output_channel);

// A_inverse_output_channel now contains the inverted matrix.
```

5.6.7. CHOLESKY MATRIX INVERSE ANALYSIS RESULTS (ASIC)

Frequency=400MhZ, Technology: 45nm

TOL	Size	Archit.	Latency	Thrhgput	Reg. Area	MUX A	Func A	Total	Runtime
5	4x4	Right	55	56	4890+306	1960	45323	52773	45.07
		Left	52	53	5043+290	2804	41433	49888	51.90
		Systolic	52	53	4896+290	2165	41407	49086	43.41
	8x8	Right	136	137	20677+741	13061	89465	125829	68.07
		Left	157	158	21335+854	15517	84979	124892	82.10
		Systolic	156	157	22458+848	14708	84786	124950	71.86
	16x16	Right	319	320	111426+1724	115182	231988	468520	400.77
		Left	313	314	111975+1691	125949	219702	471837	547.65
		Systolic	299	300	114580+1616	128054	223815	480759	446.81

5.7. Determinant (ac_determinant)

The *ac_determinant* library implementation is a fully parallelized and scalable implementation for determinant computation using template recursion functionality, with the user being able to choose between using an internally determined datatype or adding their own parameters for intermediate type precision, signedness, rounding and saturation.

5.7.1. The ac_determinant Implementation

The *ac_determinant* library provides four overloaded functions for computing the determinant of real and complex matrices. Combined, the four functions handle two datatypes, which are (a) *ac_fixed* and (b) *ac_complex<ac_fixed>*. The matrix of data elements of each type can either be passed as standard two-dimensional C++ arrays, or can be packaged in the *ac_matrix* class.

Algorithm

Determinant computation is a recursive process. This implementation leverages the recursive tendency of this function to design a hardware-efficient implementation that is scalable and is highly parallelizable. Hence, higher order matrices are reduced to 2x2 matrices by computing minors recursively on them, with each recursion reducing the row and column size by 1. A specialization is also defined for a 1x1 matrix, in which case, we just return the sole value stored in the matrix.

In order to implement a fully parallelizable and synthesizable architecture, template recursion is used, with the 2x2 matrix being the specialized case for recursion.

Internal Precision Adjustment

The *ac_determinant* implementation is such that it maintains full internal precision. To make sure that it does that, bitwidths are computed at every step of the coding.

Using full internal precision can result in a very large bitwidth, however. In case full internal precision might not be required, the user can set their own internal precision by overriding the default template parameters and providing their own internal types.

5.7.2. Function headers

The following are the overloaded function prototypes for the `ac_determinant` function to handle different datatypes:

```
template <bool override = false,
          int internal_width = 16, int internal_int = 8,
          bool internal_sign = true, ac_q_mode internal_rnd = AC_RND,
          ac_o_mode internal_sat = AC_SAT,
          unsigned M,
          int W1, int I1, bool S1, ac_q_mode q1, ac_o_mode o1,
          int W2, int I2, ac_q_mode q2, ac_o_mode o2>
void ac_determinant(
    ac_matrix <ac_fixed <W1, I1, S1, q1, o1>, M, M> &input,
    ac_fixed<W2, I2, true, q2, o2> &result
)
```

```
template <bool override = false,
          int internal_width = 16, int internal_int = 8,
          bool internal_sign = true, ac_q_mode internal_rnd = AC_RND,
          ac_o_mode internal_sat = AC_SAT,
          unsigned M,
          int W1, int I1, bool S1, ac_q_mode q1, ac_o_mode o1,
          int W2, int I2, ac_q_mode q2, ac_o_mode o2>
void ac_determinant(
    const ac_matrix<ac_complex<ac_fixed<W1,I1,S1,q1,o1> >, M, M> &input,
    ac_complex<ac_fixed<W2, I2, true, q2, o2> > &result
)
```

```
template <bool override = false,
          int internal_width = 16, int internal_int = 8,
          bool internal_sign = true, ac_q_mode internal_rnd = AC_RND,
          ac_o_mode internal_sat = AC_SAT,
          unsigned M,
          int W1, int I1, bool S1, ac_q_mode q1, ac_o_mode o1,
          int W2, int I2, ac_q_mode q2, ac_o_mode o2>
void ac_determinant(
    const ac_fixed<W1, I1, S1, q1, o1> a[M][M],
    ac_fixed<W2, I2, true, q2, o2> &result
)
```

)

```
template <bool override = false,
          int internal_width = 16, int internal_int = 8,
          bool internal_sign = true, ac_q_mode internal_rnd = AC_RND,
          ac_o_mode internal_sat = AC_SAT,
          unsigned M,
          int W1, int I1, bool S1, ac_q_mode q1, ac_o_mode o1,
          int W2, int I2, ac_q_mode q2, ac_o_mode o2>
void ac_determinant(
    const ac_complex<ac_fixed<W1, I1, S1, q1, o1> > a[M][M],
    ac_complex<ac_fixed<W2, I2, true, q2, o2> > &result
)
```

The library also provides the following two functions to allow the user to return by value.

```
template<class T_out,
         bool override = false,
         int internal_width = 16, int internal_int = 8,
         bool internal_sign = true, ac_q_mode internal_rnd = AC_RND,
         ac_o_mode internal_sat = AC_SAT,
         unsigned M, class T_in>
T_out ac_determinant(const ac_matrix <T_in, M, M> &input)
```

```
template<class T_out,
         bool override = false,
         int internal_width = 16, int internal_int = 8,
         bool internal_sign = true, ac_q_mode internal_rnd = AC_RND,
         ac_o_mode internal_sat = AC_SAT,
         unsigned M, class T_in>
T_out ac_determinant(const T_in input[M][M])
```

5.7.3. C++ Compiler

The functions use default template arguments. This requires using a C++ compiler that supports the C++11 standard, or a later standard. Failing to use such a compiler will result in compilation errors.

5.7.4. Example Function Call

The following give examples of using the determinant function with both `ac_matrix` and C style array inputs.

```
// Using the ac_matrix wrapper class
ac_matrix <ac_fixed <8, 5, true, AC_RND, AC_SAT>, 2, 2> input;
```

```
typedef ac_fixed <27, 18, true, AC_RND, AC_SAT> output_type;
output_type output;
// Assign elements to input
input(0, 0) = 1;
input(0, 1) = 1.5;
input(1, 0) = 0.5;
input(1, 1) = 1.75;
ac_determinant (input, output);
// The above function call can also be replaced by:
output = ac_determinant <output_type> (input);
```

```
// Using C-style arrays
ac_fixed <8, 5, true, AC_RND, AC_SAT> input[2][2];
typedef ac_fixed <27, 18, true, AC_RND, AC_SAT> output_type;
output_type output;
// Assign elements to input
input[0][0] = 1;
input[0][1] = 1.5;
input[1][0] = 0.5;
input[1][1] = 1.75;
ac_determinant (input, output);
// The above function call can also be replaced by:
output = ac_determinant <output_type> (input);
```

5.8. Systolic matrix multiplication (ac_matmult_systolic)

The `ac_matmult_systolic` systolic array for matrix multiplication is a specialized hardware architecture comprising a 2D grid of interconnected processing elements (PEs) that is used to efficiently compute matrix multiplication using the grid of PEs that pass data rhythmically between neighbours. This design facilitates a highly parallel and pipelined dataflow.

5.8.1. `ac_matmult_systolic` implementation

`ac_matmult_systolic` is implemented with a tiled approach as a templated C++ class (`matmult_systolic_top`) for square matrix multiplication. This involves dividing large input matrices into `TileSize` x `TileSize` submatrices, which are then processed by a `TileSize` x `TileSize` systolic array. Key loops within the design are extensively unrolled, and core computational functions are efficiently pipelining aiming for high clock frequencies and throughput.

5.8.2. Algorithm

Large input matrices are divided into `TileSize` x `TileSize` submatrices, which are then processed by a `TileSize` x `TileSize` systolic array. Systolic matrix multiplication employs an output-stationary systolic array topology,

where each processing element (process_element) accumulates partial sums for a specific output element, storing these in local memory (PE_reg) across multiple computations. Input data is streamed into the array via ac_channels and intelligently re-timed and diagonalized by line_buffer components to meet the precise systolic dataflow requirements.

5.8.3. Function header

```
template<class MACReturnType, class InputType, class OutputType, int MatrixDim,
int TileSize>
class matmult_systolic_top
{
Public:
    void systolicMatrixMult (
        ac_channel< inDataPacketType<InputType,TileSize> > &Ain_ch,
        ac_channel< inDataPacketType<InputType,TileSize> > &Bin_ch,
        ac_channel< inDataPacketType<OutputType,TileSize> > &Cout_ch
    )
};
```

5.8.4. Template arguments

InputType, OutputType, and MACReturnType can be configured to support various data precisions, including integer (ac_int) and fixed-point (ac_fixed) numbers with configurable bit widths and signedness. The overall matrix dimension (MatrixDim) and the systolic array's tile dimension (TileSize) are configurable.

5.8.5. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

5.8.6. Example function call

```
template<class MACReturnType, class InputType, class OutputType, int MatrixDim,
int TileSize>
class matmult_systolic_dut {
public:
    matmult_systolic_top<MACReturnType, InputType, OutputType, MatrixDim, Tile-
Size> dut;
    matmult_systolic_dut() {}

    void CCS_BLOCK(hardwareAcceleratedMult) (
        ac_channel< inDataPacketType<InputType,TileSize> > &Ain_ch,
        ac_channel< inDataPacketType<InputType,TileSize> > &Bin_ch,
```



```

        ac_channel< inDataPacketType<OutputType,TileSize> > &Cout_ch
    ) {
        dut.systolicMatrixMult(Ain_ch, Bin_ch, Cout_ch);
    }
};

```

5.9. Matrix Multiplication (ac_matrixmul)

The ac_matrixmul function is designed to provide the multiplication of two matrices with full precision by default and if user wants to specify his own precision then a provision is provided for this purpose.

5.9.1. The ac_matrixmul Implementation

The ac_matrixmul library provides implementation for ac_fixed and ac_complex<ac_fixed> datatypes. There is one generalized function which handles the computation for both the datatypes. A prototype of the ac_matrixmul function is shown below:

```

template<int mw = 0, int mi = 0, ac_q_mode mq = AC_RND,
        ac_o_mode mo= AC_SAT,
        int sw = 0, int si = 0, ac_q_mode sq = AC_RND,
        ac_o_mode so = AC_SAT,
        unsigned M, unsigned N, unsigned P, class T_in_A,
        class T_in_B, class T_op>
void
ac_matrixmul(
    const ac_matrix<T_in_A, M, N> &A,
    const ac_matrix<T_in_B, N, P> &A,
    ac_matrix<T_op, M, P> &C
)

```

Definition of matrices:

In the above prototype, ac_matrix is a container class that helps to define two dimensional matrices. The following is the way to define a matrix using the ac_matrix class for the ac_fixed datatype:

```
ac_matrix<ac_fixed<20, 10, true, AC_RND, AC_SAT>, M, N> A;
```

In the definition above, A is a matrix of dimension M x N where its elements are of ac_fixed datatype.

Similarly, the following is the way to define a matrix using the ac_matrix class for the ac_complex<ac_fixed> datatype:

```
ac_matrix<ac_complex<ac_fixed<40, 20, true, AC_RND, AC_SAT> >, M, N> A;
```

In the definition above, A is a matrix of dimension M x N where its elements are of ac_fixed datatype.

Default Template Arguments:

By default, the internal variables have full precision turned on. There are two internal variables mult and sum used in the computation of matrix multiplication. The code snippet where the sum and mult variables are used in the matrix multiplication block is as follows:

```
for (unsigned i=0; i<M; i++) {
    for (unsigned j=0; j<P; j++) {
        mult = 0;
        sum = 0;
        for (unsigned k=0; k<N; k++) {
            mult = A(i,k) * B(k,j);
            sum = sum + mult;
        }
        C(i,j) = sum;
    }
}
```

The default template arguments in the order which they have been mentioned in the list of template parameters in the function prototype for this block are:

- Width of the mult variable – mw
- Integer width of the mult variable – mi
- Quantization mode of the mult variable – mq
- Overflow mode of the mult variable – mo
- Width of the sum variable – sw
- Integer width of the sum variable – si
- Quantization mode of the sum variable – sq
- Overflow mode of the sum variable - so

For the ac_fixed datatype

From the above code snippet, in order to achieve full precision for the mult variable its width and integer width is calculated as the sum of the widths and the sum of the integer widths of the elements of A and B matrices respectively. For the sum variable, the full precision is dependent on the number of times it is accumulated by the mult variable which is in turn decided by the number of iterations in the for loop in which it is accumulated. As the number of iterations are N in this for loop, the width and the integer width for the sum variable is calculated as the width of the mult variable + $\log_2\text{ceil}(N)$ and the integer width of the mult variable +

$\log_2 \text{ceil}(N)$ respectively. The default quantization and overflow modes for the mult and the sum variables are defined as AC_RND and AC_SAT respectively.

For the `ac_complex<ac_fixed>` datatype

Everything above applies for the `ac_complex<ac_fixed>` datatype except that the width and integer of the mult variable is calculated as the sum of the widths + 1 and the sum of the integer widths + 1 of the elements of A and B matrices respectively. This is because when two complex numbers are multiplied for example:

$$(a_1 + b_1 i) \times (a_2 + b_2 i) = a_1 a_2 - b_1 b_2 + (a_1 b_2 + a_2 b_1) i$$

The computation of the real part and the imaginary part involves an extra addition or subtraction and thus 1 is added. The user must keep in mind that in order to allow the usage of default template arguments for functions, they must use a version of C++ that allows this usage, such as C++ 11, in case they wish to compile and execute the code for the matrixmul library, failing which an error will be thrown by the compiler.

If the user wants to specify his/her own precision for the internal variables instead of the default, then please refer to the "Example Function Call" section below for the same.

Wrong Datatypes

If any of the two inputs and the output have different datatypes from each other, then a static assert is thrown stating that *"Both arguments need to be of the same datatype"* in case of C++11 version of the compiler and in versions prior to that a compiler error is generated.

5.9.2. Example Function Call

An example of a function call to store the result matrix of the two input matrices A and B who have elements of `ac_fixed` datatype is shown below:

```
typedef ac_matrix<ac_fixed<20, 10, true, AC_RND, AC_SAT>, M, N> input_type1;
typedef ac_matrix<ac_fixed<20, 10, true, AC_RND, AC_SAT>, N, P> input_type2;
typedef ac_matrix<ac_fixed<61, 31, true, AC_RND, AC_SAT>, M, P> output_type;
input_type1 A;
input_type2 B;
output_type C;
ac_matrixmul(A, B, C);    //Function call
```

The two input matrices here are A and B and the output matrix here is C.

Changing default template parameters through function call:

- If the user wants a different precision from the default precision, then the default

template parameters have to be modified. If the user wants to modify all the template parameters then the user has to modify the above function call in the following way:

```
ac_matrixmul<mw, mi, mq, mo, sw, si, sq, so> (A, B, C);
```

The above order in which the template arguments are specified has to be maintained.

- Suppose the user only wants to modify the width of the mult variable i.e mw and keep all the other template arguments as it is, then the function call can be written as:

```
ac_matrixmul<mw> (A, B, C);
```

- But if the user wants to modify only sw for instance, then in order to do this the user has to make a function call like this:

```
ac_matrixmul<mw, mi, mq, mo, sw> (A, B, C);
```

5.9.3. Debug

Here is a provision to enable debug mode in the *ac_matrixmul.h* file. It can be done by defining the macro as

```
#define MATRIMUL_DEBUG
```

5.10. QR Decomposition (ac_qrd)

The *ac_qrd* function provides QR decomposition of square input matrix A. In linear algebra, QR decomposition is a decomposition of matrix A into product $A = QR$ where Q is an orthogonal matrix and R is an upper triangular matrix.

This function uses systolic array based implementation where each iteration performs a Given's rotation algorithm to convert given matrix into Q and R matrices. Note that this function uses *ac_matrix* container class and input and output matrices are either supplied in the form of *ac_matrix* objects or via standard AC datatype 2D arrays. This QR decomposition implementation supports *ac_fixed* and *ac_complex<ac_fixed>* datatypes. If the function is called with other datatypes, compilation error is thrown.

5.10.1. The ac_qrd Implementation

The *ac_qrd* implementation used Givens rotation algorithm to convert any given matrix into an upper triangular matrix which is R. Computation of Q requires same set of sequence of rotations and can be computed simultaneously. Givens rotation has the property of implementing the QRD decomposition in systolic manner, which makes it possible to synthesize parallel architecture and simplify the design.

The process of QR decomposition can be divided into two different functions (processing elements) that are called as off-diagonal and diagonal processing elements (PEs). Diagonal PEs are used to compute rotational parameters, whereas off-diagonal PEs are used to apply them to the rows and compute the new values of the rows in the final R and Q matrices.

Systolic Array Structure of QRD

QRD can be massively parallelized if only it performs rotation on two different set of rows.

The process of applying rotations and computing rotational parameters c and s can be divided into two different functions (processing elements) that are called as off-diagonal and diagonal processing elements

(PEs). Diagonal PEs are used to compute c and s, where as off-diagonal PEs are used to apply them to the rows and compute the new values of the rows in the final R matrix.

- **Diagonal Processing Elements** are expressed in the form of function, `diagonal_PE` which takes element to be zeroed (b) and element in the row just above that in the same column (a) as inputs and returns the rotational matrix parameters. It also takes the default Boolean parameter `isowl`, which allows user to do architectural exploration by switching between PWL functions and accurate `ac_math` functions. By default, PWL functions are used to make sure that the error is minimum, although this increases the area of the Diagonal Processing elements.
- **Off-Diagonal Processing Elements** are expressed using function `offdiagonal_PE`, which applies Givens rotation to the relevant rows of the Q and R matrices. Givens rotation for complex matrices produces a complex output for the bottom right portion of the R matrix by default. However, certain applications might require the R matrix to have real elements throughout to enable operations such as matrix inversion during downstream processing. The user can obtain such outputs by toggling the `real_diag` template parameter to "true". The calculation of the parameters required to do this final rotation can be done through PWL functions or accurate `ac_math` functions. This choice is determined via the "isowl" template parameter.

5.10.2. Function prototypes

For `ac_matrix` of `ac_fixed`:

```
template<
    bool isowl = true, unsigned M,
    int W1, int I1, ac_q_mode q1, ac_o_mode o1,
    int W2, int I2, ac_q_mode q2, ac_o_mode o2
>
void ac_qrd(
    ac_matrix<ac_fixed<W1, I1, true, q1, o1>, M, M> &A,
    ac_matrix<ac_fixed<W2, I2, true, q2, o2>, M, M> &Q,
    ac_matrix<ac_fixed<W2, I2, true, q2, o2>, M, M> &R
)
```

For `ac_matrix` of `ac_complex<ac_fixed>`:

```
template<
    bool real_diag = false, bool isowl = true, unsigned M,
    int W1, int I1, ac_q_mode q1, ac_o_mode o1,
    int W2, int I2, ac_q_mode q2, ac_o_mode o2
>
void ac_qrd(
    ac_matrix<ac_complex<ac_fixed<W1, I1, true, q1, o1>, M, M> &A,
    ac_matrix<ac_complex<ac_fixed<W2, I2, true, q2, o2>, M, M> &Q,
    ac_matrix<ac_complex<ac_fixed<W2, I2, true, q2, o2>, M, M> &R
)
```

For C++ ac_fixed array:

```
template<
    bool ispw1 = true, unsigned M,
    int W1, int I1, ac_q_mode q1, ac_o_mode o1,
    int W2, int I2, ac_q_mode q2, ac_o_mode o2
>
void ac_qrd(
    ac_fixed <W1, I1, true, q1, o1> A[M][M],
    ac_fixed <W2, I2, true, q2, o2> Q[M][M],
    ac_fixed <W2, I2, true, q2, o2> R[M][M]
)
```

For C++ ac_complex<ac_fixed> array:

```
template<
    bool real_diag = false, bool ispw1 = true, unsigned M,
    int W1, int I1, ac_q_mode q1, ac_o_mode o1,
    int W2, int I2, ac_q_mode q2, ac_o_mode o2
>
void ac_qrd(
    ac_complex<ac_fixed <W1, I1, true, q1, o1> > A[M][M],
    ac_complex<ac_fixed <W2, I2, true, q2, o2> > Q[M][M],
    ac_complex<ac_fixed <W2, I2, true, q2, o2> > R[M][M]
)
```

5.10.3. Example Function Calls

An example function call to calculate the QR decomposition of a 3x3 complex ac_matrix object is given below:

```
ac_matrix<ac_complex<ac_fixed<16, 8,false,AC_TRN,AC_WRAP> >, 3, 3> A;
ac_matrix<ac_complex<ac_fixed<32,16,false,AC_TRN,AC_WRAP> >, 3, 3> Q, R;
// Hypothetical function that initializes matrix "A"
init_matrix(A);
ac_qrd(A, Q, R);
```

An example of the same function, but for a 3x3 complex 2D C++ array is given below:

```
ac_complex<ac_fixed<16, 8,false,AC_TRN,AC_WRAP> A[3][3];
ac_complex<ac_fixed<32,16,false,AC_TRN,AC_WRAP> Q[3][3], R[3][3];
// Hypothetical function that initializes matrix A
init_matrix(A);
ac_qrd(A, Q, R);
```

By default, the design does not perform the final rotation to obtain a real bottom-right element. To make the design do that, set the "real_diag" template parameter to "true", as shown below:

```
ac_qrd<true>(A, Q, R);
```

The design also uses PWL functions by default. To enable the use of accurate ac_math functions, set the "is_pwl" template parameter to "false", as shown below:

```
ac_qrd<true, false>(A, Q, R);
```

Chapter 6: Miscellaneous Functions

The `ac_math` package also provides the following functions, in addition to the PWL, LUT and CORDIC functions:

- [Absolute Value \(`ac\_abs`\)](#)
- [Division \(`ac\_div`\)](#)
- [Square Root \(`ac\_sqrt`\)](#)
- [Shifts \(`ac\_shift\_left`/`ac\_shift\_right`\)](#)
- [Barrel Shift \(`ac\_barrel\_shift`\)](#)
- [Padded Division \(`ac\_div\_v2`\)](#)
- [LeakyReLU \(`ac\_leakyrelu`\)](#)
- [ReLU \(`ac\_relu`\)](#)
- [PReLU \(`ac\_prelu`\)](#)
- [AC Float Adder Tree Functions \(`add\_tree` / `add\_tree\_ptr` / `block\_add\_tree` / `block\_add\_tree\_ptr`\)](#)
- [Standard Floating-Point \(`ac\_std\_float`\) Fused Adder Tree Functions \(`fadd\_tree` / `fadd\_tree\_ptr`\)](#)
- [Sorting Networks \(`ac\_sort`\)](#)
- [Parallel Search \(`ac\_search`\)](#)
- [Optimized AC Float Multiplication \(`ac\_flfx\_mul`\)](#)

The following subsections describes the implementation and usage of these functions in more detail.

6.1. Absolute Value (`ac_abs`)

The absolute value (`ac_abs`) operation produces a positive result by negating values less than zero.

- Integer Signed

```
void ac_abs(ac_int<XW,true> x, ac_int<YW,false> &y)
void ac_abs(ac_int<XW,true> x, ac_int<YW,true> &y)
```

- Fixed Point Signed


```
void ac_abs(ac_fixed<XW,XI,true,XQ,XO> x, ac_fixed<YW,YI,false,YQ,YO> &y)
void ac_abs(ac_fixed<XW,XI,true,XQ,XO> x, ac_fixed<YW,YI,true,YQ,YO> &y)
```

- Float

```
void ac_abs(ac_float<XW,XI,XE,XQ> x, ac_float<YW,YI,YE,YQ> &y)
```

6.2. Division (ac_div)

The division functions are supported for integer, fixed-point, complex, and float data types. The division functions takes two inputs: dividend and divisor and computes the quotient. If the inputs are integer there is a version of the function that also computes the remainder of the division. In all cases the div function returns true if the computed remainder is nonzero.

NOTE: Division by zero triggers an assertion failure during simulation if the `ASSERT_ON_INVALID_INPUT` macro is defined.

6.2.1. Integer Division

There are several integer division functions defined. Some of the functions compute just the quotient, some compute both the quotient and the remainder. All return a bool flag to indicate whether the remainder is non zero.

The first argument is the dividend (or numerator), the second is the divider (or denominator), the third is the quotient (output argument) and the fourth (optional) argument is the remainder (output argument). All the arguments are either all signed or all unsigned. The computed quotient is equivalent to the result computed by the operator `'/'`:

```
ac_int<8,false> n;
ac_int<5,false> d;
ac_int<6,false> q;
ac_div(n, d, q);
ac_int<6,false> q1 = n/d;    // q == q1
```

While the behavior of the last two lines is identical, Catapult will produce different hardware for each case. Catapult will allocate a component from the library for the `'/'` whereas calling the `div` function will inline the function.

```
bool ac_div(
    ac_int<NW,false> dividend,
    ac_int<DW,false> divisor,
    ac_int<QW,false> &quotquotient,
    ac_int<RW,false> &remainder
);
```

```
bool ac_div(
    ac_int<NW,true> dividend,
    ac_int<DW,true> divisor,
    ac_int<QW,true> &quotquotient,
    ac_int<RW,true> &remainder
);
```

<pre>bool ac_div(ac_int<NW,false> dividend, ac_int<DW,false> divisor, ac_int<QW,false> &quotquotient);</pre>	<pre>bool ac_div(ac_int<NW,true> dividend, ac_int<DW,true> divisor, ac_int<QW,true> &quotquotient);</pre>
--	---

Table 3: Functions for Integer Division that Return Remainder != 0

For the div functions, $\text{dividend} == \text{divisor} * \text{quotient} + \text{remainder}$ provided that both the quotient and the remainder have sufficient precision so that they don't overflow.

6.2.2. Fixed-point Division

There are several fixed-point division functions defined. Each function returns a *bool* flag to indicate whether the remainder is non zero (assumes that the quotient has enough precision that it does not overflow).

The first argument is the *dividend* (or numerator), the second is the *divider* (or denominator), the third is the *quotient* (output argument). All the arguments are either all signed or all unsigned. The computed quotient takes into account the target bitwidth, integer bitwidth, quantization and overflow modes. All quantization modes work correctly since they are based on the computation of the remainder. The operator '/', on the other hand, is forced to truncate the result without knowing the data type of the target. For example, the operator '/' for *ac_fixed* returns a result that is dependent on the type of both dividend and divisor and the bits (possibly infinite bits) to the right of the result are truncated before the quantization mode of the target is known.

The hardware produced by Catapult for a call to the '/' and '/=' operators and for a call to the div function are different. In the first case, Catapult allocates a component from the library, whereas in the second case the div function is inlined.

<pre>bool ac_div(ac_fixed<NW,NI,false,NQ,NO> divi- dend, ac_fixed<DW,DI,false,DQ,DO> divisor, ac_fixed<QW,QI,false,QQ,QO> &quo- tient);</pre>	<pre>bool ac_div(ac_fixed<NW,NI,true,NQ,NO> dividend, ac_fixed<DW,DI,true,DQ,DO> divisor, ac_fixed<QW,QI,true,QQ,QO> &quotquotient);</pre>
---	--

Table 4: Functions for Fixed-Point Division that Return Remainder != 0

The *div* functions return the *bool* value remainder != 0 which is equivalent to $(\text{divisor} * \text{quotient} != \text{dividend})$, provided that the quotient does not overflow) and may be dependent on the width and integer width of the quotient. For example:

```
ac_fixed<3,3,false> q; ac_fixed<4,3> q2;
ac_fixed<2,2,false> a = 3; ac_fixed<2,2,false> b = 2;
bool nonzero_rem = ac_div(a, b, q); // nonzero_rem == true, q == 1
bool nonzero_rem2 = ac_div(a, b, q2); // nonzero_rem2 == false, q2 == 1.5
```

6.2.3. Float Division

Inlined *ac_float* division is implemented using *ac_fixed* division and has an equivalent return value. The quotient may lose precision as neither the dividend or divisor are normalized. The resulting quotient will overflow if the result of division cannot be represented in the quotient type, otherwise the result is representable and normalized.

```
bool ac_div(
    ac_float<NW,NI,NE,NQ> dividend,
    ac_float<DW,DI,DE,DQ> divisor,
    ac_float<QW,QI,QE,QQ> &quotquotient
);
```

6.2.4. Complex Division

The *ac_div* library also provides for the division of complex inputs.

```
void ac_div(
    ac_complex<NT> dividend,
    ac_complex<DT> divisor,
    ac_complex<QT> &quotquotient
)
```

6.2.5. Output Saturation for Zero Inputs

By default, the hardware synthesized for *ac_div* saturates the output if the input is zero. This may result in coverage holes if no zero inputs are supplied, in addition to the extra logic the output saturation requires. If this is undesirable, the user may define the `AC_DIV_DISABLE_SATURATION_IF_INPUT_IS_ZERO` macro, which will eliminate the saturation logic. The user must take into account the fact that defining this macro will disable output saturation for all calls to *ac_div*.

6.3. Square Root (ac_sqrt)

The square root function has only input argument and one output argument both of which are unsigned. Integral, fixed-point and floating-point versions are provided.

6.3.1. Integer square root

The integer square root computes the largest integer value such that when squared it is equal or less than the argument. The prototype of the function is given as follows:

```
void ac_sqrt(
    ac_int<XW,false> x,
    ac_int<OW,false> &sqrt
)
```

6.3.2. Fixed-point square root

The fixed-point *ac_sqrt* function allows for full flexibility on both the input and the output precision. The *bitwidth*, *integer bitwidth*, *quantization* and *overflow modes* of the target (second argument) determine the bits of precision for computing the square root and the quantization and overflow that is performed on the result. All quantization modes are computed exactly based on the remainder of the square root computation. The prototype of the function is given as follows:

```
void ac_sqrt(
    ac_fixed<XW,XI,false,XQ,XO> x,
    ac_fixed<OW,OI,false,OQ,OO> &sqrt
)
```

An example of how the square root functions is used, consider the case where the input value is *ac_fixed<4,4, false>* (4 bit unsigned integer), and the required precision for the result is *ac_fixed<8,3, false>* (3 bit integral, 5 fractional: xxx.xxxxx), the call code would look like:

```
ac_fixed<4,4,false> x = 13;
ac_fixed<8,3,false> sqrt_x;
ac_sqrt(x, sqrt_x); // sqrt_x == (ac_fixed<8,3,false>) sqrt( x.to_double() );
```

6.3.3. Floating-point square root

The *ac_sqrt* function also allows for *ac_float*, *ac_std_float* and *ac_ieee_float* inputs and outputs. The *ac_std_float* and *ac_ieee_float* implementations are a wrapper around the *ac_float* implementation, with temporary variables present to ensure compatibility with the *ac_float* implementation. The *ac_float* implementation calculates the square root of the input mantissa, stores it in a temporary variable and then factors in the input exponent value to calculate the final output, in accordance to the equations below:

```
sqrt(floating_point_input) = sqrt(mantissa*2^exp)
                           = sqrt(mantissa)*2^(exp/2)
                           = sqrt(mantissa)*2^(exp>>1)
```

For odd exponent values, we can still use $2^{(exp>>1)}$ to factor in $2^{(exp/2)}$, provided that we left-shift the mantissa by 1 first. For instance, consider an input with $exp = 9$.

```
sqrt(floating_point_input) = sqrt(mantissa*2^9)
                           = sqrt(mantissa*2*2^8)
                           = sqrt((mantissa<<1)*2^8)
                           = sqrt(mantissa<<1) * 2^(8>>1)
                           = sqrt(mantissa<<1) * 2^(9>>1)
                           (Since 8>>1 = 9>>1 = 4)
```

The *ac_fixed* temporary variable used to store the value of $\text{sqrt}(\text{mantissa})$ (or, in the case of odd exponents, $\text{sqrt}(\text{mantissa}<<1)$) bases its fractional bits and rounding mode on default template parameters, as seen in the prototype:

```
template<bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
```

```
int XW, int XI, int XE, ac_q_mode XQ, int OW, int OI, int OE, ac_q_
mode OQ>
void ac_sqrt(
    ac_float<XW,XI,XE,XQ> x,
    ac_float<OW,OI,OE,OQ> &sqrt
)
```

If the OR_TF variable is set to false (default value), the temporary variables use the same fractional bitwidth as the output mantissa, if not, they use the fractional bitwidth supplied by the TF_ parameter (32 by default). The rounding mode for temp variables is set to that supplied by the TQ parameter (AC_TRN by default). An example ac_float function call that uses a temporary variable with 16 fractional bits and AC_RND as the rounding mode is given below:

```
ac_float<25, 2, 8> input, output;
ac_sqrt<true, 16, AC_RND>(input, output);
```

The ac_std_float and ac_ieee_float implementations use the same default template parameters, which they in turn supply to the ac_float implementation. The prototypes are given below:

```
template<bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        int XW, int XE, int OW, int OE>
void ac_sqrt(
    ac_std_float<XW, XE> x,
    ac_std_float<OW, OE> &sqrt
)

template<bool OR_TF = false, int TF_ = 32, ac_q_mode TQ = AC_TRN,
        ac_ieee_float_format Format, ac_ieee_float_format outFormat>
void ac_sqrt(
    ac_ieee_float<Format> x,
    ac_ieee_float<outFormat> &sqrt
)
```

6.3.4. Special input handling

The ac_std_float and ac_ieee_float implementations can also handle negative and NaN inputs:

Input	Output
-0.0	-0.0
Negative non-zero	-nan
+nan	+nan
-nan	-nan

This handling mirrors that provided by the sqrt() function in the math.h standard C++ library. While negative zero handling is provided by default, the AC_SQRT_NAN_SUPPORTED macro must be defined to enable NaN output generation (for negative non-zero and +/-nan inputs).

6.4. Shifts (ac_shift_left/ac_shift_right)

The *ac_shift* functions allow the specification of precision and quantization and overflow modes of the target. This gives a simple way to get around the issue of the fixed-point shift operations `>>` and `<<` for *ac_fixed* returning the type of the first operand. The shifting provided by these functions is “arithmetic” in nature, that is, these functions provide the same result as multiplying the input by $2^{(\text{shift_count})}$, all the while keeping in mind the quantization and overflow modes of the target. In this sense, they allow for saturation and rounding, something that is not provided by the `>>` and `<<` operators.

6.4.1. Bidirectional shifts

For *ac_fixed* data-types, both the right shift `>>` and the left shift `<<` operators shift in the opposite direction when the shift value is negative. The equivalent functionality is provided by the following functions:

<pre>void ac_shift_right(ac_fixed<XW,XI,false,XQ,XO> x, int n, ac_fixed<OW,OI,false,OQ,OO> &sr)</pre>	<pre>void ac_shift_right(ac_fixed<XW,XI,true,XQ,XO> x, int n, ac_fixed<OW,OI,true,OQ,OO> &sr)</pre>
<pre>void ac_shift_left(ac_fixed<XW,XI,false,XQ,XO> x, int n, ac_fixed<OW,OI,false,OQ,OO> &sl)</pre>	<pre>void ac_shift_left(ac_fixed<XW,XI,true,XQ,XO> x, int n, ac_fixed<OW,OI,true,OQ,OO> &sl)</pre>

Table 5: Functions for Fixed-Point Bidirectional Shifts

6.4.2. Unidirectional shifts

If the shift value is know to be non-negative, it is best to cast it to (unsigned int) so that the following functions are inlined. These functions will deliver better quality of results during Catapult synthesis.

<pre>void ac_shift_right(ac_fixed<XW,XI,false,XQ,XO> x, unsigned int n, ac_fixed<OW,OI,false,OQ,OO> &sr)</pre>	<pre>void ac_shift_right(ac_fixed<XW,XI,true,XQ,XO> x, unsigned int n, ac_fixed<OW,OI,true,OQ,OO> &sr)</pre>
<pre>void ac_shift_left(ac_fixed<XW,XI,false,XQ,XO> x, unsigned int n, ac_fixed<OW,OI,false,OQ,OO> &sl)</pre>	<pre>void ac_shift_left(ac_fixed<XW,XI,true,XQ,XO> x, unsigned int n, ac_fixed<OW,OI,true,OQ,OO> &sl)</pre>

Table 6: Functions for Fixed-Point Unidirectional Shift

6.4.3. Complex shifts

Wrapper functions for the unidirectional *ac_shift_left* and *ac_shift_right* are provided for complex types to perform the corresponding shift for the real and imaginary parts of the underlying type.

<pre>void ac_shift_right(ac_complex<XT> x, unsigned int n, ac_complex<OT> &sr)</pre>	<pre>void ac_shift_left(ac_complex<XT> x, unsigned int n, ac_complex<OT> &sl)</pre>
--	---

Table 7: Functions for Complex Unidirectional Shift

6.5. Barrel Shift (ac_barrel_shift)

Barrel shifter is a digital circuit that can shift a data word by a specified number of bits without the use of any sequential logic, only pure combinational logic, i.e. it inherently provides a binary operation. The way barrel shifter implemented here is as a sequence of multiplexers where the output of one multiplexer is connected to the input of the next multiplexer in a way that depends on the shift distance. A barrel shifter is often used to shift and rotate n-bits in modern microprocessors, typically within a single clock cycle.

6.5.1. Generic Block Diagram

Block diagram below show the mux based implementation of 4 bit barrel shifter for example, take a four-bit number inputs (MSB)ABCD(LSB) table below show rotation in each stage of 4 bit barrel shift. Shifting in various stages based on control bits are illustrated in table below.

SHIFT	Shift = 0		Shift = 1		Shift = 2		Shift = 3	
INPUT	C1=0	C0=0	C1=0	C0=1	C1=1	C0=0	C1=0	C0=0
A	A	A	A	D	C	C	C	B
B	B	B	B	A	D	D	D	C
C	C	C	C	B	A	A	A	D
D	D	D	D	C	B	B	B	A

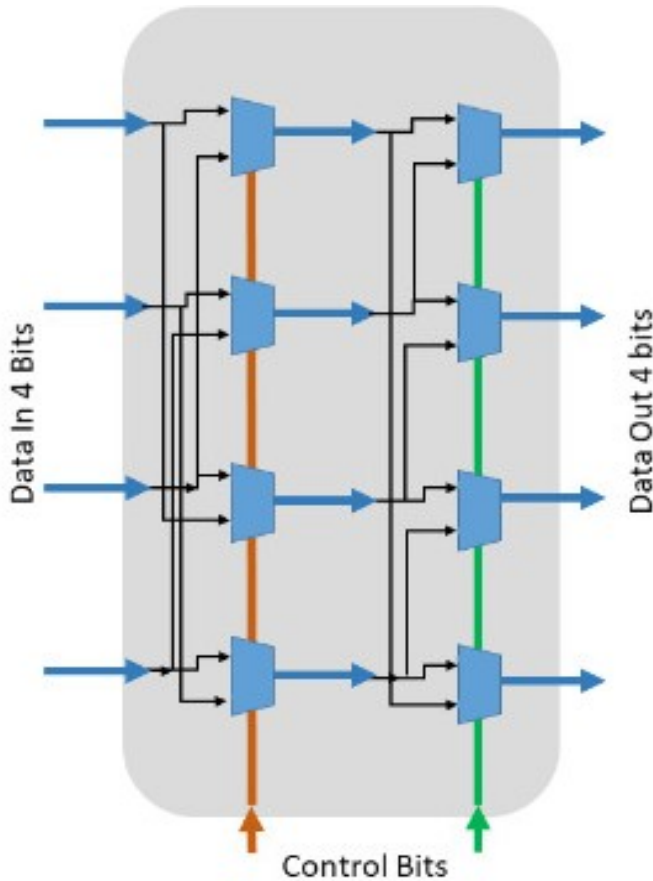


Illustration 1 Mux 2:1 based implementation for 4 bit Barrel Shifter

6.5.2. Implementation Details

Barrel shifter implemented in ac_math.h. Barrel shifter is can be used as combo C-Core or a top design based on user requirements.

Architectural Details

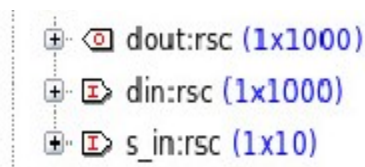
We need to reduce clock overhead to 0 which make sure use of complete cycle for the combo logic. Directive below is to be used set clock overhead.

```
directive set /barrel_shift<1000>::run/core -CLOCK_OVERHEAD 0.000000
```

As design is divided into combinational ccore to reduce the runtime catapult will synthesize each stage in the barrel shifter.

Port Descriptions

Barrel Shifter interface for N bits input is input port N bit along with a select input of width $\lceil \log_2(N) \rceil$ bits and similarly output port of N bit. Illustration below shows ports for N =1000 bits.I.



6.5.3. Using ac_barrel_shift

- To instantiate barrel_shifter in design user have include ac_math.h and follow the below steps.

```
#include<ac_math.h>
```

- Call of ac_barrel_shifter providing number of bits as template parameter

```
data_ot = ac_barrel_shift<1000>(data_in, ctrl_in) ;
```

as ac_barrel_shift only provide ability to rotate right with the value on control port i.e. if control port have value of 1 that impose input LSB become output MSB and rest change accordingly. User can modify control input or input itself to match the output as rotate left. Create variables for input output and control ports.

```
const CTL_BITS = CTRL_BITS ac::nbits<NUM_BITS-1>::val;
ac_int<NUM_BITS,false> data_in;
ac_int<CTL_BITS,false> ctrl_in;
ac_int<NUM_BITS,false> data_ot;
```

- After assigning the data and control bit, user can run the barrel shifter instance by calling. Is the top interface as well for shifter. data_ot will collect the output data.

6.5.4. Output

Output of the core is right rotated version of input word. Design is feed forward makes F-max only lemmatized by the technology library used.

Design limitation

- Core is designed to rotate right as rotate left can be derived from rotate right itself.
- Design is tested under 1024 bits.

Scheduling, Resource and Area Reports

Below are the RTL synthesis results of Barrel shifter of 1000 bits based on 45nm lib area is 30630.7 μm^2

Solution /	Area	Delay	Slack	AreaCo...	AreaSeq	TID	MinClkPrd	InputDe...	InputSe...	R2RDelay	R2RSetup
barrel_shift<1000>::run.v1 (extract)	30630.70	0.15	0.38	21555.04	9075.65	Oasys19.1-s007	0.62	0.43	0.04	0.58	0.04

6.6. Padded Division (ac_div_v2)

This division functions is supported for unsigned and signed integer data types only. Functions takes two inputs: dividend and divisor and computes the outputs: quotient and remainder. The range of values for the divisor and the accuracy of computing the quotient depends on the parameter value of PADDING_WIDTH (PW).

NOTE: Division by zero and with a value that is not supported in the range allowed triggers an assertion failure with description during simulation.

6.6.1. Design Arguments

The first argument is the dividend (or numerator), the second is the divisor (or denominator), the third is the quotient (output argument) and the fourth argument is the remainder (output argument). All the arguments can be signed or unsigned, however make sure to keep all argument types consistent. The width of each of the argument is defined by the template parameters passed to the design.

For Unisgned input/outputs

```
ac_int<NW,false> dividend,
ac_int<DW,false> divisor,
ac_int<QW,false> &quotquotient,
ac_int<RW,false> &remainder
```

For signed Input/Outputs

```
ac_int<NW,true> dividend,
ac_int<DW,true> divisor,
ac_int<QW,true> &quotquotient,
ac_int<RW,true> &remainder
```

6.6.2. Design Parameters

The following table describes the design parameters:

PARAMETER	Description
Numerator Width (NW)	Width of the numerator to be supplied
Denominator Width (NW)	Width of the denominator to be supplied
Quotient Width (NW)	Width of the output quotient expected
Remainder Width (NW)	Width of the output remainder expected
Padding Width (PW)	Number of bits to be padded to the Numerator, so we could increase the range of divisor supported while controlling QofR of the design. NOTE: PW should be withing 1 to DW for Unisgned Or 2 to DW for Signed input

Table 8: Parameter description of the design

Padding width bits are zero value bits that get padded to the numerator, such that the resulting quotient could accommodate all value from the division, with best accuracy. This parameter helps us set the range of divisor value allowed to be passed as the divisor, for example in case of unsigned numbers, if padding value is 1 and the divisor width (DW) is 5, the design supports only values 16 and above up-till 32 (maximum) for the divisor, and if padding value is 5, than divisor can be all possible values from 1 to 32. Similarly for signed values, if the DW is 5 and PW is 2, supported values for divisor excludes values from -9 to 8, and if the PW=DW, than all the values supported by Divisor bit width from -16 to +15 are supported by the synthesized design. This

padding parameter, gives the user an option to optimize the design for area and latency against the range of values supported for design.

Architectural Constraints

The `AC_DIV_CORE_LOOP` loop in the `ac_div_v2` function determines the QofR of the design. By default, this loop is set to be unrolled, which reduces the latency of the design. However, for an area efficient design (at the cost of latency cycles), you could disable the unrolling of this loop by setting the `UNROLL` directive to `no` on the `AC_DIV_CORE_LOOP` loop:

- To disable the core loop unrolling,

```
directive set /rtest_ac_div/core/AC_DIV_CORE_LOOP -UNROLL no
```

- To Enable the core loop unrolling,

```
directive set /rtest_ac_div/core/AC_DIV_CORE_LOOP -UNROLL yes
```

Thus you could efficiently use the unrolling options and the padding parameter to set the range of value and constrain the design for required QofR.

Usage Example For Unsigned Values

1. To Instantiate `ac_div_v2` in the design user have to include `ac_math/ac_div_v2.h` and follow the below steps.

```
#include <ac_math/ac_div_v2.h>
```

2. Define the Numerator, Denominator, Quotient and Remainder(if required) widths along with suitable Padding width to set the range of values supported

```
enum {
    MAX_NW = 5,
    MAX_DW = 3,
    QUO_W  = 5,
    REM_W  = 5,
    PAD_W  = 1, // Value to be padded to the numerator width, that varies
               // the range of value divisor can accept.
};
```

3. Provide the input values for numerator and denominator.

```
typedef ac_int<MAX_NW,false> num_type; // Unsigned division
typedef ac_int<MAX_DW,false> den_type;
typedef ac_int<QUO_W,false> quo_type;
typedef ac_int<REM_W,false> rem_type

num_type num = 32;
```

```
den_type den = 6;
quo_type quo;
rem_type rem;
```

4. Call the `ac_div_v2` function with variable widths as template parameters,, and check for the results in the Quotient and Remainder variables.

```
ac_div<NW,DW,QW,RW,PW>(num, den, quo, rem);
```

For the div functions, $\text{dividend} == \text{divisor} * \text{quotient} + \text{remainder}$ provided that both the quotient and the remainder have sufficient precision so that they don't overflow.

6.7. LeakyReLU (`ac_leakyrelu`)

The `ac_leakyrelu` library provides an implementation of the LeakyReLU activation function for *ac_fixed* inputs with minimal inaccuracy.

6.7.1. Function Templates

The following is the function prototype for the `ac_leakyrelu` function for *ac_fixed* datatypes:

```
template<int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_leakyrelu(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The `ac_leakyrelu` function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        class T_in>
T_out ac_leakyrelu(const T_in &input);
```

6.7.2. Example Function Calls

The following code snippet shows the user how to call the `ac_leakyrelu` function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;
typedef ac_fixed<15, 2, false> alpha_type;
```

```
input_type x = 2.5;
output_type y;
alpha_type alpha;

// Returns y = leakyrelu(x), and returns by reference.
ac_leakyrelu (x, y, alpha);

// The following line returns by value instead of by reference.
y = ac_leakyrelu <output_type> (x, alpha);
```

6.8. ReLU (**ac_relu**)

The *ac_relu* library provides an implementation of the ReLU activation function for *ac_fixed* inputs with minimal inaccuracy.

6.8.1. Function Templates

The following is the function prototype for the *ac_relu* function for *ac_fixed* datatypes:

```
template<int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_relu(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The *ac_relu* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        class T_in>
T_out ac_relu(const T_in &input);
```

6.8.2. Example Function Calls

The following code snippet shows the user how to call the *ac_relu* function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 32, false, AC_RND, AC_SAT> output_type;
```

```
input_type x = 2.5;
output_type y;

// Returns y = relu(x), and returns by reference.
ac_relu (x, y);

// The following line returns by value instead of by reference.
y = ac_relu <output_type> (x);
```

6.9. PReLU (ac_prelu)

The *ac_prelu* library provides an implementation of the PReLU activation function for *ac_fixed* inputs with minimal inaccuracy.

6.9.1. Function Templates

The following is the function prototype for the *ac_prelu* function for *ac_fixed* datatypes:

```
template<int W, int I, bool S, ac_q_mode Q, ac_o_mode O,
        int outW, int outI, ac_q_mode outQ, ac_o_mode outO>
void ac_prelu(
    const ac_fixed<W, I, S, Q, O> &input,
    ac_fixed<outW, outI, false, outQ, outO> &output
)
```

Returning by Value

The *ac_prelu* function can return its output by value as well as by reference. In order to return the value, the user must pass the information of the output type to the function as a template argument. For an example of how to do this, please refer to the following Example Function Calls section below. The prototype for the function which returns by value is shown below:

```
template<class T_out,
        class T_in>
T_out ac_prelu(const T_in &input);
```

6.9.2. Example Function Calls

The following code snippet shows the user how to call the *ac_prelu* function.

```
typedef ac_fixed<9, 4, false, AC_RND, AC_SAT> input_type;
typedef ac_fixed<64, 8, false, AC_RND, AC_SAT> output_type;
typedef ac_fixed <15, 2, false> alpha_type;
```

```
input_type x = 2.5;
output_type y;
alpha_type alpha;

// Returns y = prelu(x), and returns by reference.
ac_prelu (x, y, alpha);

// The following line returns by value instead of by reference.
y = ac_prelu <output_type> (x, alpha);
```

6.10. AC Float Adder Tree Functions (add_tree / add_tree_ptr / block_add_tree / block_add_tree_ptr)

The AC Float adder tree functions provide an implementation for an arbitrary floating point adder tree which doesn't do intermediate normalization and hence saves on hardware. All the functions accept an array of ac_float inputs, and produce either ac_fixed or ac_float outputs, depending on the implementation selected. All the implementations are contained in the header file ac_float_add_tree.h .

6.10.1. Adder Tree

The input array for the adder tree functions can be passed by reference or by degrading them to a pointer. Both variants support ac_fixed and ac_float outputs.

```
// Passing by reference, ac_fixed output:
template<
    int AccExtraLSBs = 0,
    bool use_ac_sl = true,
    int WA, int IA, bool SA, ac_q_mode QA, ac_o_mode OA,
    int WE, int IE, int EE, ac_q_mode QE,
    int N_ELEMS
>
void add_tree(
    const ac_float<WE,IE,EE,QE> (&x) [N_ELEMS],
    ac_fixed<WA,IA,SA,QA,OA> &acc
)

// Passing by reference, ac_float output:
template<
    int AccExtraLSBs = 0,
    int WA, int IA, int EA, ac_q_mode QA,
    int WE, int IE, int EE, ac_q_mode QE,
    int N_ELEMS
>
```

```
void add_tree(
    const ac_float<WE,IE,EE,QE> (&x) [N_ELEMS],
    ac_float<WA,IA,EA,QA> &acc
)

// Passing by pointer, ac_fixed output:
template<
    int N_ELEMS,
    int AccExtraLSBs = 0,
    bool use_ac_sl = true,
    int WA, int IA, bool SA, ac_q_mode QA, ac_o_mode OA,
    int WE, int IE, int EE, ac_q_mode QE
>
void add_tree_ptr(
    const ac_float<WE,IE,EE,QE> x[N_ELEMS],
    ac_fixed<WA,IA,SA,QA,OA> &acc
)

// Passing by pointer, ac_float output:
template<
    int N_ELEMS,
    int AccExtraLSBs = 0,
    int WA, int IA, int EA, ac_q_mode QA,
    int WE, int IE, int EE, ac_q_mode QE
>
void add_tree_ptr(
    const ac_float<WE,IE,EE,QE> x[N_ELEMS],
    ac_float<WA,IA,EA,QA> &acc
)
)
```

The core implementation for all these variants uses recursive templates to add all the inputs, and the recursive templates model the different stages of the adder tree. For every pairwise addition, the mantissa of the input with the lower exponent value is right-shifted to align it with the other input's mantissa. Due to the right-shifting that happens during this alignment, some precision might be lost. If both inputs have the same exponent, no shifting takes place. If a zero input is added with a non-zero input, the former's exponent value is ignored and neither of the inputs are right-shifted.

The pass-by-pointer adder tree functions have the suffix "_ptr" attached to their names to clearly differentiate them from the pass-by-reference functions and avoid any ambiguity between the variants.

Since the passing the array as a pointer involves a loss of information regarding the array dimension, the user must explicitly specify N_ELEMS template parameter while calling the pass-by-pointer functions. The pass-by-reference functions can infer the dimension values, and the N_ELEMS parameter need not be explicitly specified in such a case.

To reduce the loss of precision that may happen during the mantissa alignment, the user may pad extra LSBs to the accumulator variable via the `AccExtraLSBs` template parameter. This parameter is set to zero by default, meaning no extra LSBs are added by default.

The `use_ac_sl` template parameter specifies whether to use the `ac_shift_left` function in `ac_shift.h`, when using the functions with a fixed-point output. More details on this parameter are given in `Using ac_shift_left`. By default, this parameter is set to true, i.e. `ac_shift_left` is used.

Usage examples for these functions are provided below:

```
// foo_default: Uses default template parameters.
// foo_odefault_fxpt: Overrides all default template parameters for the adder
// tree implementation with fixed-point outputs.
// foo_odefault_flpt: Overrides all default template parameters for the adder
// tree implementation with floating-point outputs.
// Since the input array used in all the functions mentioned above will
// undergone pointer decay, we must call add_tree_ptr() and explicitly specify
// N_ELEMS for the function call.

template <int N_ELEMS, class T1, class T2>
void foo_default(const T1 in_arr[N_ELEMS], const T2 &out) {
    ac_math::add_tree_ptr<N_ELEMS>(in_arr, out);
}

template <int N_ELEMS, int AccExtraLSBs, bool use_ac_sl, class T1, class T2>
foo_odefault_fxpt(const T1 in_arr[N_ELEMS], const T2 &out) {
    ac_math::add_tree_ptr<N_ELEMS, AccExtraLSBs, use_ac_sl>(in_arr, out);
}

template <int N_ELEMS, int AccExtraLSBs, class T1, class T2>
foo_odefault_flpt(const T1 in_arr[N_ELEMS], const T2 &out) {
    ac_math::add_tree_ptr<N_ELEMS, AccExtraLSBs>(in_arr, out);
}

void foo() {
    constexpr int N_ELEMS = 16;
    constexpr int AccExtraLSBs = 8;
    constexpr bool use_ac_sl = false;

    ac_float<25, 2, 8> in_arr[N_ELEMS];
    // Code to initialize in_arr.

    typedef ac_float<25, 2, 9> T_out_fl;
    typedef ac_fixed<64, 32, true, AC_TRN, AC_SAT> T_out_fx;

    // Using the pass-by-reference functions first. N_ELEMS will be inferred.
```

```
// Using default template parameters.
T_out_fl out_fl;
ac_math::add_tree(in_arr, out_fl);

// Overriding defaults for fixed point output functions.
T_out_fx out_fx;
ac_math::add_tree<AccExtraLSBs, use_ac_sl>(in_arr, out_fx);

// Overriding defaults for floating point output functions.
T_out_fl out_fl2;
ac_math::add_tree<AccExtraLSBs>(in_arr, out_fl2);

// Using the pass-by-pointer versions now. N_ELEMS is explicitly specified.

// Using default template parameters.
T_out_fl out_fl3;
foo_default<N_ELEMS>(in_arr, out_fl3);

// Overriding defaults for fixed point output functions.
T_out_fx out_fx2;
foo_odefault_fxpt<N_ELEMS, AccExtraLSBs, use_ac_sl>(in_arr, out_fx2);

// Overriding defaults for floating point output functions.
T_out_fl out_fl4;
foo_odefault_flpt<N_ELEMS, AccExtraLSBs>(in_arr, out_fl4);
}
```

6.10.2. Block Adder Tree

The input array for the block adder tree functions can also be passed by reference or by degrading them to a pointer. Both variants support `ac_fixed` and `ac_float` outputs.

```
// Passing by reference, ac_fixed output:
template<
    int AccExtraLSBs = 0,
    bool ZeroHandling = true,
    bool use_ac_sl = true,
    int WA, int IA, bool SA, ac_q_mode QA, ac_o_mode OA,
    int WE, int IE, int EE, ac_q_mode QE,
    int N_ELEMS
>
void block_add_tree(
    const ac_float<WE,IE,EE,QE> (&x) [N_ELEMS],
    ac_fixed<WA,IA,SA,QA,OA> &acc
)
```

```
// Passing by reference, ac_float output:
template<
    int AccExtraLSBs = 0,
    bool ZeroHandling = true,
    bool norm = true,
    int WA, int IA, int EA, ac_q_mode QA,
    int WE, int IE, int EE, ac_q_mode QE,
    int N_ELEMS
>
void block_add_tree(
    const ac_float<WE,IE,EE,QE> (&x) [N_ELEMS],
    ac_float<WA,IA,EA,QA> &acc
)

// Passing by pointer, ac_fixed output:
template<
    int N_ELEMS,
    int AccExtraLSBs = 0,
    bool ZeroHandling = true,
    bool use_ac_sl = true,
    int WA, int IA, bool SA, ac_q_mode QA, ac_o_mode OA,
    int WE, int IE, int EE, ac_q_mode QE
>
void block_add_tree_ptr(
    const ac_float<WE,IE,EE,QE> x[N_ELEMS],
    ac_fixed<WA,IA,SA,QA,OA> &acc
)

// Passing by pointer, ac_float output:
template<
    int N_ELEMS,
    int AccExtraLSBs = 0,
    bool ZeroHandling = true,
    bool norm = true,
    int WA, int IA, int EA, ac_q_mode QA,
    int WE, int IE, int EE, ac_q_mode QE
>
void block_add_tree_ptr(
    const ac_float<WE,IE,EE,QE> x[N_ELEMS],
    ac_float<WA,IA,EA,QA> &acc
)
)
```

The core implementation for all these variants finds the maximum exponent and aligns all the input mantissas at the very start with regard to the maximum exponent value. Once the values are aligned, all the mantissas are added in an unrolled for loop, which is synthesized as an adder tree. The output of the adder tree is then

combined with the maximum exponent value obtained earlier to produce the final output. This means that the adder tree itself operates purely on fixed point values, without requiring the extra hardware needed to do intermediate alignments. This bypasses the use of MUXes and other alignment-related hardware in the datapath of the adder tree, resulting in more area savings and a shorter critical path. However, the use of the block adder tree may also result in some precision being sacrificed, due to the absence of intermediate alignment stages.

Like with the adder tree functions mentioned in Adder Tree, the pass-by-pointer functions have the suffix "_ptr" attached to their names to clearly differentiate them from the pass-by-reference functions and avoid any ambiguity between the variants. The pass-by-pointer functions for the adder tree must also have the N_ELEMS template parameter explicitly specified in the function call.

To reduce the loss of precision that may happen during the mantissa alignment, the user may pad extra LSBs to the accumulator variable via the AccExtraLSBs template parameter. This parameter is set to zero by default, meaning no extra LSBs are added by default.

The ZeroHandling template parameter specifies whether zero-handling logic is incorporated in the design. The block adder tree includes special handling for zero inputs to make sure that their exponent does not override that of small, non-zero inputs and result in unwanted precision loss while right-shifting. This results in bit-masking logic to set the exponent values for zero inputs to the minimum value ($-2^{(E-1)}$ where E is the number of exponent bits) while doing comparisons to find the maximum exponent. If the user decides that the potential precision loss is tolerable and wants to avoid the extra logic involved in zero-handling, they can set the ZeroHandling template parameter to false. By default, this parameter is set to true, i.e. zero handling logic is incorporated in the design.

The use_ac_sl template parameter specifies whether to use the ac_shift_left function in ac_shift.h, when using the functions with a fixed-point output. More details on this parameter are given in Using ac_shift_left. By default, this parameter is set to true, i.e. ac_shift_left is used.

The norm template parameter is used in the final stage of the floating-point output functions, where the output of the adder tree is combined with the maximum exponent value to produce the final output of the block_add_tree function. For the floating-point output functions, this is done via an ac_float constructor:

```
acc = ac_float<WA, IA, EA, QA>(acc_fx, max_exp, norm);
```

acc is the final floating point output, acc_fx is the output of the fixed-point adder tree, max_exp is the maximum exponent value, and norm is the aforementioned template parameter. If norm is set to true--as is the default--extra normalization hardware will be used to normalize the output mantissa. To avoid this hardware, the user can set the norm template parameter to false.

Usage examples for the block adder tree functions are provided below:

```
// foo_default: Uses default template parameters.
// foo_odefault_fxpt: Overrides all default template parameters for the
//   fixed-point output implementation.
// foo_odefault_flpt: Overrides all default template parameters for the
//   floating-point output implementation..
// Since the input array used in all the functions mentioned above will
// undergone pointer decay, we must call add_tree_ptr() and explicitly specify
// N_ELEMS for the function call.
```

```
template <int N_ELEMS, class T1, class T2>
void foo_default(const T1 in_arr[N_ELEMS], const T2 &out) {
    ac_math::block_add_tree_ptr<N_ELEMS>(in_arr, out);
}

template <int N_ELEMS, int AccExtraLSBs, bool ZeroHandling, bool use_ac_sl,
class T1, class T2>
foo_odefault_fxpt(const T1 in_arr[N_ELEMS], const T2 &out) {
    ac_math::block_add_tree_ptr<N_ELEMS, AccExtraLSBs, ZeroHandling,
use_ac_sl>(in_arr, out);
}

template <int N_ELEMS, int AccExtraLSBs, bool ZeroHandling, bool norm,
class T1, class T2>
foo_odefault_flpt(const T1 in_arr[N_ELEMS], const T2 &out) {
    ac_math::block_add_tree_ptr<N_ELEMS, AccExtraLSBs, ZeroHandling, norm>(in-
_arr, out);
}

void foo() {
    constexpr int N_ELEMS = 16;
    constexpr int AccExtraLSBs = 8;
    constexpr bool use_ac_sl = false;
    constexpr bool ZeroHandling = false;
    constexpr bool norm = false;

    ac_float<25, 2, 8> in_arr[N_ELEMS];
    // Code to initialize in_arr.

    typedef ac_float<25, 2, 9> T_out_fl;
    typedef ac_fixed<64, 32, true, AC_TRN, AC_SAT> T_out_fx;

    // Using the pass-by-reference functions first. N_ELEMS will be inferred.

    // Using default template parameters.
    T_out_fl out_fl;
    ac_math::block_add_tree(in_arr, out_fl);

    // Overriding defaults for fixed point output functions.
    T_out_fx out_fx;
    ac_math::block_add_tree<AccExtraLSBs, ZeroHandling, use_ac_sl>(in_arr, out-
_fx);

    // Overriding defaults for floating point output functions.
```

```
T_out_fl out_fl2;
ac_math::block_add_tree<AccExtraLSBs, ZeroHandling, norm>(in_arr, out_fl2);

// Using the pass-by-pointer versions now. N_ELEMS is explicitly specified.

// Using default template parameters.
T_out_fl out_fl3;
foo_default<N_ELEMS>(in_arr, out_fl3);

// Overriding defaults for fixed point output functions.
T_out_fx out_fx2;
foo_odefault_fxpt<N_ELEMS, AccExtraLSBs, ZeroHandling, use_ac_sl>(in_arr,
out_fx2);

// Overriding defaults for floating point output functions.
T_out_fl out_fl4;
foo_odefault_flpt<N_ELEMS, AccExtraLSBs, ZeroHandling, norm>(in_arr, out-
_fl4);
}
```

6.10.3. Using ac_shift_left

All the adder tree functions internally produce a mantissa and exponent value which is combined to produce the final output. The fixed-point output functions must convert these two values to the final output by left-shifting the mantissa by the exponent value. Since floating-point types can typically be used to store a larger range of values than their fixed-point counterparts, this conversion stage must accommodate different quantization and overflow modes to store data that's outside the range supported by the fixed-point output as accurately as possible. There are two ways to do this:

1. Using the `ac_shift_left` function mentioned [here](#).
2. Carrying out the shifting on a large enough type to store all left-shifted values regardless of what the exponent value is.

The bitwidth of the intermediate type used in the second solution increases exponentially as the width of the exponent field increases. This can result in a very large shifter being synthesized to enable the conversion. On the other hand, the bitwidth of the intermediate type used in the first solution does not see any such exponential increase. However, the first solution requires extra masking hardware to support saturation and rounding (if enabled), and may actually result in a larger area cost for smaller exponent fields, as compared to the second solution.

By default, the first solution is picked by setting the `use_ac_sl` template parameter to true. The user can use the second solution by setting this template parameter to false.

For either solution, the quantization and overflow handling implemented depends on the quantization and overflow modes of the output fixed point type.

6.11. Standard Floating-Point (ac_std_float) Fused Adder Tree Functions (fadd_tree / fadd_tree_ptr)

The fused adder tree functions for standard floating-point datatypes provide an implementation for a floating-point adder that receives an arbitrary number of inputs and generates the result in the same datatype as the inputs. The inputs are provided as an array of elements which can be passed either as a pointer or an array of references depending on the implementation selected. All the implementations are contained in the header file `ac_std_float_add_tree.h`.

6.11.1. Function Outline

The fused adder can be called either by the `fadd_tree` function, which takes an array of references as an input or the `fadd_tree_ptr` function that takes a pointer as an input. Both functions are implemented for all datatypes supported by the `ac_std_float.h` library, namely `ac_std_float`, `ac_ieee_float` and `ac::bfloat16`. All these different versions are implemented as overload functions which share the same core implementation. The available versions along with their parameters are summarized in Table 1.

Table 9: Summary of the fused addition tree function

Code	Description
<pre>template< ac_q_mode QR, bool No_SubNormals, int AccExtraLSBs, int W, int E, int N_ELEMS > void fadd_tree (const ac_std_float<W,E> (&x) [N_ELEMS], ac_std_float<W,E> &acc)</pre>	W, E define the width and exponent width of the <code>ac_std_float</code> datatype. The values are being inferred from the input datatype. Format defines the <code>ac_ieee_float</code> format. Valid values: <code>binary16</code>, <code>binary32</code>, <code>binary64m</code>, <code>binary128</code>, <code>binary256</code>. The value is inferred from the input datatype.
<pre>template< ac_q_mode QR, bool No_SubNormals, int AccExtraLSBs, ac_ieee_float_format Format, int N_ELEMS > void fadd_tree (const ac_ieee_float<Format> (&x) [N_ELEMS], ac_ieee_float<Format> &acc)</pre>	N_ELEMS defines the number of input operands. This parameter is only required in the pointer version. QR is the rounding mode. Currently the supported modes are the ones supported by the rest of <code>ac_std_float</code> operators:
<pre>template< ac_q_mode QR, bool No_SubNormals, int AccExtraLSBs, int N_ELEMS > void fadd_tree (const ac::bfloat16 (&x) [N_ELEMS], ac::bfloat16 &acc)</pre>	<pre>AC_RND_CONV (default), AC_TRN_ZERO (default for ac::bfloat16), AC_RND_INF, AC_RND_CONV_ODD</pre>
<pre>template<int N_ELEMS, ac_q_mode QR, bool No_SubNormals, int AccExtraLSBs,</pre>	If No_SubNormals == true then subnormal operands/result are flushed to zero. The default value is false. AccExtraLSBs is used to define the number

Code

```
int W, int E
> void fadd_tree_ptr (
    const ac_std_float<W,E> x[N_ELEMS],
    ac_std_float<W,E> &acc )

template< int N_ELEMS,
    ac_q_mode QR, bool No_SubNormals,
    int AccExtraLSBs,
    ac_ieee_float_format Format
> void fadd_tree_ptr (
    const ac_ieee_float<Format> x[N_ELEMS],
    ac_ieee_float<Format> &acc )

template< int N_ELEMS,
    ac_q_mode QR, bool No_SubNormals,
    int AccExtraLSBs
> void fadd_tree_ptr (
    const ac::bfloat16 x[N_ELEMS],
    ac::bfloat16 &acc )
```

Description

of extra LSBs that will be used. This value affects the accuracy of the result. The default value is AccExtraLSBs =3.

The core implementation of the fused addition performs a single parallel alignment to all the input operands and the addition of the mantissas is implemented in a tree structure. By skipping intermediate normalizations and having a single rounding step at the end, the function aims to save both area and latency.

Here the overall architecture of the fused addition operator is summarized. First the special values and the {sign, exponent, mantissa} part of each input operand are getting extracted. Then, the maximum value of the exponents is selected, using a recursive template implementation. For each input's exponent, the difference to the maximum value is computed and this result is used to align the mantissas. The aligned mantissas are converted to two's complement representation, and they get accumulated. The accumulation is designed as a tree structure of fixed-point additions. The result of the addition gets normalized and rounded, and finally the result is packed again as an ac_std_float datatype.

6.11.2. Special Values

As part of the ac_std_float library, the inputs and the outputs of the fused addition operator support special values as defined in the IEEE-754 standard. An input operand can have one of the following values: Infinite (positive or negative), not-a-number, Subnormal or Normal value. The output result can also have one of these values, which derive from the following scenarios:

- If one or more inputs are positive Infinite (+Inf), the result is also positive infinite (+Inf)
- If one or more inputs are negative infinite (-Inf), the result is also negative infinite (-Inf)
- If there is no infinite input, but the exponent is overflowed to the maximum value, the result is also infinite, having the sign of the computed result.

In the case of rounding mode AC_TRN_ZERO, this value is saturated to the maximum normal number

- If there are both positive (+Inf) and negative (-Inf) infinite inputs, the result is not-a-number.
- If one or more inputs is not-a-number, the result is also a not-a-number.
- In the case of No_SubNormals==true, all subnormal values for inputs and output are flushed to zero.

In any other case, the result can be any normal or subnormal value.

6.11.3. Accuracy of the result

The template parameter "AccExtraLSBs" is used to define the number of LSBs that will be preserved during the addition and the rounding.

In floating-point addition, when two operands get aligned, the smaller one is shifted towards right according to the difference of their exponents, thus discarding the LSB part of that number. To maintain the accuracy of the result, the 3 most significant bits of the discarded part of the number are preserved and form the Guard, Round and Sticky (GRS) bits. While Guard and Round are the last two bits of the number that were shifted out, the sticky bit is the result of an OR operation between the rest of the bits that have been discarded due to right shifting. By default, the value of AccExtraLSBs is 3 to represent the GRS bits.

Changing the number of bits that will be preserved, the accuracy of the result as well as the area requirements for the operator are affected, as the bit width of the operands is changed. Selecting bigger "AccExtraLSBs" values, both the accuracy of the result and the area of the hardware are increased. Table 10 shows how the different values of the "AccExtraLSBs" parameter affect the preserved bits of the operator.

For this example, we assume that the number

```
1010101100110101
1010101100110101
```

must be shifted 6 bits to the right.

Table 10: Example of how the operand changes for different values of the parameter "AccExtraLSBs"

	Aligned part	ExtraLSBs	Discarded bits	Final Size of the operand
1010101100110101				
GRSBits=3	0000001010101100	111	0101	19 bits
GRSBits=0	0000001010101101	-	110101	16 bits
GRSBits=1	0000001010101100	1	10101	17 bits
GRSBits=5	0000001010101100	11011	1	21 bits
GRSBits=6	0000001010101100	110101	-	22 bits

6.11.4. Usage Example

Here are some examples of using the fadd_tree and the fadd_tree_ptr functions with the default or overridden template parameters and for different datatypes.

```
#include <ac_std_float_add_tree.h>

typedef ac_std_float<32,8> std_flt;
typedef ac_ieee_float<binary32> ieee_flt;
typedef ac::bfloat16 b_flt;

void foo () {
    const int N = 8;
    const ac_q_mode ROUND = AC_TRN_ZERO;
    const bool NO_SUBNORMALS = true;
    const int AccExtraLSBs = 1;

    std_flt in0[INPUTS];
    ieee_flt in1[INPUTS];
    b_flt in2[INPUTS];
    // code to initialize arrays

    std_flt default_out0, ovr_out0;
    ieee_flt default_out1, ovr_out1;
    b_flt default_out2, ovr_out2;

    // Example call for ac_std_float datatype
    // Using the default template parameters
    ac_math::fadd_tree(in0, default_out0);
    ac_math::fadd_tree_ptr<N>(in0, default_out0);

    // Overriding the template parameters
    ac_math::fadd_tree<ROUND, NO_SUBNORMALS, AccExtraLSBs>(in0, ovr_out0);
    ac_math::fadd_tree_ptr<N, ROUND, NO_SUBNORMALS, AccExtraLSBs>(in0,
    ovr_out0);

    // Example call for ac_ieee_float datatype
    // Using the default template parameters
    ac_math::fadd_tree(in1, default_out1);

    // Overriding the template parameters
    ac_math::fadd_tree<ROUND, NO_SUBNORMALS, AccExtraLSBs>(in1, ovr_out1);

    // Example call for ac_bfloat16 datatype
    // Using the default template parameters
    ac_math::fadd_tree(in2, default_out2);

    // Overriding the template parameters
    ac_math::fadd_tree<ROUND, NO_SUBNORMALS, AccExtraLSBs>(in2, ovr_out2);
```

6.11.5. Fused Addition Accuracy and Area

The `fadd_tree` function is a many-term floating-point addition operation that follows a fused architecture to improve latency and area. As a fused operation, the output of this function may differ from a value that has been computed using multiple add operations that are compliant to the standard.

Here is an example of a 3-operand addition with a rounding-to-nearest-even mode:

```
typedef ac_std_float<32,8> f1t;

f1t in[3] = {-3.281355e-37, -0.000983646, -0.03996174};

//A: Using 2 input standard add operations
f1t a = in[0] + in[1];
// the value of a is -0.000983646 because in[0] has been completely shifted out
f1t r0 = a + in[2];
the value of r0 is -0.040945385

//B: Using a 3-input fused addition
f1t r1;
ac_math::fadd_tree<3>(in, r1);
// the value of r1 is -0.04094539
```

The two results are different because in the first case, the sticky bit that was produced by the shifting of the "in[0]" operand could not affect the rounding during the addition of "in[0]" and "in[1]". Additionally, in the addition of "a" with "in[2]", the "in[2]" operand is shifted towards right, generating a round and a sticky bit, again not affecting the rounding. In the second case, however, the sticky bit of "in[0]" and the round and sticky bits of "in[2]" lead to the generation of a guard bit, that affects the rounding by adding +1 to the result. This proves that the result of a fused addition is not equal to the one generated by sequential use of standard add operations, however, the result of the fused addition is closer to the real value of the addition.

The default extra number of bits that participate in the computation and are used for accuracy during the normalization and rounding is 3, the guard, round, and sticky bits. The template parameter "AccExtraLSBs" offers the ability to change the number of LSBs that will be preserved to improve the accuracy of the result. Preserving more LSBs increases the number of operands in the accumulation and subsequently in the shifting operations.

The following illustration presents the accuracy of the fused addition for different number of inputs and for arbitrary number of extra LSBs. The accuracy is compared to the accurate result of the addition computed using the full precision of the operands during the addition and with a single rounding at the end of the operation. Each bar shows the number of tests (in percentage) that may have an error only at the LSB, meaning that the rest of the number's bits are completely equal to the accurate result. The darker part of each bar represents the percentage of the results that are completely equivalent to the accurate value.

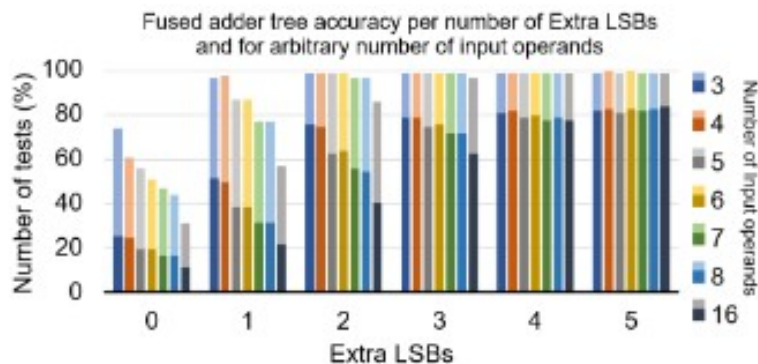


Figure 1: Accuracy of the fused addition compared to full precision addition

The dark colored bars show the percentage of test that were completely accurate. The light-colored part of the bars shows the percentage of tests that have an error only at the LSB. The combined bars show the percentage of tests that may only have an error at the LSB.

The previous illustration indicates that using 2 or 3 extra LSBs (3 is the default number of extra LSBs of the operator), the result can be very accurate, with less than 1% possibility to have an error in a higher order bit than the LSB. For a 16-input addition the accuracy is slightly degraded which is normal, due to the increased number of inputs. In general, when the number of input operands is highly increase, more extra LSBs are required to maintain high accuracy of the result.

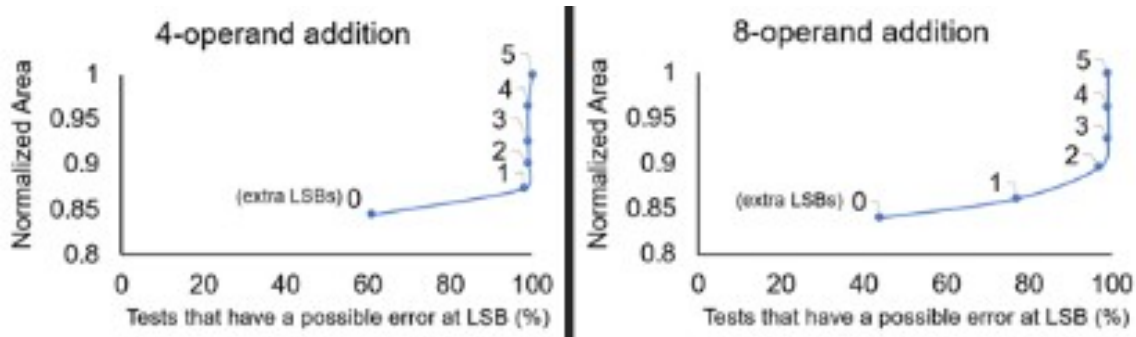


Figure 2: Area versus accuracy curves that prove the optimal number of extra LSBs for the 4- and 8-input operand fused addition

The previous illustration shows the accuracy versus normalized area tradeoff for the 4- and the 8-input operand fused addition. For the 4-input adder, using more than 1 extra LSBs has a bigger impact on the area with minimal increase in the accuracy of the result. A similar point for the 8-input fused adder would be at the 2 extra LSBs.

6.12. Sorting Networks (`ac_sort`)

The `ac_sort` library implementation provides sorting network implementations that use fixed sequence of compare-and-swap operations for structured parallel sorting to achieve predictable execution time and allow full unrolling in hardware.

6.12.1. `ac_sort` implementation

The `ac_sort` library implements three distinct sorting algorithms:

- Bubble sort
- Brick sort
- Bitonic sort

These sorting algorithms can be selected at synthesis time through directive configuration. The implementation allows customization of data types through template parameters, supporting both integer (`ac_int`) and fixed-point (`ac_fixed`) numbers with configurable bit widths and signedness. The design

parameters, including the number of elements to be sorted (*ArraySize*), sorting direction (*Ascending*) for each sorting network implementation are configurable.

6.12.2. Algorithm

Sorting network is a comparison network which is a set of comparators interconnected by wires. Each comparator input is connected to a wire that is either one of the network's n input wires ($a_1, a_2, \dots, a_n$) or is connected to the output of another comparator. Similarly, each comparator output is connected a wire that is either one the network's n output wires ($b_1, b_2, \dots, b_n$) or is connected to the input of another comparator.

6.12.3. Function header

```
// BUBBLE sort
template <typename DataType = char,
          uint32_t ArraySize = 16,
          bool Ascending = true>
class ac_bubble_sort : public ac_sort_base<DataType> {
public:
void hardwareAcceleratedSort(
    ac_array<DataType, ArraySize> &in_array,
    ac_array<DataType, ArraySize> &out_array)
};

// BRICK sort
template <typename DataType = char,
          uint32_t ArraySize = 16,
          bool Ascending = true>
class ac_brick_sort : public ac_sort_base<DataType> {
public:
void hardwareAcceleratedSort(
    ac_array<DataType, ArraySize> &in_array,
    ac_array<DataType, ArraySize> &out_array)
};

// BITONIC sort
template <typename DataType = char,
          uint32_t ArraySize = 32,
          bool Ascending = true,
          bool CCore = true>
class ac_bitonic_sort : public ac_sort_base<DataType> {
public:
void hardwareAcceleratedSort(
    ac_array<DataType, ArraySize> &in_array,
    ac_array<DataType, ArraySize> &out_array)
};
```

6.12.4. Template arguments

Design supports both integer (*ac_int*) and fixed-point (*ac_fixed*) data types with configurable bit-width and signedness. The *ArraySize* parameter is used to configure number of elements to be sorted (for bitonic sort number of elements must be power of 2). Ascending parameter is used to configure sorting direction. For bitonic sorting network *CCore* parameter is used to enable/ disable CCORE generation conditional swap logic to improve runtime for sorting networks with large number of input elements.

6.12.5. C++ compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, The user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

6.12.6. Example function call

```
template <typename dataType, uint32_t ArraySize, bool Ascending,
          bool CCORE_, int SORT_ALGORITHM>
class sort_wrapper {
public:
    sort_wrapper() {}

    void CCS_BLOCK(hardwareSorter) (
        ac_array<dataType, ArraySize> &in_array,
        ac_array<dataType, ArraySize> &out_array)
    {
        ac_math::ac_bitonic_sort<dataType, ArraySize, Ascending, CCORE_> sorter;
        sorter.hardwareAcceleratedSort(in_array, out_array);
    }
};
```

6.13. Parallel Search (*ac_search*)

The *ac_search* provides parallel search network implement a fixed pattern of comparisons. , these networks provide deterministic execution time and can be fully unrolled in hardware, enabling single-cycle throughput for streaming applications. The structured nature of parallel search also simplifies timing analysis and facilitates efficient pipelining.

6.13.1. *ac_search* implementation

ac_search is implemented as a templated C++ class. The design features configurable data types, supporting any comparable type through template parameterization, enabling efficient hardware mapping for different data representations. The array size is configurable but must be a power of 2 (such as 2, 4, 8, 16, etc.) to maintain the binary tree reduction structure. Additionally, the search direction can be configured at compile time to either find the maximum or minimum value in the input array.

6.13.2. Algorithm

A parallel search network is constructed as a binary tree of comparators, where each comparator performs either a maximum or minimum operation based on the search configuration. The network processes n input elements ($a_1, a_2, \dots, a_n$) through $\log_2(n)$ stages, where each stage reduces the number of active elements by half through parallel comparisons.

In each stage, pairs of elements are compared simultaneously, with the winner (maximum or minimum) propagating to the next stage. The binary tree structure ensures that after $\log_2(n)$ stages, the result emerges at the root of the tree.

6.13.3. Function header

```
template <typename DataType = char,
          uint32_t ArraySize = 16,
          bool FindMax = true>
class ac_parallel_search : public ac_search_base<DataType> {
public:
    void hardwareAcceleratedSearch(ac_array<DataType, ArraySize> &in_array,
                                   DataType &op_value)
};
```

6.13.4. Template arguments

Design supports both integer (*ac_int*) and fixed-point (*ac_fixed*) data types with configurable bit-width and signedness. the *ArraySize* parameter is used to configure number of input elements (must be a power of 2 to maintain the binary tree reduction structure). The *FindMax* parameter can be configured at compile time to either find the maximum or minimum value in the input array.

6.13.5. C++ Compiler

The functions use default template arguments. In order to use a C++ compiler that supports this functionality, the user must use C++11 or a later standard as the standard for their compilation, failing which a compile-time error is thrown.

6.13.6. Example function call

```
template <typename DataType, uint32_t ArraySize, bool FindMax>
class parallel_search_wrapper {
public:
    parallel_search_wrapper() {}

    ac_parallel_search<DataType, ArraySize, FindMax> searcher;

    void CCS_BLOCK(hardwareAcceleratedSearch) (
        ac_array<DataType, ArraySize> &in_array,
        DataType &op_value)
```

```
{
    searcher.hardwareAcceleratedSearch(in_array, op_value);
}
};
```

6.14. Optimized AC Float Multiplication (*ac_flfx_mul*)

The *ac_flfx_mul()* functions enable the instantiation of multiplication operations with mixed *ac_fixed* and *ac_float* datatypes. The purpose of these functions is to alleviate the overhead of converting between *ac_fixed* and *ac_float* datatypes when mixed datatype designs require multiplications between those two datatypes.

These functions support the following three implementations:

- Operation *ac_float* * *ac_fixed* = *ac_float*
- Operation *ac_float* * *ac_fixed* = *ac_fixed*
- Operation *ac_float* * *ac_float* = *ac_fixed*

These three variations are implemented by the templated function *ac_flfx_mul()*. The function has different implementations with different template parameters that allow the use of the appropriate variation depending on the input and output datatypes. For each variation of the function, there are code examples that show how the function is being called.

6.14.1. Operation *ac_float* * *ac_fixed* = *ac_float*

The function receives an *ac_fixed* and an *ac_float* input and returns an *ac_float* output. The user can interchange the order of the fixed- and floating-point inputs in the function call. Both functions should give the same output values and result in the same hardware.

```
template <
    bool subn_support = false,
    int W1, int I1, int E1, ac_q_mode Q1,
    int W2, int I2, bool S, ac_q_mode Q2, ac_o_mode O,
    int outW, int outI, int outE, ac_q_mode outQ
>
void ac_flfx_mul(
    const ac_float<W1, I1, E1, Q1> &in1,
    const ac_fixed<W2, I2, S, Q2, O> &in2,
    ac_float<outW, outI, outE, outQ> &out
)

template <
    bool subn_support = false,
    int W1, int I1, bool S, ac_q_mode Q1, ac_o_mode O,
    int W2, int I2, int E2, ac_q_mode Q2,
```



```
int outW, int outI, int outE, ac_q_mode outQ
>
void ac_flfx_mul(
    const ac_fixed<W1, I1, S, Q1, O> &in1,
    const ac_float<W2, I2, E2, Q2> &in2,
    ac_float<outW, outI, outE, outQ> &out
)
```

The `subn_support` template parameter determines whether subnormal floating point inputs/outputs are supported. Please refer to Subnormal Support for more details.

Only outputs where `outQ = AC_TRN` are supported.

Since the target output precision is known via template deduction, it can be fully leveraged during the computation of the output mantissa, output exponent (including all the necessary masking+saturation logic) and the final conversion to the floating point input via an `ac_float` constructor, all to produce optimal hardware.

Subnormal Support

By default, subnormal inputs are not handled. This means that if the floating point input and/or the expected output is subnormal, the actual output will be zero. Assuming that the floating point input and output are of type `ac_float<14, 2, 6>` and the fixed point input is of type `ac_fixed<10, 2, true>`, the expected and actual outputs for four different combinations of input values are given in the table below.

Float Input	Fixed Input	Expected Output	Actual Output	Reason for Actual Output
0.75e2-32	1.75	1.3125e2-32	0e2+0	<code>ac_float</code> input is subnormal. Actual output is hence zero.
1e2-32	0.75	.75e2-32	0e2+0	Expected output is subnormal. Actual output is hence zero.
.75e2-32	0.75	.5625e2-32	0e2+0	Both <code>ac_float</code> input and expected output are subnormal. Actual output is hence zero.
1.75e2-32	0.75	1.3125e2-32	1.3125e2-32	Both <code>ac_float</code> input and expected output are normal. Actual and expected outputs are hence the same.

Not having subnormal support can lead to discrepancies like those seen between the first and last rows, where the actual output differs despite the expected output being the same. This is because only floating point values can be subnormal. If this discrepancy is tolerable and the user decides to use `subn_support = false`, they can avail of the reduction in normalization hardware as we only need to fully normalize the fixed point input. This normalized fixed point value gets multiplied directly with the floating point input mantissa to generate the "mantissa product", or "*mprod*". At this stage, since the `ac_float` input mantissa and normalized `ac_fixed` input are both assumed to be normal (with the exception of zero inputs), *mprod* is mostly normalized and we only need to calculate the `leading_sign()` output by traversing 3 or 2 MSBs of *mprod* if the fixed point input is *signed* or *unsigned*, respectively. This results in a smaller `leading_sign` operation for this stage and a smaller shifter, as the value to be shifted by is at most 2 or 1 for *signed* or *unsigned* `ac_fixed` inputs, respectively. In addition to this, we do not have to right-shift the output mantissa for subnormal outputs, which also saves on a shifter and further reduces area.

If, however, the discrepancy is not tolerable, the user can choose to make the actual and expected outputs always match by setting *subn_support* to *true*. In this case, the *ac_float* input mantissa is directly multiplied with the *ac_fixed* input and all of *mprod* is treated as not normalized. The *leading_sign()* operator and shifter required to normalize *mprod* are hence significantly bigger. In addition to that, an extra shifter is needed to right-shift the output mantissa for subnormal outputs, further increasing area.

Note: Regardless of whether *subn_support* is true or false, the design will output zero if one or both of the inputs are zero, as one would expect of a multiplier.

Example Function Calls

The snippet below has example function calls for a few possible use cases.

```
// Define input and output types.
typedef ac_float<14, 2, 6> fl_in_type, out_type,
typedef ac_fixed<10, 2, true> fx_in_type;

// Initialize input variables.
fl_in_type fl_in = 1.5;
fx_in_type fx_in = 0.5;
// Declare output variables.
out_type out1, out2, out3, out4;

// ac_float input first, subn_support = false (default).
ac_math::ac_flfx_mul(fl_in, fx_in, out1); // out1 = 1.5e2-1
// ac_fixed output first, subn_support = false (default).
ac_math::ac_flfx_mul(fx_in, fl_in, out2); // out2 = 1.5e2-1

// ac_float input first, subn_support = true.
ac_math::ac_flfx_mul<true>(fl_in, fx_in, out3); // out3 = 1.5e2-1
// ac_fixed output first, subn_support = true.
ac_math::ac_flfx_mul<true>(fx_in, fl_in, out4); // out4 = 1.5e2-1
```

The floating point input and output are both big enough to be expressed as normalized values so there's no difference in the output value produced when *subn_support* = *true* and *subn_support* = *false*. However, a detailed explanation of how subnormal inputs/outputs are handled is given in Subnormal Support.

6.14.2. Operation *ac_float* * *ac_fixed* = *ac_fixed*

The function receives an *ac_fixed* and an *ac_float* input and returns an *ac_fixed* output. The user can interchange the order of the fixed- and floating-point inputs in the function call. Both functions should give the same output values and result in the same hardware.

```
template <
    bool subn_support = false,
    int W1, int I1, int E1, ac_q_mode Q1,
    int W2, int I2, bool S2, ac_q_mode Q2, ac_o_mode O2,
    int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO
```

```
>
void ac_flfx_mul(
    const ac_float<W1, I1, E1, Q1> &in1,
    const ac_fixed<W2, I2, S2, Q2, O2> &in2,
    ac_fixed<outW, outI, outS, outQ, outO> &out
)

template <
    bool subn_support = false,
    int W1, int I1, bool S1, ac_q_mode Q1, ac_o_mode O1,
    int W2, int I2, int E2, ac_q_mode Q2,
    int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO
>
void ac_flfx_mul(
    const ac_fixed<W1, I1, S1, Q1, O1> &in1,
    const ac_float<W2, I2, E2, Q2> &in2,
    ac_fixed<outW, outI, outS, outQ, outO> &out
)
```

The *subn_support* template parameter determines whether subnormal floating-point inputs are supported. Like the previous case, the default value is *false*. A detailed explanation of how subnormal inputs/outputs are handled is given in Subnormal Support.

Note: In this case, only the input can be subnormal as the output is a fixed-point datatype. This means that when *subn_support* = *false*, the operator has increased area as it utilizes extra gates to identify a subnormal input and drive the result to a zero value.

Example Function Calls

The snippet below has example function calls for a few possible use cases.

```
// Define input and output types.
typedef ac_float<14, 2, 6> fl_in_type;
typedef ac_fixed<10, 2, true> fx_in_type;
typedef ac_fixed<87,35,true> out_type;

// Initialize input variables.
fl_in_type fl_in = 1.5;
fx_in_type fx_in = 0.5;
// Declare output variables.
out_type out1, out2, out3, out4;

// ac_float input first, subn_support = false (default).
ac_math::ac_flfx_mul(fl_in, fx_in, out1); // out1 = 0.75

// ac_fixed input first, subn_support = false (default).
ac_math::ac_flfx_mul(fx_in, fl_in, out2); // out2 = 0.75
```

```
// ac_float input first, subn_support = true.
ac_math::ac_flfx_mul<true>(fl_in, fx_in, out3); // out3 = 0.75

// ac_fixed input first, subn_support = true.
ac_math::ac_flfx_mul<true>(fx_in, fl_in, out4); // out4 = 0.75
```

6.14.3. Operation `ac_float * ac_float = ac_fixed`

The function receives two *ac_float* inputs and returns an *ac_fixed* output.

```
template <
    bool subn_support = false,
    int W1, int I1, int E1, ac_q_mode Q1,
    int W2, int I2, int E2, ac_q_mode Q2,
    int outW, int outI, bool outS, ac_q_mode outQ, ac_o_mode outO
>

void ac_flfx_mul(
    const ac_float<W1, I1, E1, Q1> &in1,
    const ac_float<W2, I2, E2, Q2> &in2,
    ac_fixed<outW, outI, outS, outQ, outO> &out
)
```

The function performs an *ac_float* multiplication “*”. The datatype of the product is in full data width precision so that there will be no normalization. The product of the mantissas is then shifted according to exponent value and the result is casted to the output fixed-point datatype. If the value is shifted more than the available bit width, the value gets saturated.

The *subn_support* template parameter is also available in this implementation. Like the *ac_fixed * ac_float = ac_fixed* case, *subn_support = false* result in increase hardware due to the extra logic to identify a subnormal value. A detailed explanation of how subnormal inputs/outputs are handled is given in Subnormal Support.

Example Function Calls

The snippet below has example function calls for a few possible use cases.

```
// Define input and output types.
typedef ac_float<14, 2, 6> fl_in1_type;
typedef ac_float<6, 2, 3> fl_in2_type;
typedef ac_fixed<90,38,true> out_type;

// Initialize input variables.
fl_in1_type fl_in = 1.5;
fl_in2_type fx_in = 0.5;
// Declare output variables.
out_type out1, out2;
```

```
// subn_support = false (default).  
ac_math::ac_flfx_mul(fl_in, fx_in, out1); // out1 = 0.75  
  
// subn_support = true.  
ac_math::ac_flfx_mul<true>(fl_in, fx_in, out2); // out2 = 0.75
```